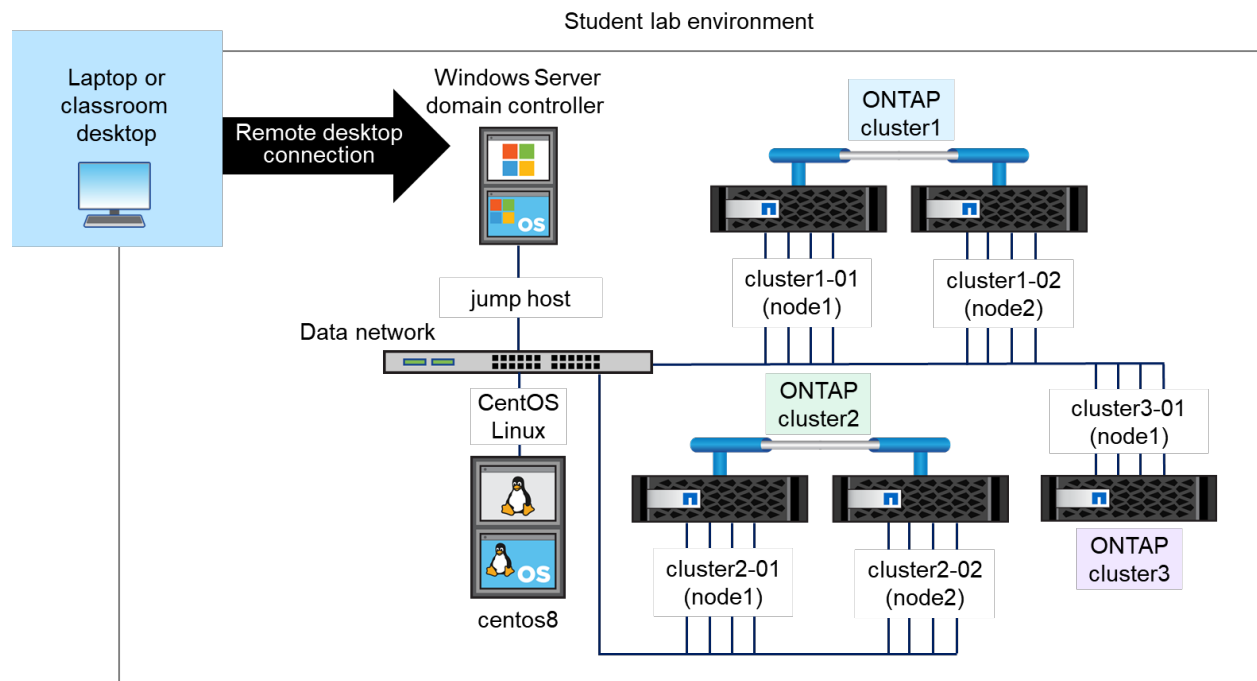


Module 2: SnapMirror for Disaster Recovery and Backup

Exercise Equipment Diagram



System	Host Name	IP Addresses	User Name	Password
Windows Server 2022 Datacenter	JUMPHOST	192.168.0.5	DEMO\Administrator	Netapp1!
ONTAP cluster-management LIF (Cluster1)	Cluster1	192.168.0.101	admin (case-sensitive)	Netapp1!
node 1 (Cluster1)	cluster1-01-console	192.168.0.111	admin (case-sensitive)	Netapp1!
node 2 (Cluster1)	cluster1-02-console	192.168.0.112	admin (case-sensitive)	Netapp1!
ONTAP cluster-management LIF (Cluster2)	Cluster2	192.168.0.102	admin (case-sensitive)	Netapp1!
node 1 (Cluster2)	cluster2-01-console	192.168.0.113	admin (case-sensitive)	Netapp1!
node 2 (Cluster2)	cluster2-02-console	192.168.0.114	admin (case-sensitive)	Netapp1!
ONTAP cluster-management LIF (Cluster3)	Cluster3	192.168.0.103	admin (case-sensitive)	Netapp1!
node 1 (Cluster3)	cluster3-01-console	192.168.0.115	admin (case-sensitive)	Netapp1!
Linux Server	centos8	192.168.0.21	root	Netapp1!

Exercise 1(a): Using SnapMirror to Mirror FlexVol Volumes

In this exercise, you create and implement a SnapMirror relationship between a source volume on Cluster1 and a destination volume on Cluster2. You follow a SnapMirror implementation workflow to implement the solution. Then you verify data transfer.

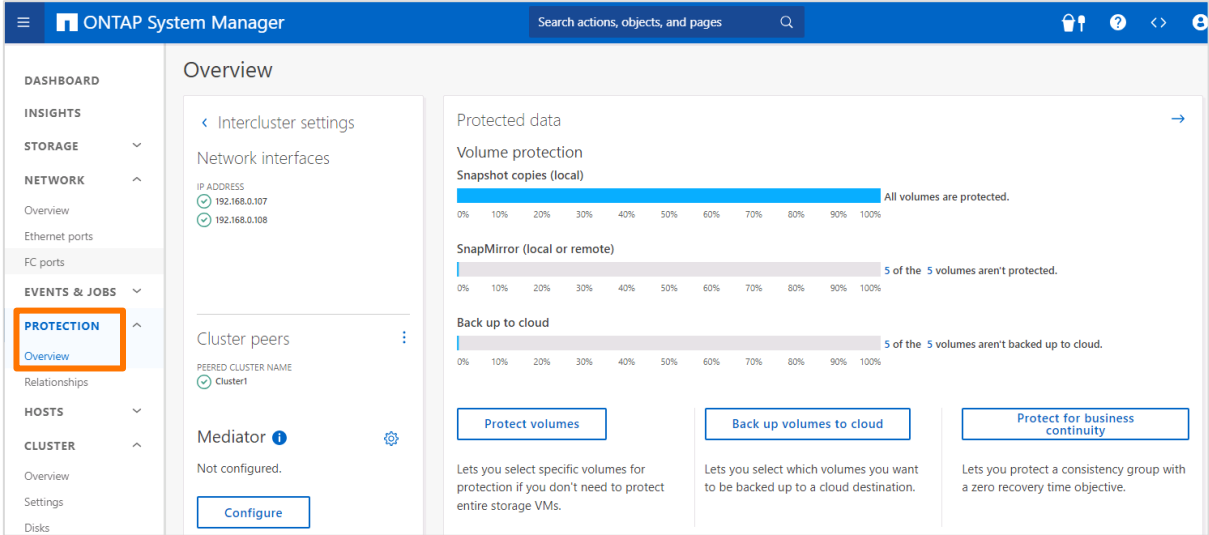
Objectives

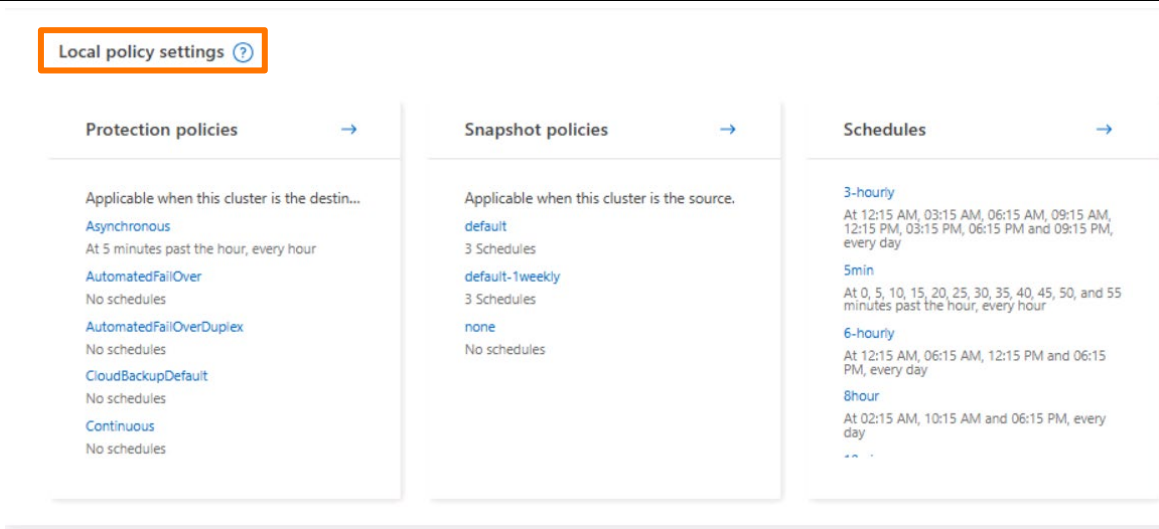
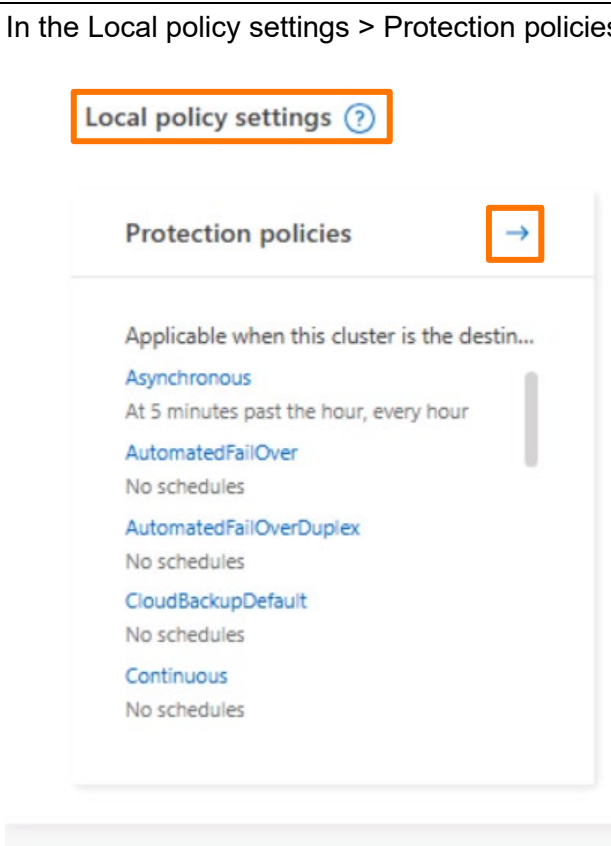
This exercise focuses on enabling you to do the following:

- Create a SnapMirror policy
- Create a SnapMirror relationship
- Perform an initial SnapMirror transfer
- Perform a manual SnapMirror update
- Schedule automatic SnapMirror updates
- Verify data transfer

Task 1: Create a SnapMirror Policy and Configure a SnapMirror Relationship

In this task, you use NetApp ONTAP System Manager to create a SnapMirror policy and relationship, and you configure SnapMirror updates.

Step	Action
1-1	 Cluster2 is the destination cluster for the protection relationship. You create the custom SnapMirror policy on the destination cluster.
1-2	In the System Manager navigation pane for Cluster2, select Protection > Overview. 
1-3	Scroll down to the Local policy settings section and then expand Local policy settings .

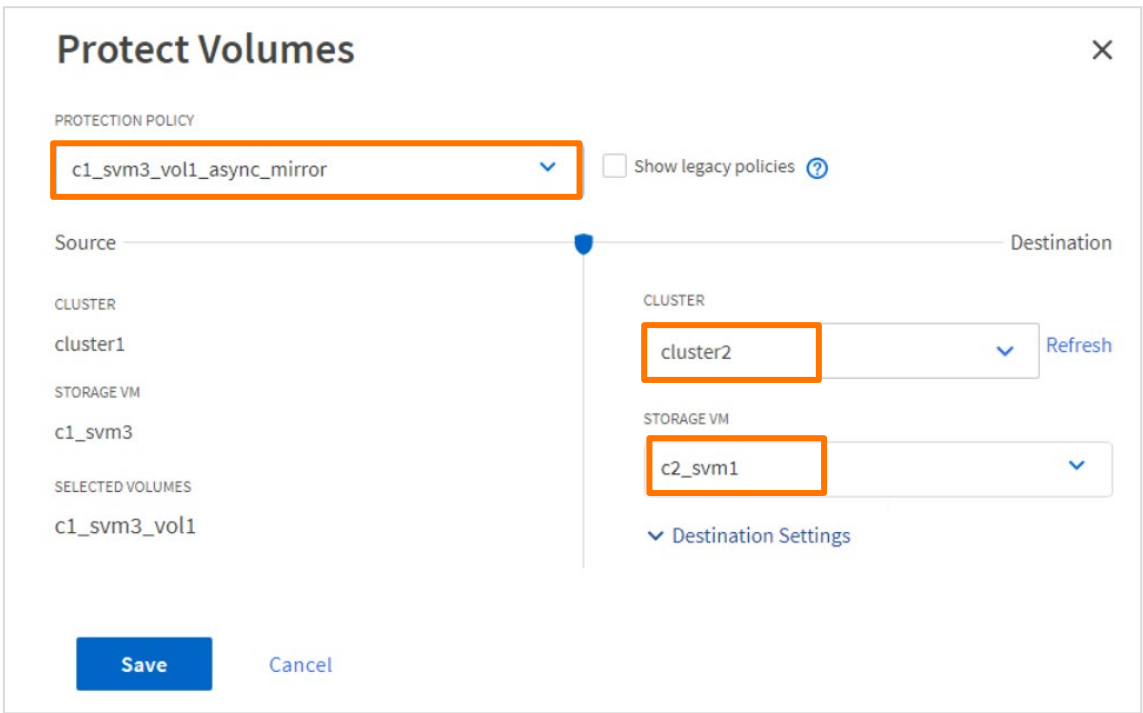
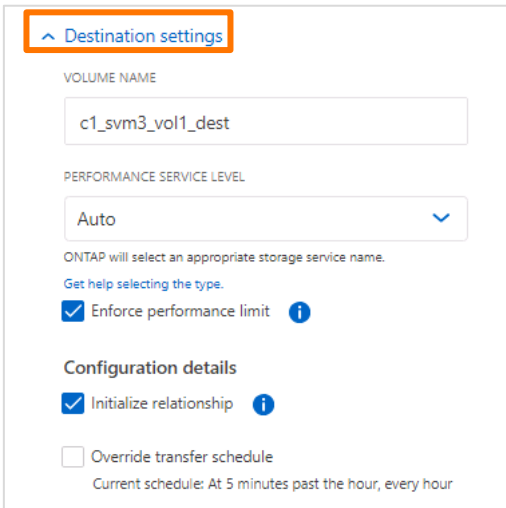
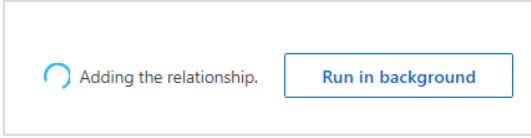
Step	Action
	
1-4	In the Local policy settings > Protection policies pane, click the arrow in the upper-right corner. 
1-5	In the Policies > Protection policies pane, click Add to create a policy.

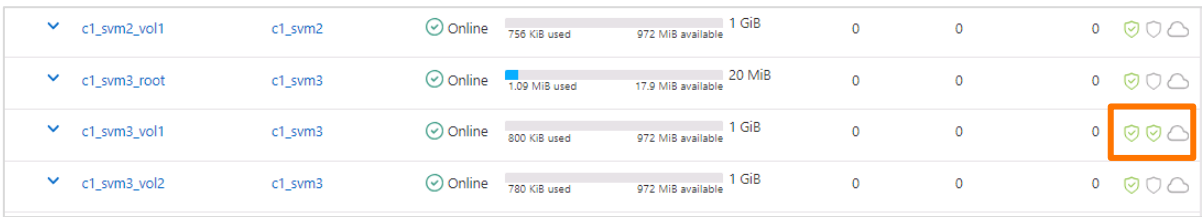
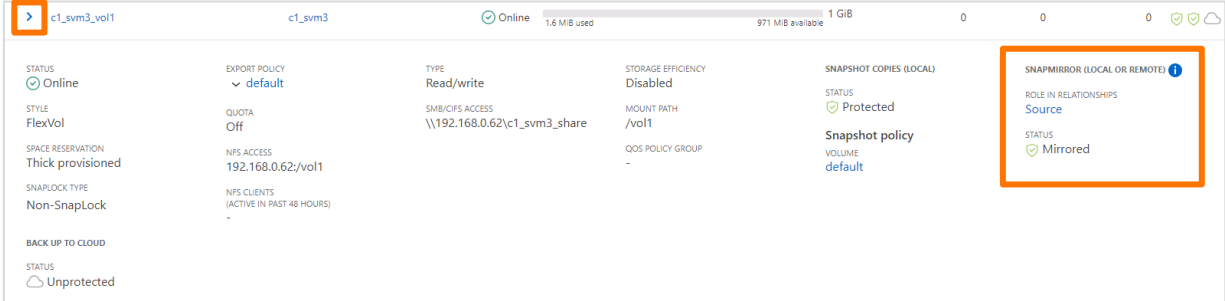
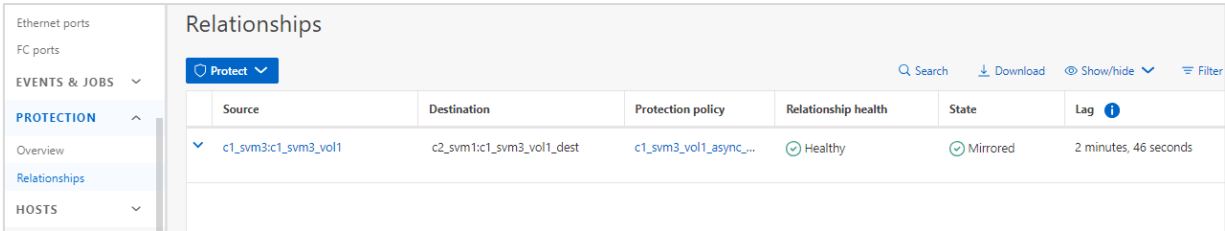
Step	Action																												
	Policies Protection overview Protection policies Snapshot policies + Add <table><thead><tr><th></th><th>Name</th><th>Description</th><th>Policy type</th></tr></thead><tbody><tr><td>▼</td><td>Asynchronous</td><td>A unified Asynchronous SnapMirror and SnapVault policy for mirroring ...</td><td>Asynchronous</td></tr><tr><td>▼</td><td>AutomatedFailOver</td><td>Policy for SnapMirror Synchronous with zero RTO guarantee where clie...</td><td>Synchronous</td></tr><tr><td>▼</td><td>CloudBackupDefault</td><td>Vault policy with daily rule.</td><td>Asynchronous</td></tr><tr><td>▼</td><td>Continuous</td><td>Policy for S3 bucket mirroring.</td><td>Continuous</td></tr><tr><td>▼</td><td>DailyBackup</td><td>Vault policy with a daily rule and a daily transfer schedule.</td><td>Asynchronous</td></tr><tr><td>▼</td><td>DPDefault</td><td>Asynchronous SnapMirror policy for mirroring all Snapshot copies and ...</td><td>Asynchronous</td></tr></tbody></table>		Name	Description	Policy type	▼	Asynchronous	A unified Asynchronous SnapMirror and SnapVault policy for mirroring ...	Asynchronous	▼	AutomatedFailOver	Policy for SnapMirror Synchronous with zero RTO guarantee where clie...	Synchronous	▼	CloudBackupDefault	Vault policy with daily rule.	Asynchronous	▼	Continuous	Policy for S3 bucket mirroring.	Continuous	▼	DailyBackup	Vault policy with a daily rule and a daily transfer schedule.	Asynchronous	▼	DPDefault	Asynchronous SnapMirror policy for mirroring all Snapshot copies and ...	Asynchronous
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1-6	In the Add protection policy dialog box, specify the following values: - Policy Name: c1_svm3_vol1_async_mirror - Policy Description: cluster1:svm3_vol1 – Async Mirror policy - Policy Scope: Cluster - Policy type: Asynchronous - Transfer Schedule: daily - Enable network compression: <i>Clear checkbox</i> - Identify preserve: <i>Clear checkbox</i> - Copy all source Snapshot copies: <i>Select checkbox</i>																												

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1-9	i Cluster1 is the source cluster in the protection relationship. Next, you protect the volume c1_svm3_vol1 on c1_svm3 with the SnapMirror policy that you created.																																																
1-10	In the System Manager navigation pane for Cluster1, select Storage > Volumes .																																																
1-11	Dashboard Insights Storage Overview Volumes LUNs NVMe namespaces Consistency groups Shares Buckets Qtrees Quotas Storage VMs Tiers Volumes + Add ⋮ More <table><thead><tr><th></th><th>Name</th><th>Storage VM</th><th>Status</th><th>↓</th><th>Protection</th></tr></thead><tbody><tr><td>▼</td><td>c1_svm2_root</td><td>c1_svm2</td><td>Online</td><td></td><td> ✓ 🛡 ☁ </td></tr><tr><td>▼</td><td>c1_svm3_root</td><td>c1_svm3</td><td>Online</td><td></td><td> ✓ 🛡 ☁ </td></tr><tr><td>▼</td><td>c1_svm2_vol1</td><td>c1_svm2</td><td>Online</td><td></td><td> ✓ 🛡 ☁ </td></tr><tr><td>▼</td><td>c1_svm3_vol1</td><td>c1_svm3</td><td>Online</td><td></td><td> ✓ 🛡 ☁ </td></tr><tr><td>▼</td><td>c1_svm3_vol2</td><td>c1_svm3</td><td>Online</td><td></td><td> ✓ 🛡 ☁ </td></tr><tr><td>▼</td><td>c1_svm1_root</td><td>c1_svm1</td><td>Online</td><td></td><td> ✓ 🛡 ☁ </td></tr><tr><td>▼</td><td>c1_svm1_vol1</td><td>c1_svm1</td><td>Online</td><td></td><td> ✓ 🛡 ☁ </td></tr></tbody></table>		Name	Storage VM	Status	↓	Protection	▼	c1_svm2_root	c1_svm2	Online		✓ 🛡 ☁	▼	c1_svm3_root	c1_svm3	Online		✓ 🛡 ☁	▼	c1_svm2_vol1	c1_svm2	Online		✓ 🛡 ☁	▼	c1_svm3_vol1	c1_svm3	Online		✓ 🛡 ☁	▼	c1_svm3_vol2	c1_svm3	Online		✓ 🛡 ☁	▼	c1_svm1_root	c1_svm1	Online		✓ 🛡 ☁	▼	c1_svm1_vol1	c1_svm1	Online		✓ 🛡 ☁
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1-12	i In the Protection column, the first icon in the column indicates protection with Snapshot copies, the second icon indicates protection with SnapMirror (local or remote), and the third icon indicates protection with backup to cloud.																																																
1-13	Click the three vertical dots to the right of c1_svm3_vol1 and then select Protect .																																																

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1-14	Alternatively, select the c1_svm3_vol1 checkbox and then click Protect at the top of the Volumes list.																								

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☒	c1_svm3_vol1	c1_svm3	Online																														
☐	c1_svm3_vol2	c1_svm3	Online																														
1-15	In the Protect volume dialog box, click the down arrow and then select c2_svm1 as the destination storage VM (storage virtual machines, also known as SVM). Protect volume SOURCE CLUSTER Cluster1 PROTECTION POLICY Asynchronous SOURCE VOLUME c1_svm3_vol1 DESTINATION CLUSTER Cluster2 Refresh STORAGE VM c2_svm1 More options Cancel Protect																																
1-16	Click More options .																																
1-17	In the Protect Volumes pane, specify the following values: - Protection Policy: c1_svm3_vol1_async_mirror - Destination > Cluster: cluster2 - Destination > Storage VM: c2_svm1																																

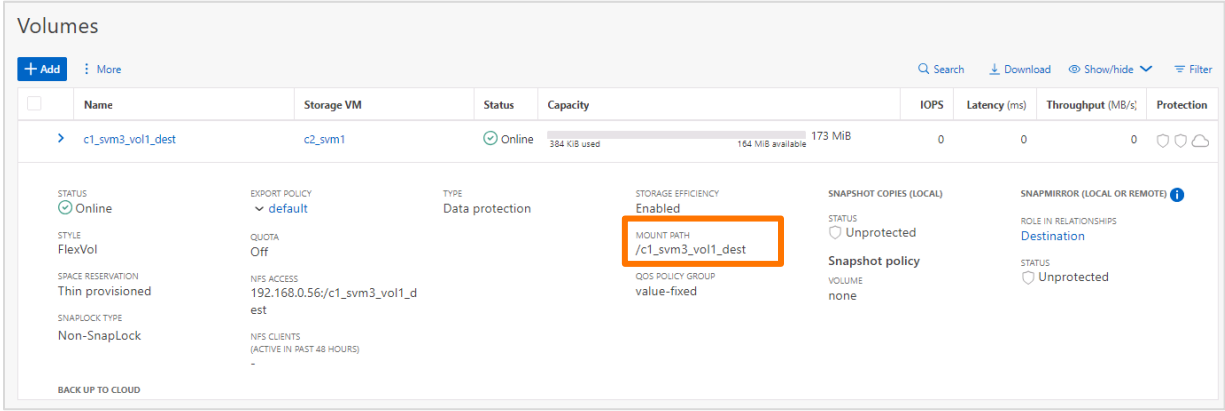
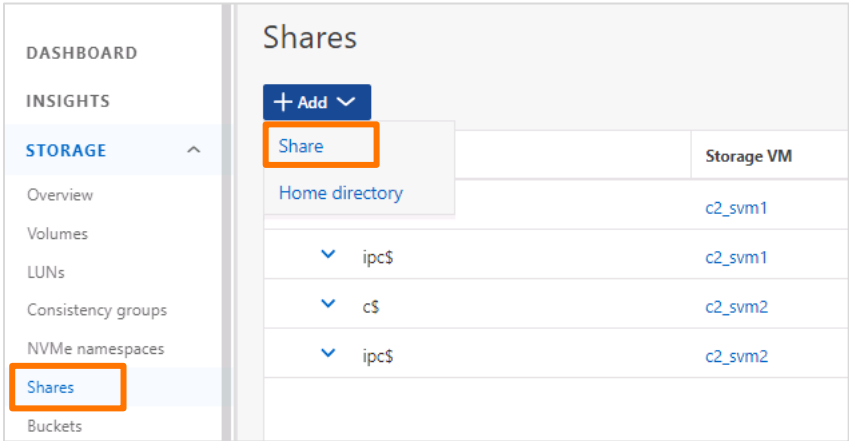
Step	Action
	
1-18	Expand Destination settings. 
1-19	Review the Destination settings values, but do not make any changes.
1-20	Click Save .
1-21	Notice the appearance of the Adding the Relationship status box for a few seconds or click Run in background. 
1-22	In the Volumes pane, notice the Protection status for c1_svm3_vol1.

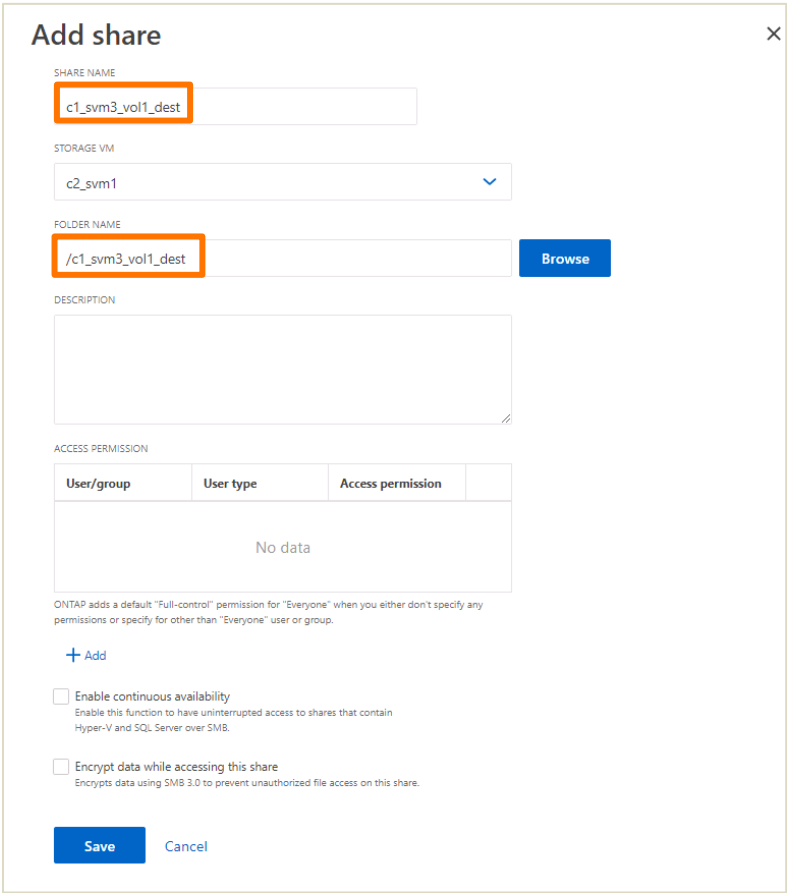
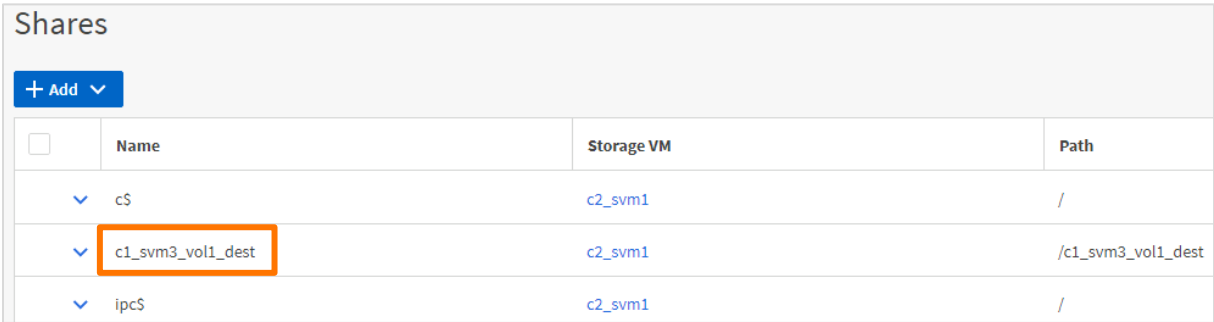
Step	Action
	
1-23	In the Volumes pane, click the down arrow to the left of volume c1_svm3_vol1 to view details and verify that the protection relationship was created successfully. 
1-24	Return to System Manager on Cluster2.
1-25	In the navigation pane, select Protection > Relationships .
1-26	To verify the SnapMirror status, expand the relationship that is listed. 
1-27	Answer the following questions: - Is the relationship healthy? _____ - What is the relationship state? _____ - What is the policy type? _____

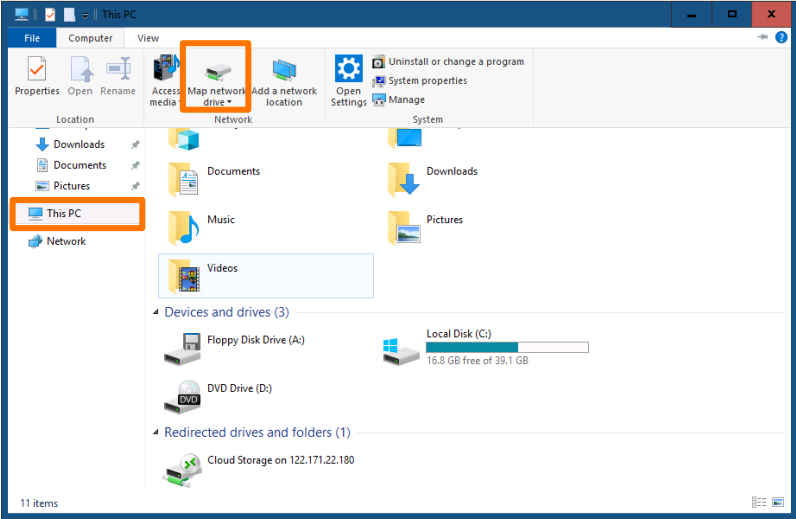
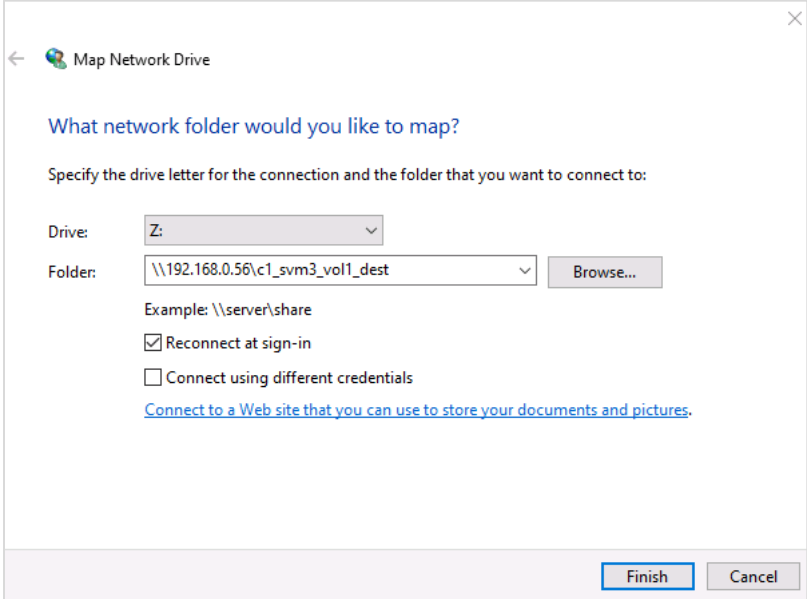
Task 2: Verify Data Transfer

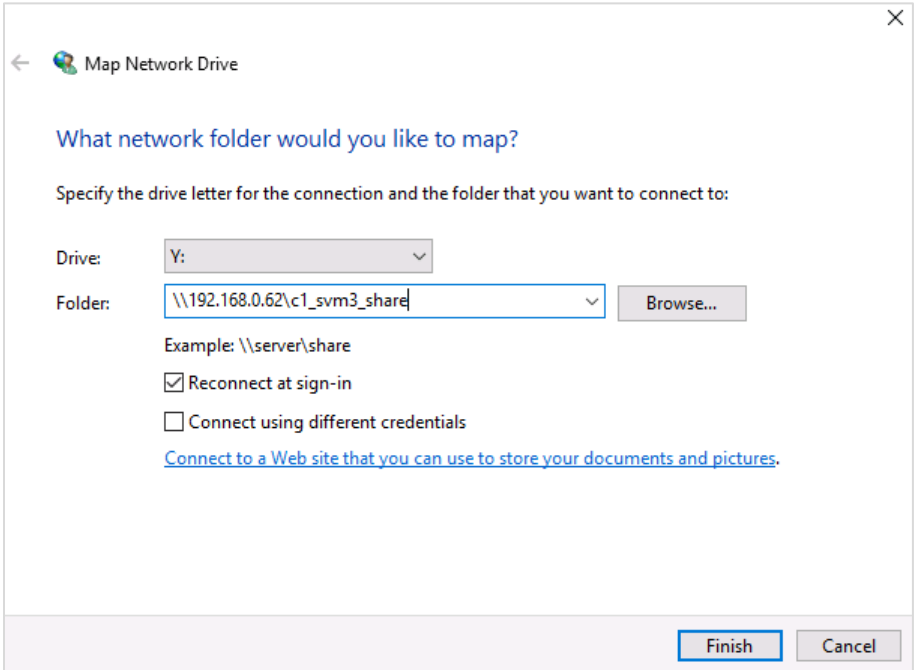
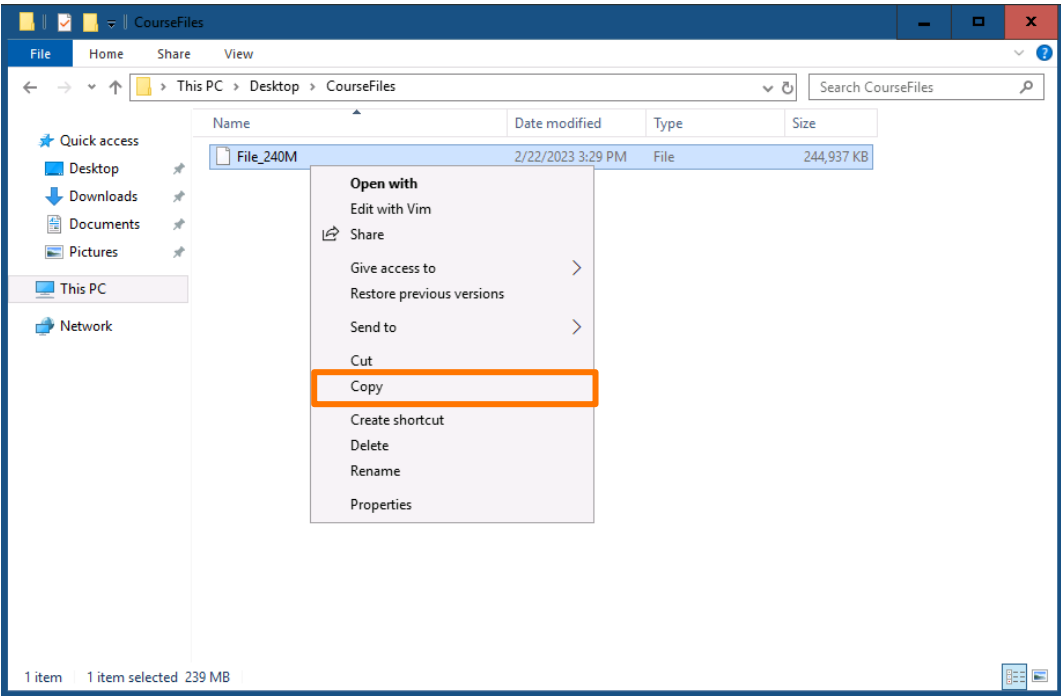
In this task, you first look at the SnapMirror destination volume to see the initialized data. You then create data on the primary volume. You perform a manual update and then verify that the new data was transferred to the SnapMirror destination volume.

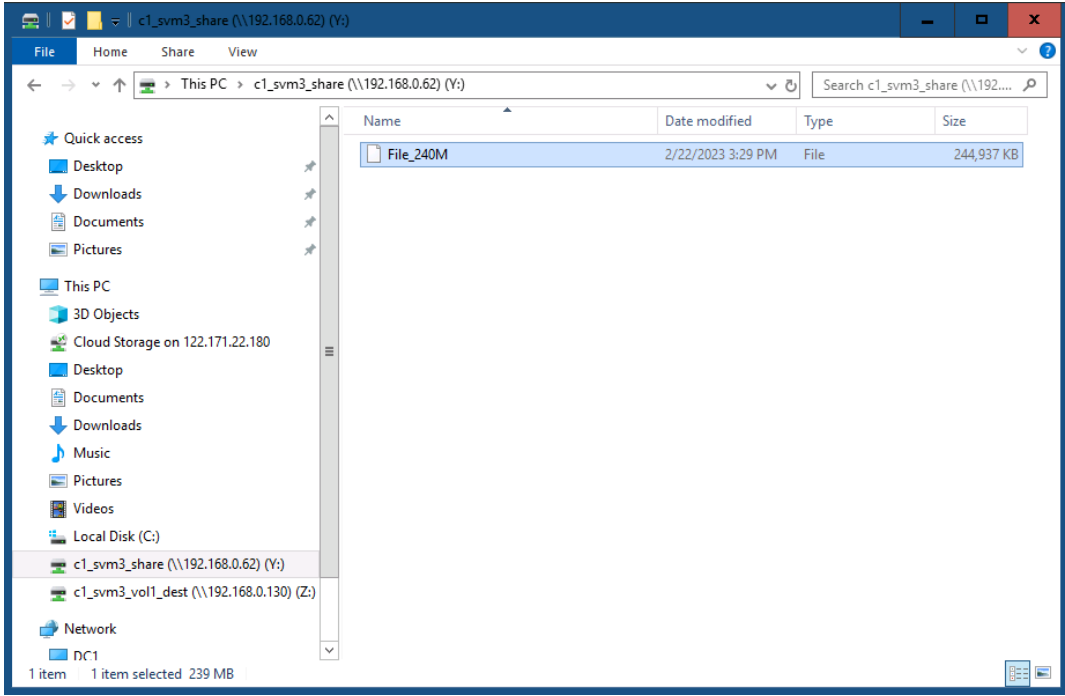
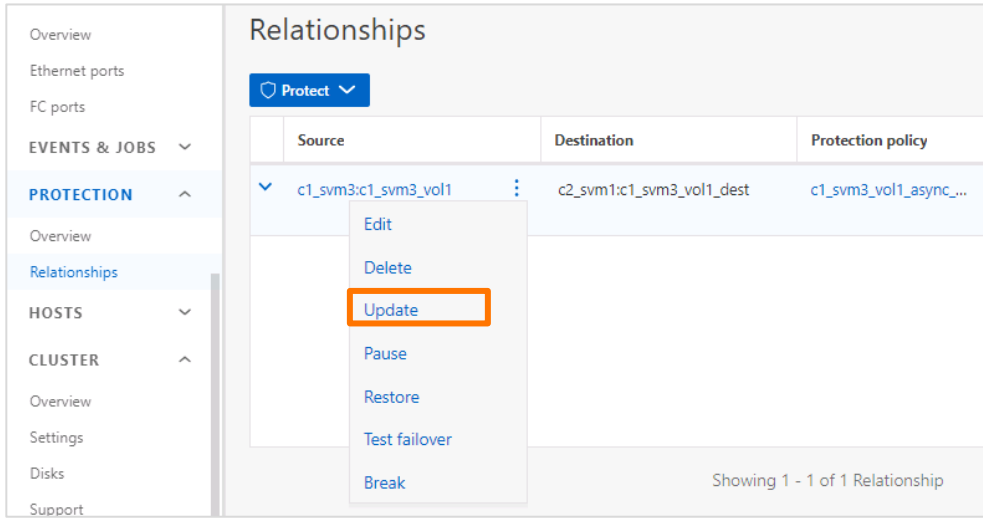
Step	Action																																
2-1	i During the creation and initialization of the SnapMirror relationship, the SnapMirror secondary volume is not automatically mounted. To see the data that is copied to the SnapMirror secondary volume, you must first mount the secondary volume in the cluster namespace. In the following steps, you mount the secondary volume in the Cluster2 namespace.																																
2-2	In the System Manager navigation pane for Cluster2, select Storage > Volumes .																																
2-3	Click the three vertical dots to the right of c1_svm3_vol1_dest and then select Edit > Mount path. Volumes + Add : More <table><tr><th>☐</th><th>Name</th><th></th><th>Storage VM</th></tr><tr><td>☒</td><td>c1_svm3_vol1_dest</td><td> Edit Resize Delete Clone Take offline Move Protect </td><td>c2_svm1</td></tr><tr><td>☒</td><td>c2svm2_root</td><td></td><td>Volume</td></tr><tr><td>☒</td><td>c2_svm3_root</td><td></td><td>Cloud tier settings</td></tr><tr><td>☒</td><td>c2_svm4_root</td><td></td><td>Export policy</td></tr><tr><td>☒</td><td>FlexGroup1</td><td></td><td>Mount path</td></tr><tr><td>☒</td><td>svm0_root</td><td></td><td>c2_svm2</td></tr><tr><td>☒</td><td></td><td></td><td>c2_svm1</td></tr></table>	☐	Name		Storage VM	☒	c1_svm3_vol1_dest	Edit Resize Delete Clone Take offline Move Protect	c2_svm1	☒	c2svm2_root		Volume	☒	c2_svm3_root		Cloud tier settings	☒	c2_svm4_root		Export policy	☒	FlexGroup1		Mount path	☒	svm0_root		c2_svm2	☒			c2_svm1
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☒			c2_svm1																														
2-4	In the Edit mount path pane, select the Mount checkbox and then click Save. Edit mount path VOLUME c1_svm3_vol1_dest ☒ Mount PATH /c1_svm3_vol1_dest Browse Save Cancel																																

Step	Action
2-5	In the Volumes pane, expand c1_svm3_vol1_dest and view its mount path. 
2-6	 In the following steps, you create a share for the volume that you added to the namespace.
2-7	In the System Manager navigation pane for Cluster2, select Storage > Shares.
2-8	In the Shares pane, click Add and then select Share. 

Step	Action
2-9	In the Add share pane, specify the following values: - Share Name: c1_svm3_vol1_dest - Folder Name: /c1_svm3_vol1_dest (or click Browse and then select the folder) - Description: <i>Clear field</i> 
2-10	Click Save and then verify that the new share is listed in the Shares pane. 
2-11	 Next, on the jump host, you map a drive to the destination volume.
2-12	On the jump host, open File Explorer.

Step	Action
2-13	Click This PC and then, in the ribbon, select Computer > Network > Map network drive. 
2-14	To map a drive to the destination volume on cluster2\c2_svm1, use the folder path \\192.168.0.56\c1_svm3_vol1_dest. 
2-15	Click Finish .

Step	Action
2-16	On the jump host, use File Explorer to map a drive to the source volume, and use the folder path <code>\\192.168.0.62\c1_svm3_share</code>. 
2-17	Click Finish .
2-18	Notice that both the source drive and destination drive are empty.
2-19	In File Explorer, navigate to C: > CourseFiles and then copy the file File_240M. 
2-20	Alternatively, you can find the file on the Desktop.

Step	Action
2-21	Paste the File_240M file to the mapped drive c1_svm3_share (\\192.168.0.62). 
2-22	In the System Manager navigation pane for Cluster2, select Protection > Relationships.
2-23	In the Relationships pane, click the three vertical dots to the right of the source volume and then select Update to perform a manual update. 

Step

Action

2-24

In the Update relationship dialog box, click **Update** without changing any settings.

Update relationship

×

Updates the data from the source volume to the destination volumes.
This operation will continue to run in the background. You can check the transfer status in the relationship.

Source

STORAGE VM
c1_svm3
VOLUME
c1_svm3_vol1

As per policy ⓘ

Select the Snapshot copy

Destination

STORAGE VM
c2_svm1
VOLUME
c1_svm3_vol1_dest

Cancel

Update

2-25

Expand the relationship and wait for the Transfer Status value to change from Transferring to Success.

Relationships

Protect

Source

c1_svm3:c1_svm3_vol1

Destination

c2_svm1:c1_svm3_vol1_dest

Protection policy

c1_svm3_vol1_async_mirror

Relationship health

Healthy

State

Mirrored

SOURCE CLUSTER
Cluster1

EXPORTED SNAPSHOT COPY
snapmirror.058e3793-ec41-11ee-9a76-005056b0d71e_2154036400.2024-04-01_123129

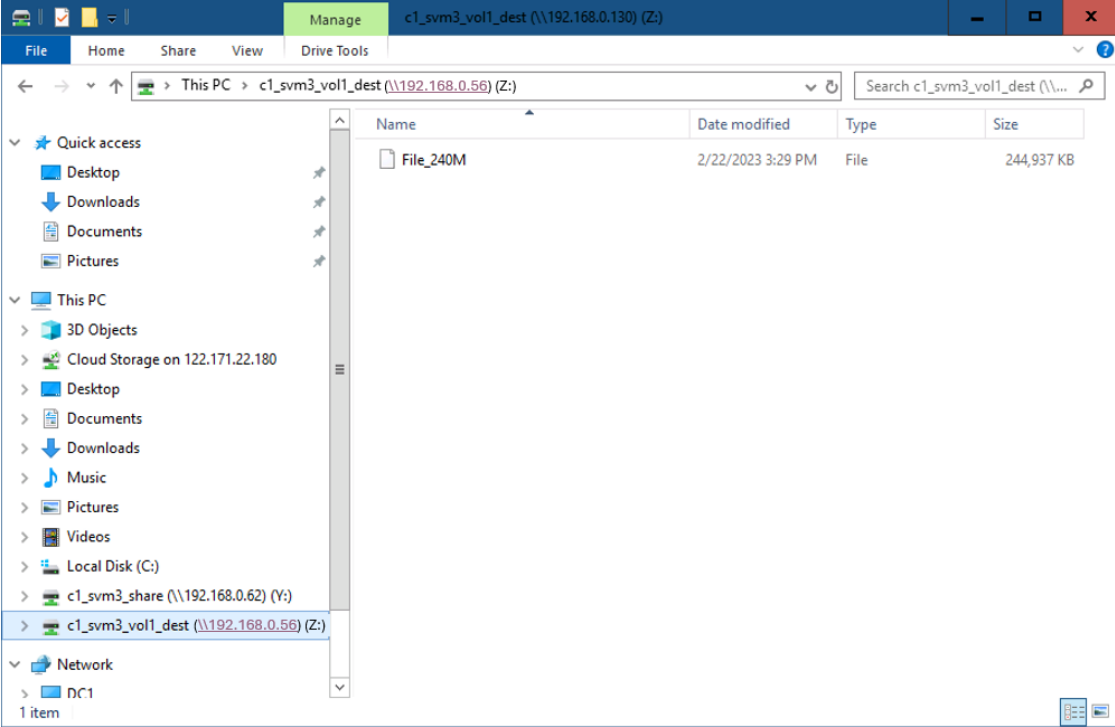
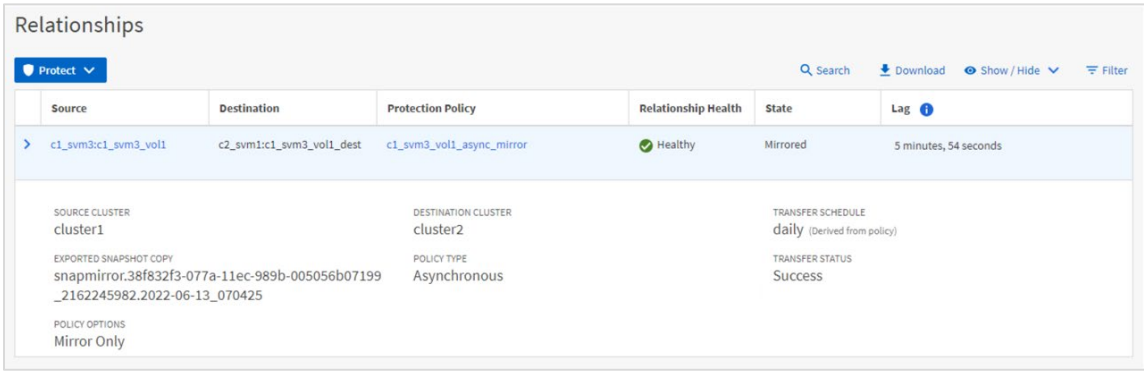
POLICY OPTIONS
Mirror only

DESTINATION CLUSTER
Cluster2

POLICY TYPE
Asynchronous

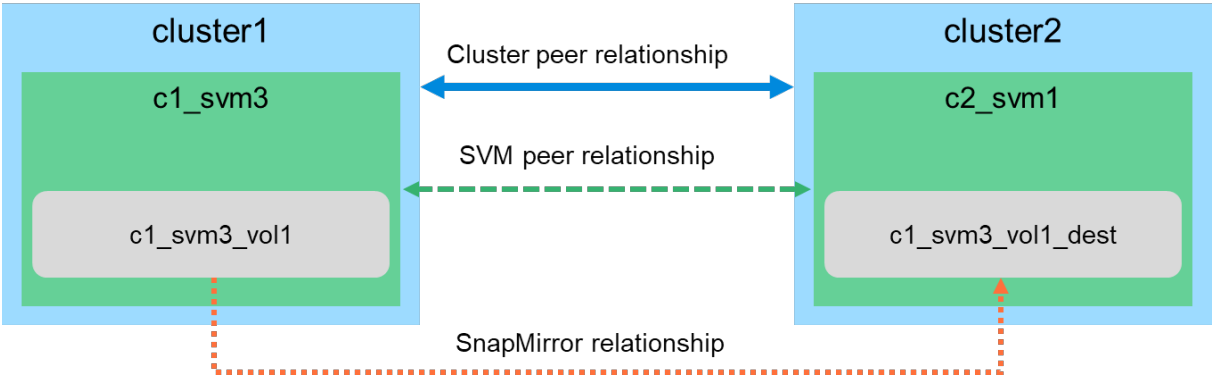
TRANSFER SCHEDULE
hourly (Derived from policy)

TRANSFER STATUS
Success

Step	Action
2-26	In File Explorer, open the mapped drive c1_svm3_vol1_dest (\\192.168.0.56) and verify that the SnapMirror relationship transferred the file to the destination volume. 
2-27	In the System Manager navigation pane for Cluster2, select Protection > Relationships.
2-28	In the Relationships pane, monitor the status of SnapMirror data transfers to verify that the transfers occur on the specified schedule and that the following statements are true: - The Relationship Health value is Healthy. - The State field value is Mirrored. - The Lag time value is no greater than the transfer schedule interval. 

Task 3: Review the Exercise Environment

In this task, you review the exercise environment that you created in the previous tasks.

Step	Action
3-1	Verify the configuration of the SnapMirror relationship. 

End of exercise

Exercise 1(b): Performing SnapMirror Disaster Recovery

In this exercise, you simulate a disaster on the source volume. You make the destination volume data accessible to clients. You then repair the simulated disaster and return the SnapMirror relationship to its original configuration.

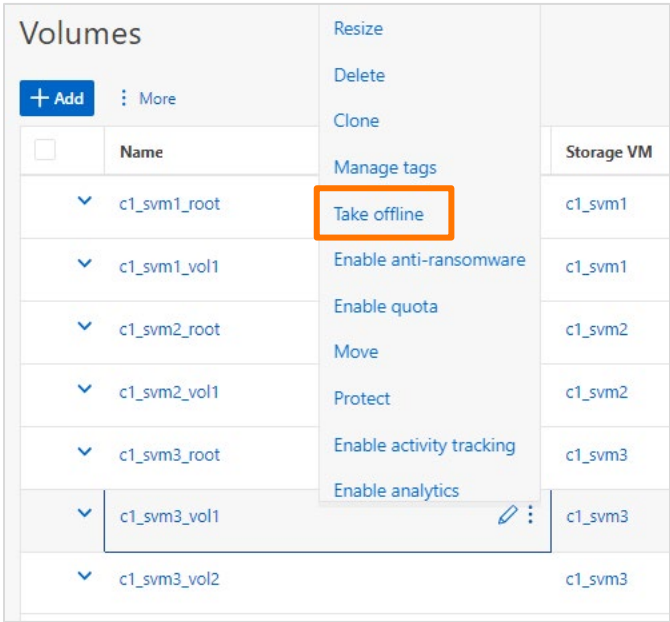
Objectives

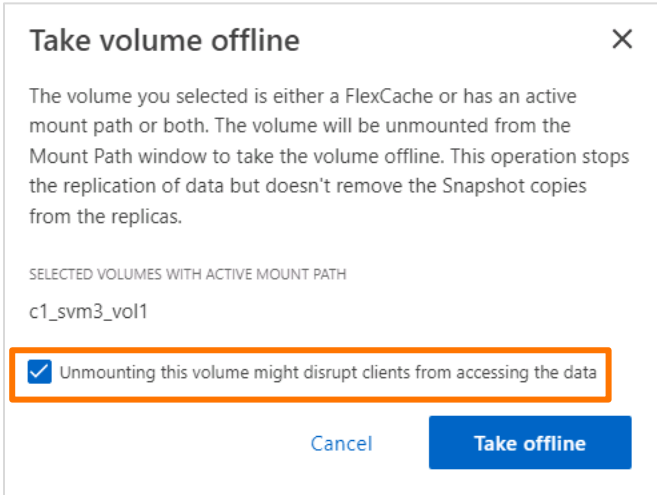
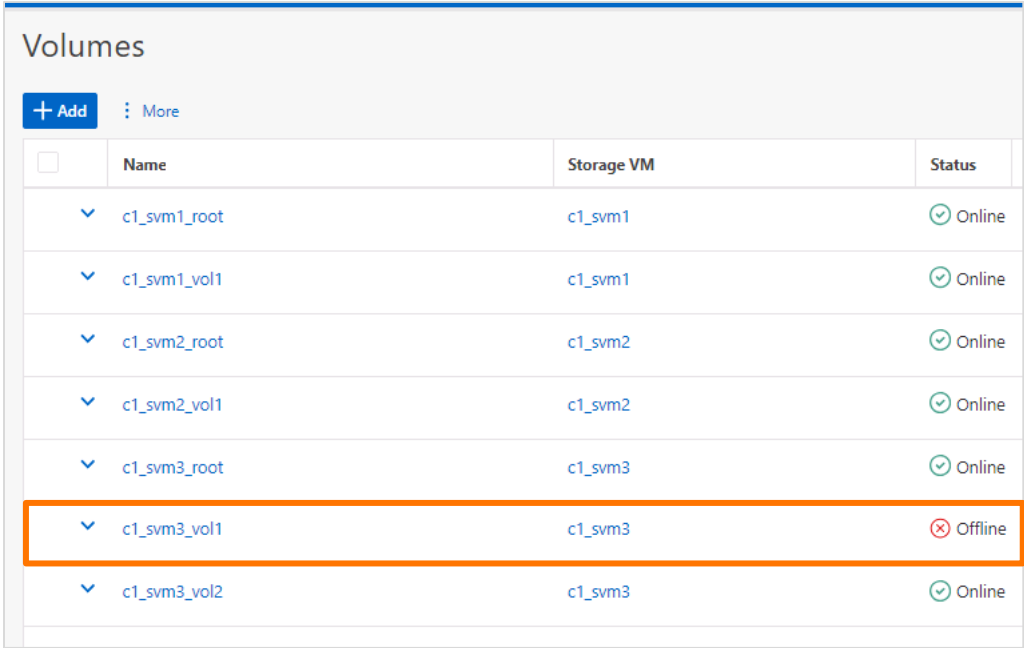
This exercise focuses on enabling you to do the following:

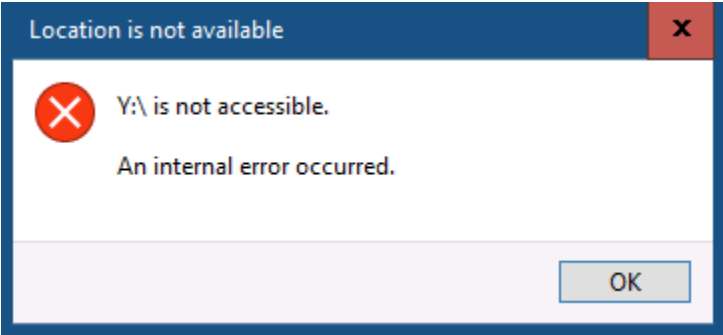
- Take a source volume offline
- Activate a destination volume
- Reactivate the original source volume
- Restore the original SnapMirror relationship

Task 1: Take the Source Volume Offline

In this task, you take the source volume on Cluster1 offline. Before you can take a shared volume offline, you must remove the volume from the namespace.

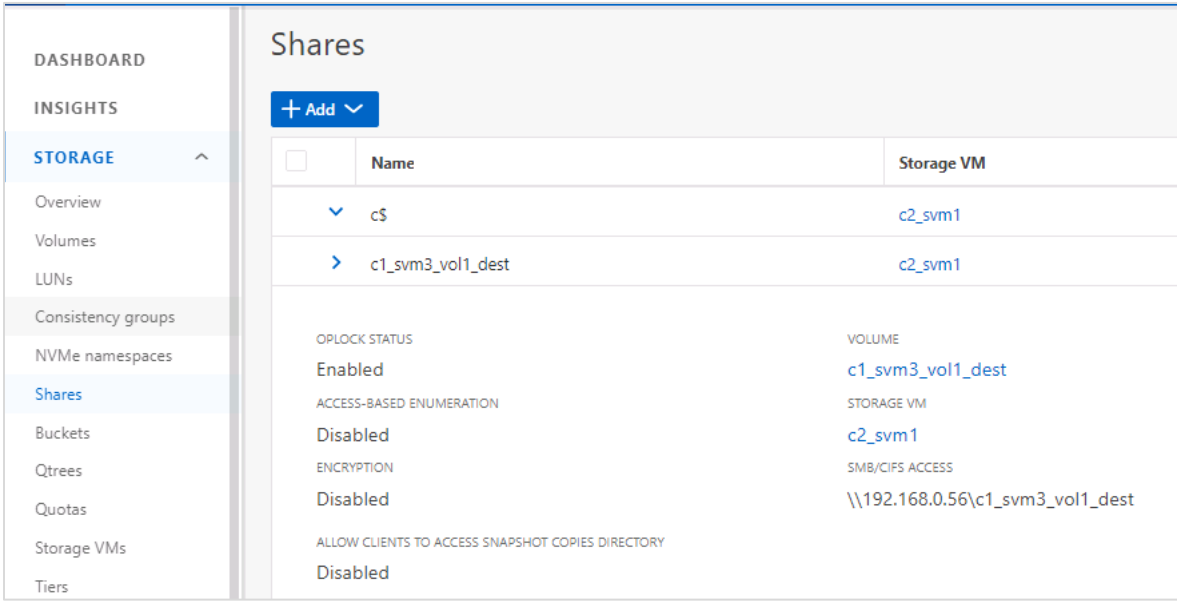
Step	Action
1-1	In the System Manager navigation pane for Cluster1, select Storage > Volumes .
1-2	In the Volumes pane, click the three vertical dots to the right of c1_svm3_vol1 and then select Take offline. 

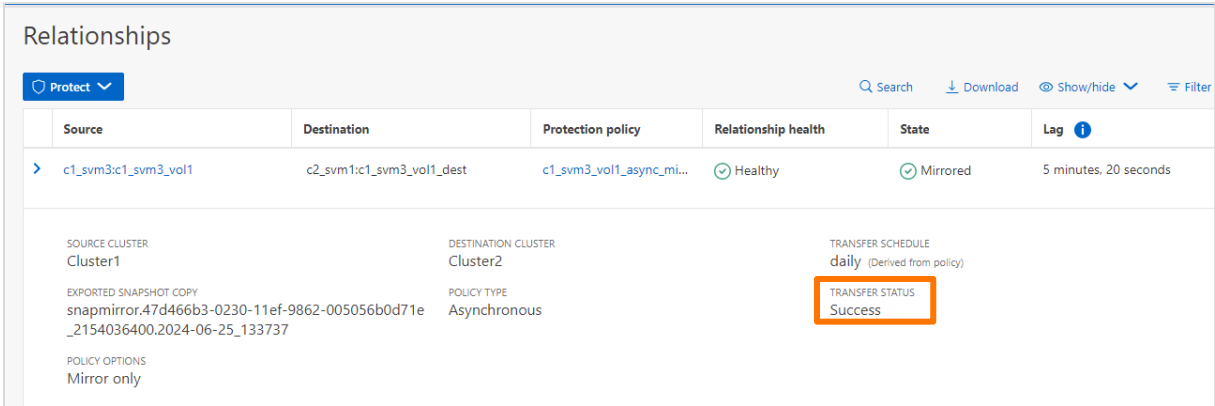
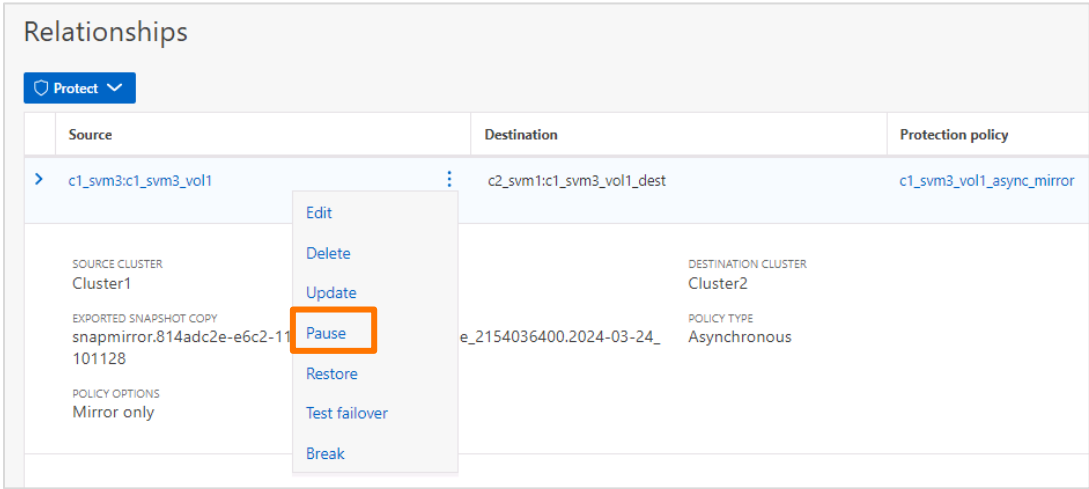
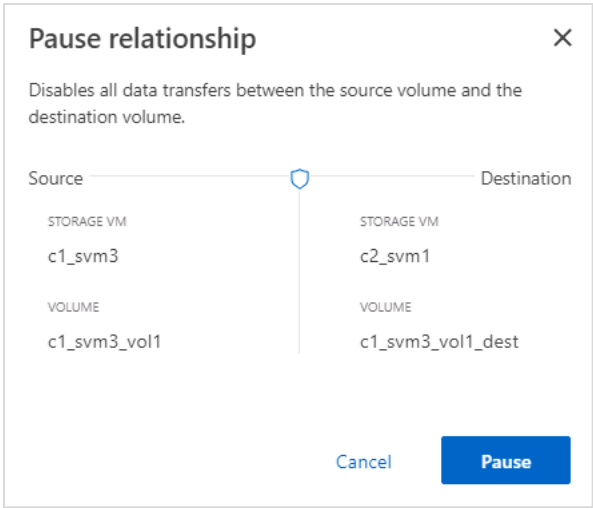
Step	Action
1-3	In the Take volume offline dialog box, select the checkbox to unmount the volume and then click Take offline. 
1-4	Verify that the status of the volume is Offline, as indicated by the red status icon. 
1-5	On your Windows desktop, try to access the mapped drive c1_svm3_share (\\192.168.0.62).

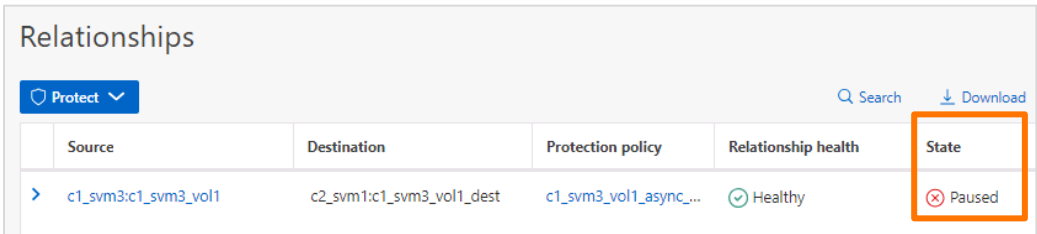
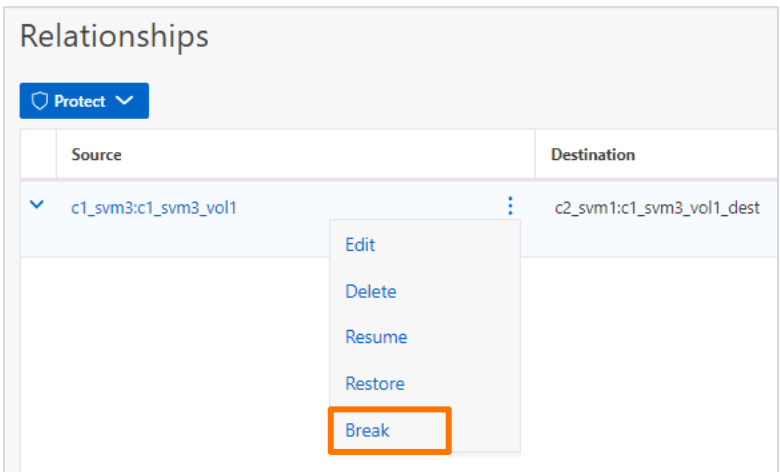
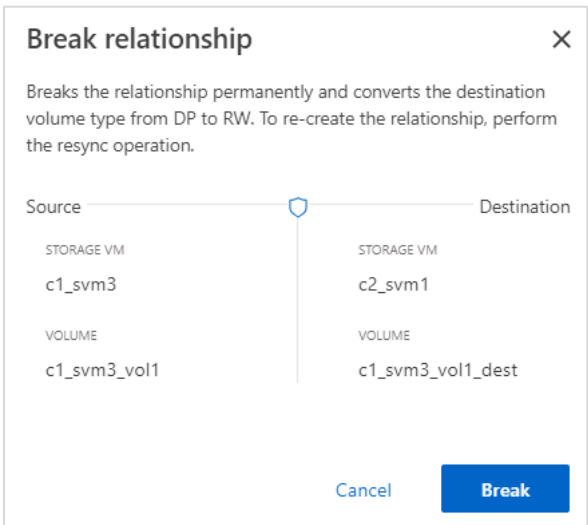
Step	Action
1-6	After some time, Windows should display an error message that the drive is not accessible. Example: 

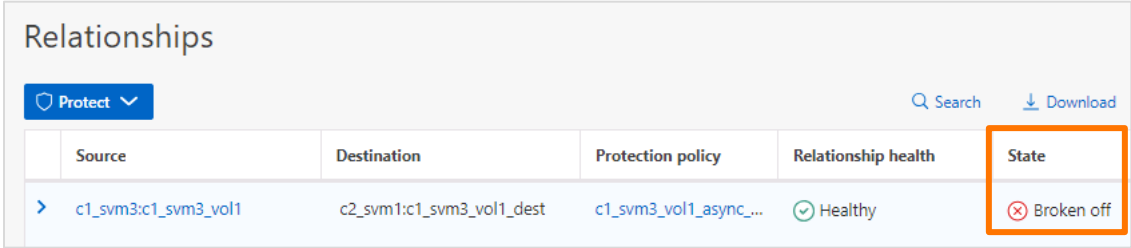
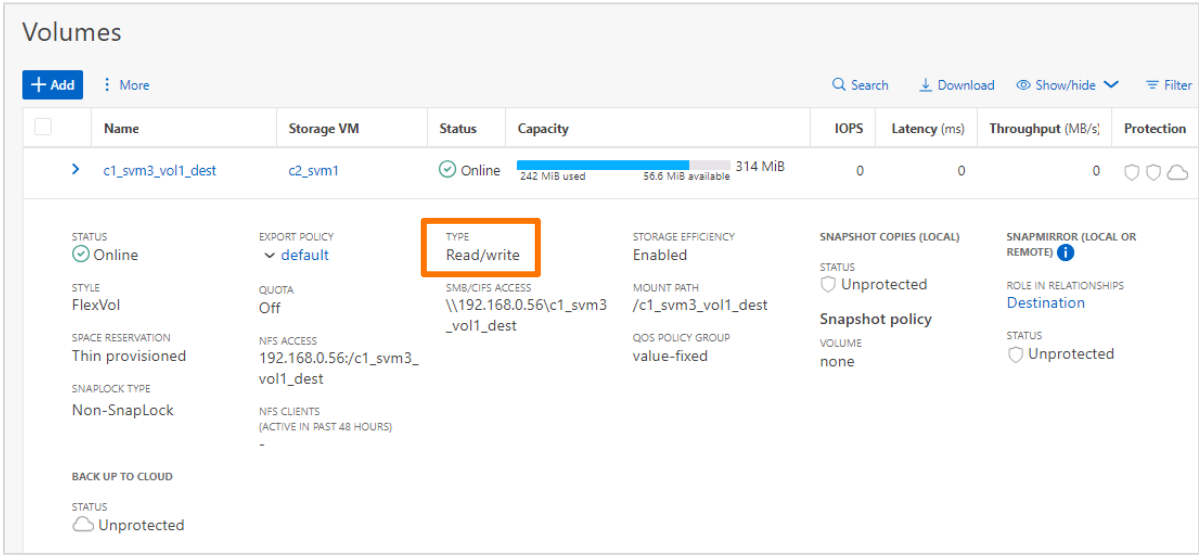
Task 2: Activate the Destination Volume

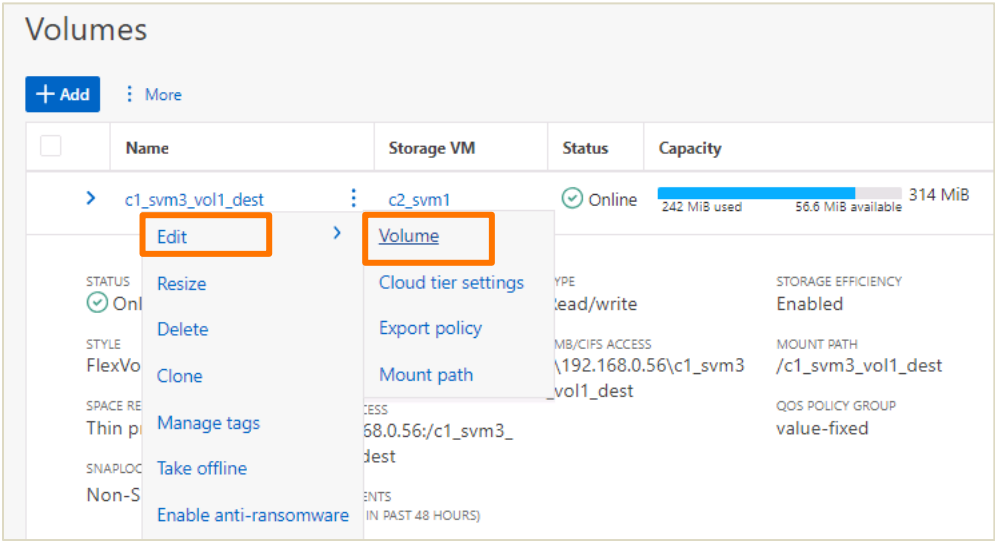
In the previous task, you took the source volume offline and rendered the shared data unavailable. In this task, you activate the destination volume and use a best practices workflow to configure the volume for data access.

Step	Action
2-1	 The first steps of the volume disaster workflow include verifying the source volume status, quiescing and breaking the SnapMirror relationship, and verifying the destination volume status.
2-2	 You must always break the SnapMirror relationship from the destination volume.
2-3	In the System Manager navigation pane for Cluster2, select Storage > Shares .
2-4	Expand c1_svm3_vol1_dest and verify that the volume is mounted on the destination storage VM namespace. 

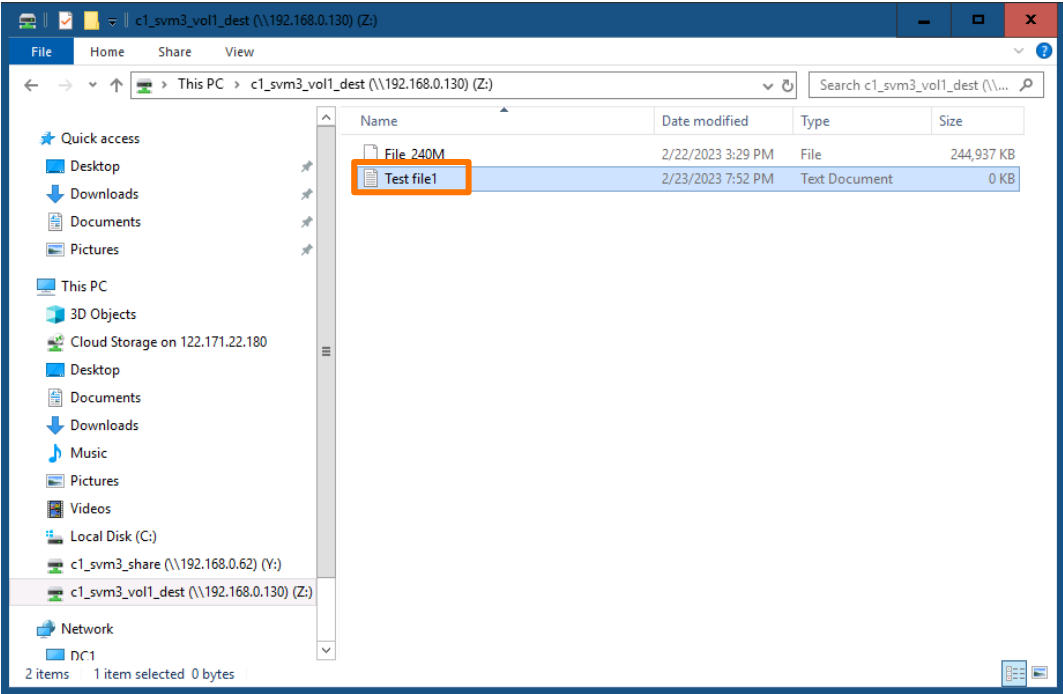
Step	Action
2-5	In the navigation pane, select Protection > Relationships .
2-6	Expand the relationship and verify that the Transfer Status is Success.  The screenshot shows the 'Relationships' page in a web interface. At the top, there's a 'Protect' dropdown menu. Below it is a table with columns: Source, Destination, Protection policy, Relationship health, State, and Lag. The first row shows a relationship between 'c1_svm3:c1_svm3_vol1' and 'c2_svm1:c1_svm3_vol1_dest' with a 'c1_svm3_vol1_async_mi...' policy, 'Healthy' status, 'Mirrored' state, and a lag of '5 minutes, 20 seconds'. Below the table, there's a detailed view of the relationship. It shows 'SOURCE CLUSTER: Cluster1' and 'DESTINATION CLUSTER: Cluster2'. Under 'SOURCE CLUSTER', it lists 'EXPORTED SNAPSHOT COPY: snapmirror.47d466b3-0230-11ef-9862-005056b0d71e_2154036400.2024-06-25_133737' and 'POLICY OPTIONS: Mirror only'. Under 'DESTINATION CLUSTER', it lists 'POLICY TYPE: Asynchronous'. To the right, it shows 'TRANSFER SCHEDULE: daily (Derived from policy)' and 'TRANSFER STATUS: Success', which is highlighted with an orange box.
2-7	To disable future data transfers, click the three vertical dots to the right of the source volume and then select Pause .  The screenshot shows the 'Relationships' page with a context menu open over the source volume 'c1_svm3:c1_svm3_vol1'. The menu options are: Edit, Delete, Update, Pause (highlighted with an orange box), Restore, Test failover, and Break. The background shows the same relationship details as in the previous screenshot, but the 'Pause' option is the focus.
2-8	In the Pause relationship dialog box, click Pause .  The screenshot shows the 'Pause relationship' dialog box. It has a title bar with a close button (X). The text inside says 'Disables all data transfers between the source volume and the destination volume.' Below this, there's a diagram showing 'Source' and 'Destination' volumes connected by a line with a shield icon. The 'Source' side lists 'STORAGE VM: c1_svm3' and 'VOLUME: c1_svm3_vol1'. The 'Destination' side lists 'STORAGE VM: c2_svm1' and 'VOLUME: c1_svm3_vol1_dest'. At the bottom, there are two buttons: 'Cancel' and 'Pause' (highlighted with an orange box).

Step	Action
2-9	 You must ensure that the transfer status of the SnapMirror relationship is paused before you break the SnapMirror relationship. If a transfer is in progress, the quiesce operation might take a long time.
2-10	In the Relationships pane, wait a few seconds and verify that the relationship state is Paused. 
2-11	Click the three vertical dots to the right of the source volume and then select Break. 
2-12	In the Break relationship dialog box, click Break. 

Step	Action
2-13	Verify that the relationship state is Broken off. 
2-14	 During typical SnapMirror operations, the secondary volume is a data protection volume of the read-only type. In the next step, you look at the volume type of the secondary volume to determine whether it is still read-only or has changed to read/write.
2-15	In the System Manager navigation pane, select Storage > Volumes.
2-16	
2-17	 The SnapMirror relationship is now broken. The SnapMirror primary volume is offline and unavailable to read and write requests from clients. The SnapMirror secondary volume is now writable.

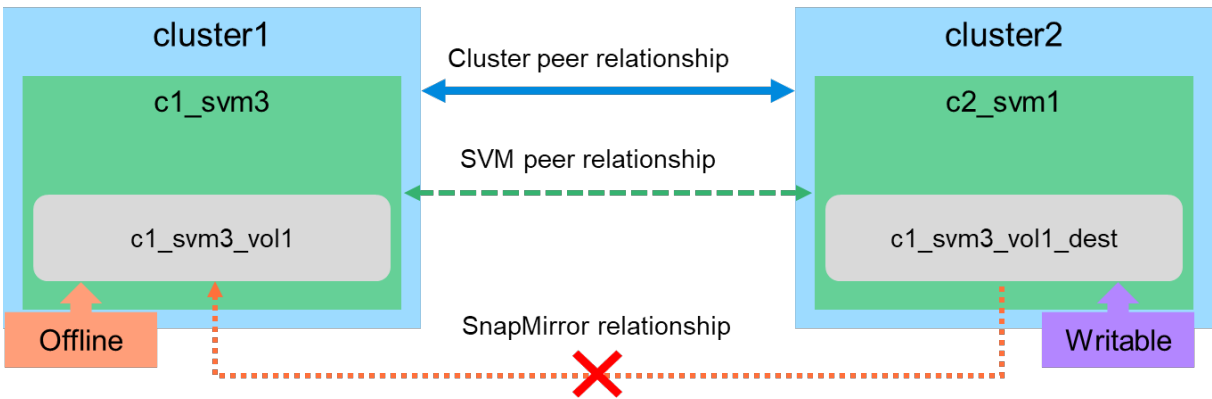
Step	Action
2-18	In the Volumes pane, click the three vertical dots to the right of c1_svm3_vol1_dest and then select Edit > Volume. 

Step	Action																				
2-19	In the Edit volume pane, scroll down to the Snapshot copies (local) settings section, and then specify the following values: - Schedule Snapshot copies: <i>Select checkbox</i> - Snapshot Policy: default - Show the Snapshot copies directory to clients: <i>Select checkbox</i> Snapshot (local) settings Snapshot reserve % 5 Existing snapshot reserve 51.2 MiB ☒ Schedule snapshots Snapshot policy default <table><thead><tr><th>Schedule na...</th><th>Maximum snapshots</th><th>Schedule</th><th>SnapMirror label</th><th>SnapLock retention period</th></tr></thead><tbody><tr><td>hourly</td><td>6</td><td>At 5 minutes past the hour, every hour</td><td>-</td><td>0 second</td></tr><tr><td>daily</td><td>2</td><td>At 12:10 AM, every day</td><td>daily</td><td>0 second</td></tr><tr><td>weekly</td><td>2</td><td>At 12:15 AM, only on Sunday</td><td>weekly</td><td>0 second</td></tr></tbody></table> ☐ Enable snapshot locking ⓘ Enables the ability to lock snapshots that were created either manually or by snapshot policies. The snapshots are locked only when a retention period is specified. ☐ Automatically delete older snapshots ☒ Show the snapshots directory to clients Client systems will be able to display and access the snapshots directory.	Schedule na...	Maximum snapshots	Schedule	SnapMirror label	SnapLock retention period	hourly	6	At 5 minutes past the hour, every hour	-	0 second	daily	2	At 12:10 AM, every day	daily	0 second	weekly	2	At 12:15 AM, only on Sunday	weekly	0 second
Schedule na...	Maximum snapshots	Schedule	SnapMirror label	SnapLock retention period																	
hourly	6	At 5 minutes past the hour, every hour	-	0 second																	
daily	2	At 12:10 AM, every day	daily	0 second																	
weekly	2	At 12:15 AM, only on Sunday	weekly	0 second																	
2-20	Scroll down and then click Save .																				
2-21	On your Windows desktop, open the mapped drive c1_svm3_vol1_dest (\\192.168.0.56) .																				

Step	Action
2-22	Create a text document in the folder to verify that you can write to the volume. 

Task 3: Review the Exercise Environment

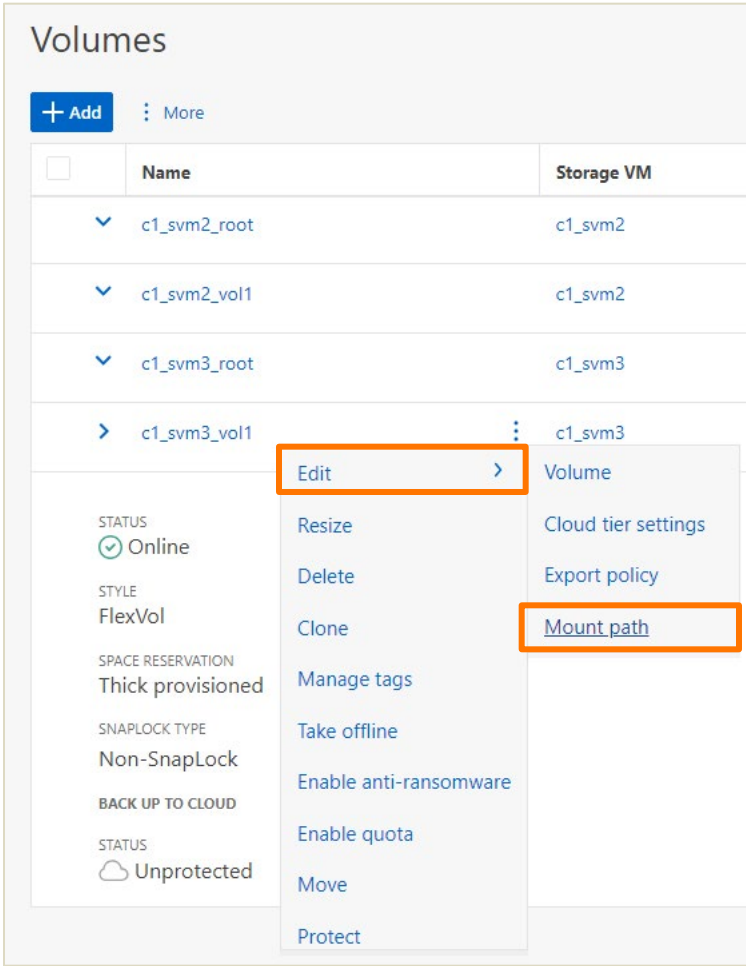
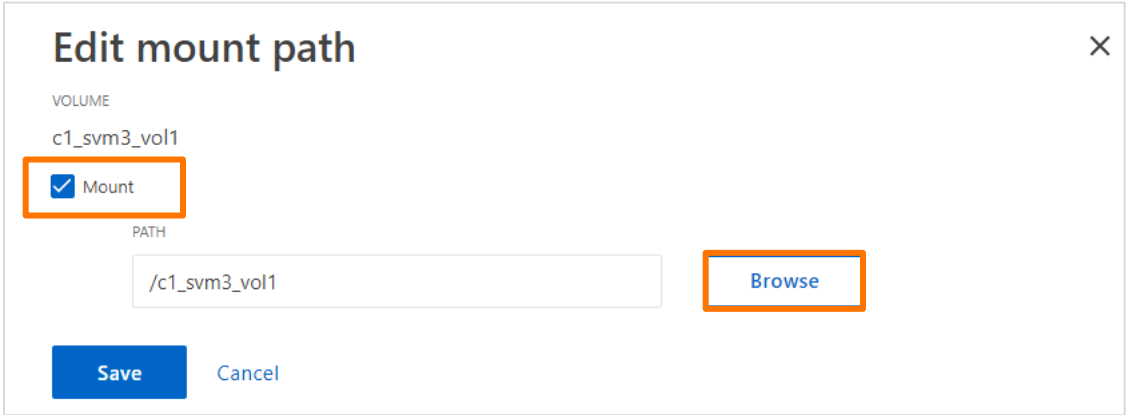
In this task, you review the exercise environment that you created in the previous tasks.

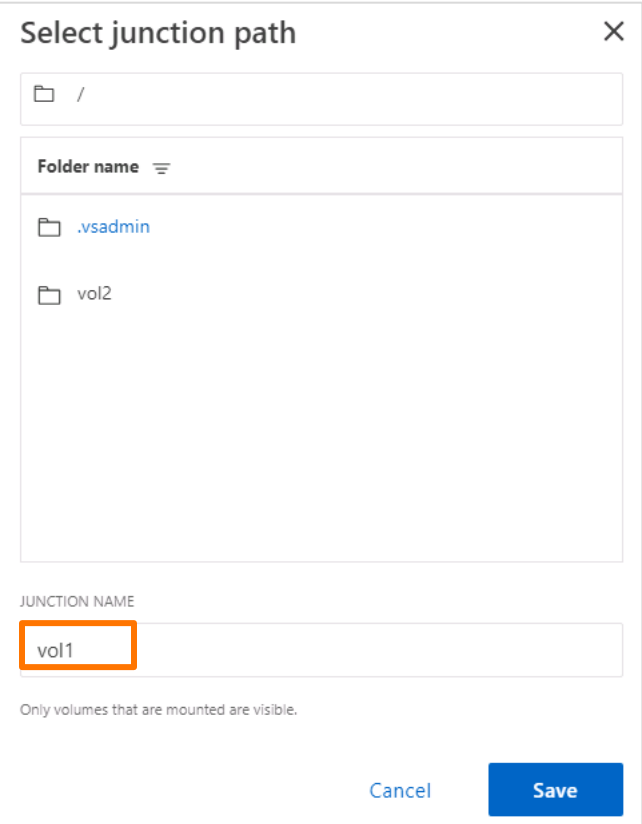
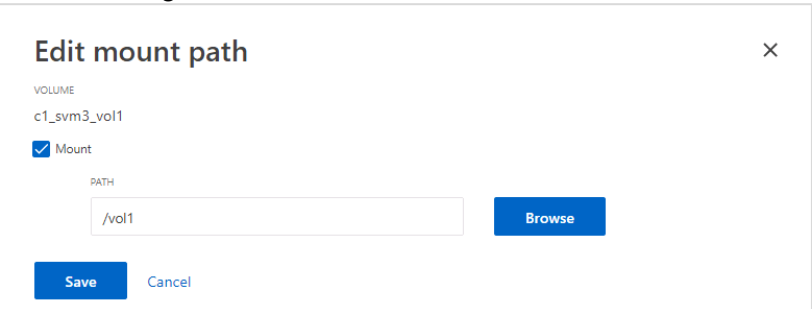
Step	Action
3-1	Verify that the SnapMirror relationship is broken and that the destination volume is in a read/write state. 

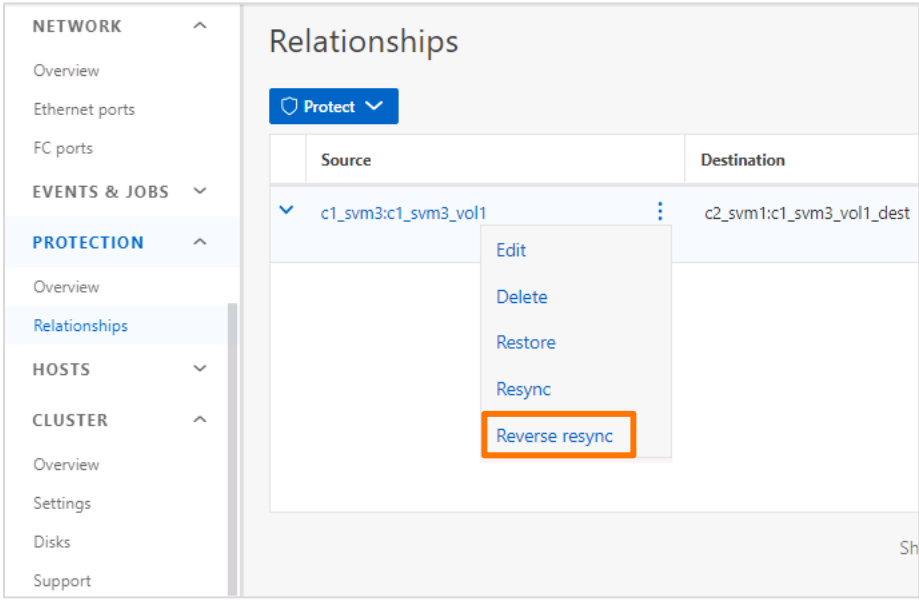
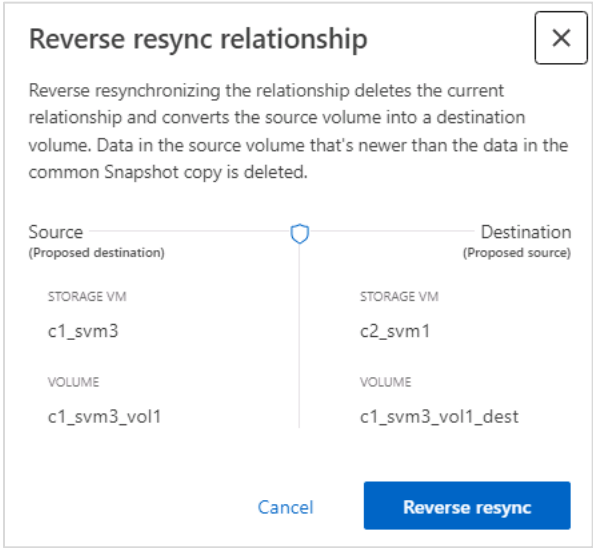
Task 4: Reactivate the Original Source Volume

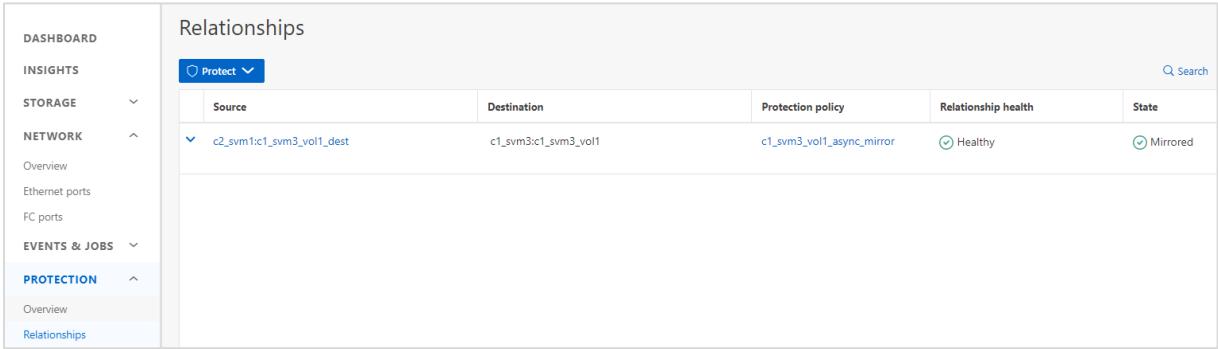
In this task, you bring the primary volume back online. Before normal operations can be restored, you must run a SnapMirror reverse resynchronization to update the primary volume with data that was written to the secondary volume.

Step	Action																																								
4-1	i When the source volume becomes available, you must resynchronize the data from the destination volume to the source volume, update any modifications after the resynchronization, and then activate the source volume. First, you bring the original source volume online.																																								
4-2	In the System Manager navigation pane for Cluster1, select Storage > Volumes .																																								
4-3	Click the three vertical dots to the right of c1_svm3_vol1 and then select Bring online. Volumes + Add : More <table><tr><th>☐</th><th>Name</th><th>Storage VM</th><th>Status</th><th>Capa</th></tr><tr><td>☒</td><td>c1_svm1_root</td><td>c1_svm1</td><td>Online</td><td>972 K</td></tr><tr><td>☒</td><td>c1_svm1_vol1</td><td>c1_svm1</td><td>Online</td><td>2.02</td></tr><tr><td>☒</td><td>c1_svm2_root</td><td>c1_svm2</td><td>Online</td><td>964 K</td></tr><tr><td>☒</td><td>c1_svm2_vol1</td><td>c1_svm2</td><td>Online</td><td>1.94</td></tr><tr><td>☒</td><td>c1_svm3_root</td><td>c1_svm3</td><td>Online</td><td>976 K</td></tr><tr><td>☒</td><td>c1_svm3_vol1</td><td>c1_svm3</td><td>Offline</td><td></td></tr><tr><td>☒</td><td>c1_svm3_vol2</td><td>c1_svm3</td><td>Online</td><td>1.86</td></tr></table> Edit Delete Bring online Enable activity tracking Enable analytics	☐	Name	Storage VM	Status	Capa	☒	c1_svm1_root	c1_svm1	Online	972 K	☒	c1_svm1_vol1	c1_svm1	Online	2.02	☒	c1_svm2_root	c1_svm2	Online	964 K	☒	c1_svm2_vol1	c1_svm2	Online	1.94	☒	c1_svm3_root	c1_svm3	Online	976 K	☒	c1_svm3_vol1	c1_svm3	Offline		☒	c1_svm3_vol2	c1_svm3	Online	1.86
☐	Name	Storage VM	Status	Capa																																					
☒	c1_svm1_root	c1_svm1	Online	972 K																																					
☒	c1_svm1_vol1	c1_svm1	Online	2.02																																					
☒	c1_svm2_root	c1_svm2	Online	964 K																																					
☒	c1_svm2_vol1	c1_svm2	Online	1.94																																					
☒	c1_svm3_root	c1_svm3	Online	976 K																																					
☒	c1_svm3_vol1	c1_svm3	Offline																																						
☒	c1_svm3_vol2	c1_svm3	Online	1.86																																					
4-4	Wait for a few seconds for c1_svm3_vol1 to be updated and then verify that the volume is online.																																								

Step	Action
4-5	Click the three vertical dots to the right of c1_svm3_vol1 and then select Edit > Mount path. 
4-6	In the Edit mount path pane, select the Mount checkbox, and then click Browse next to the Path field. 

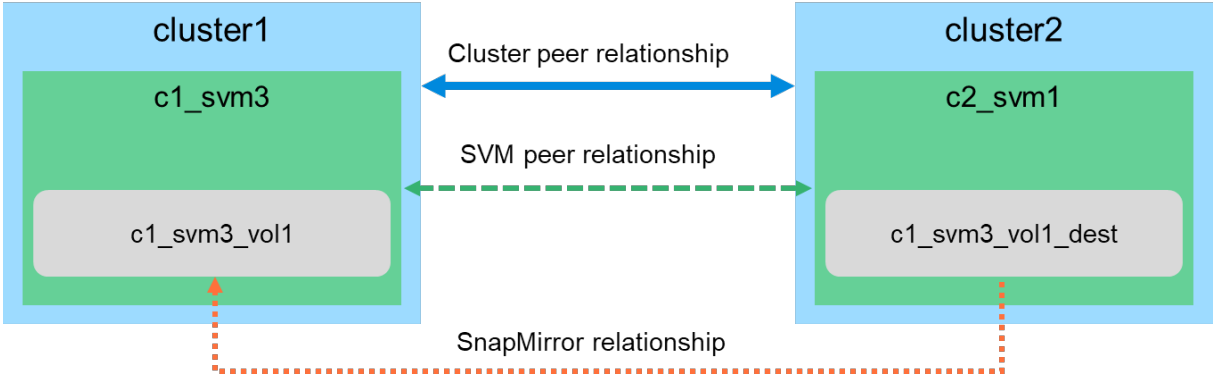
Step	Action
4-7	In the Junction Name field, enter vol1, and then click Save. 
4-8	Click Save again. 
4-9	 The primary volume is online. In the next step, on Cluster2, you check the SnapMirror status.
4-10	In the System Manager navigation pane for Cluster2, select Protection > Relationships.
4-11	 The status of the SnapMirror relationship is Broken Off. The primary volume is online. However, data is written to the secondary volume. In the next step, you reverse the direction of the SnapMirror relationship to copy data that was written to the secondary volume back to the primary volume. You can run the Reverse resync operation only from the SnapMirror destination SVM.

Step	Action
4-12	In the Relationships pane, click the three vertical dots to the right of the source volume, and then select Reverse resync. 
4-13	In the Reverse resync relationship dialog box, review the provided information, and then click Reverse resync. 
4-14	Notice that the SnapMirror relationship is removed from the relationships list on Cluster2.

Step	Action
4-15	In the System Manager navigation pane for Cluster1, select Protection > Relationships, and then find the reversed relationship that is listed on the page. 
4-16	 In the Volume Relationships pane for Cluster1, the original source is now the destination, and the original destination is now the source.

Task 5: Review the Exercise Environment

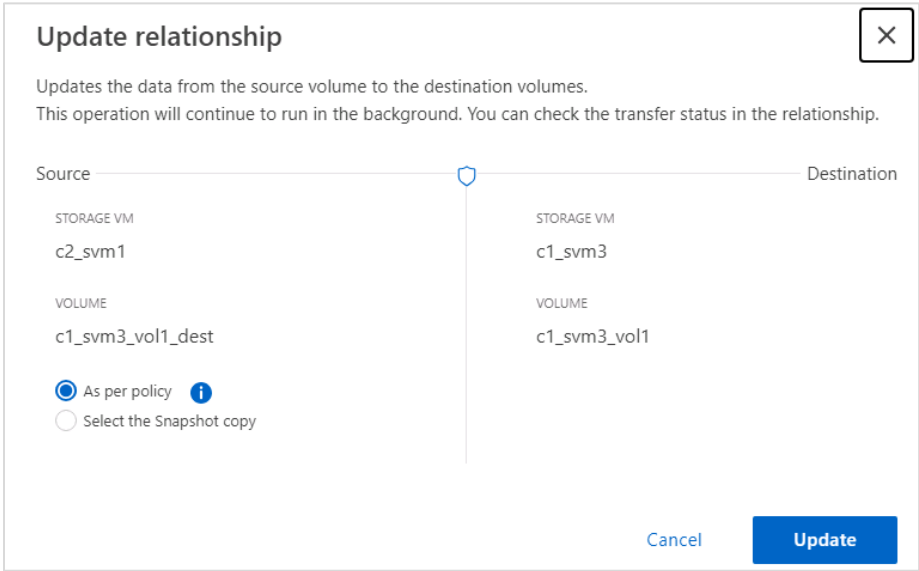
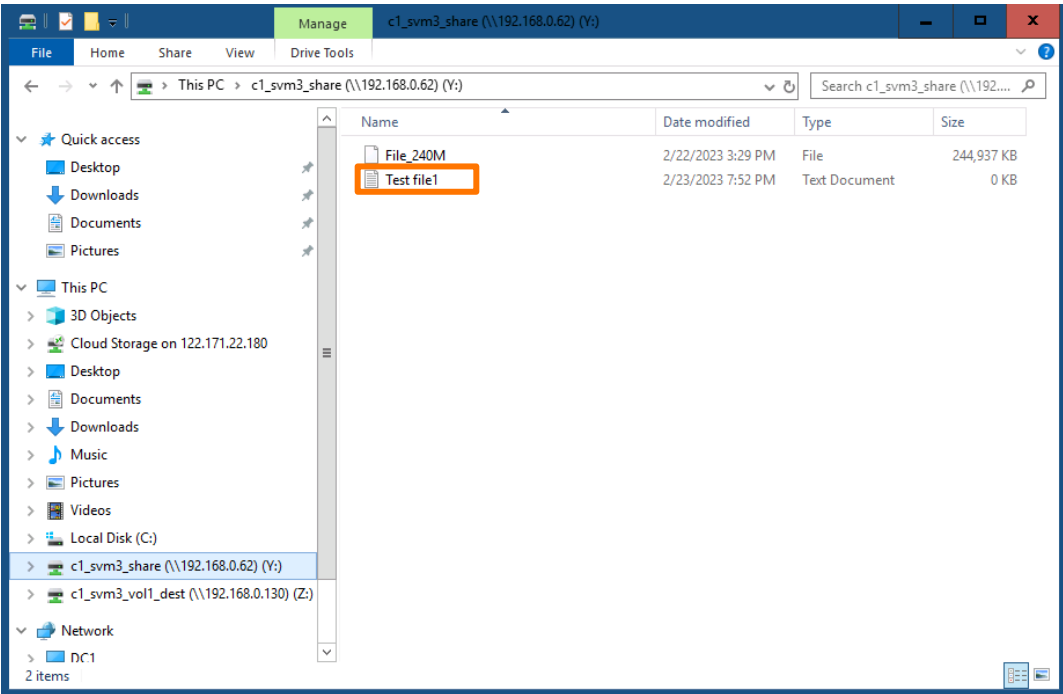
In this task, you review the exercise environment that you created in Task 4.

Step	Action
5-1	Verify that the original source volume is now configured as the SnapMirror destination and that the original destination is now the SnapMirror source volume. If the relationship stays in this configuration for a long time, you might need to reset the mirror policy and update schedule until you return the SnapMirror relationship to its original configuration. 

Task 6: Restore the Original SnapMirror Relationship

The data that was written to the SnapMirror secondary volume has been updated to the primary volume. In this task, you restore the original relationship between the two SnapMirror volumes.

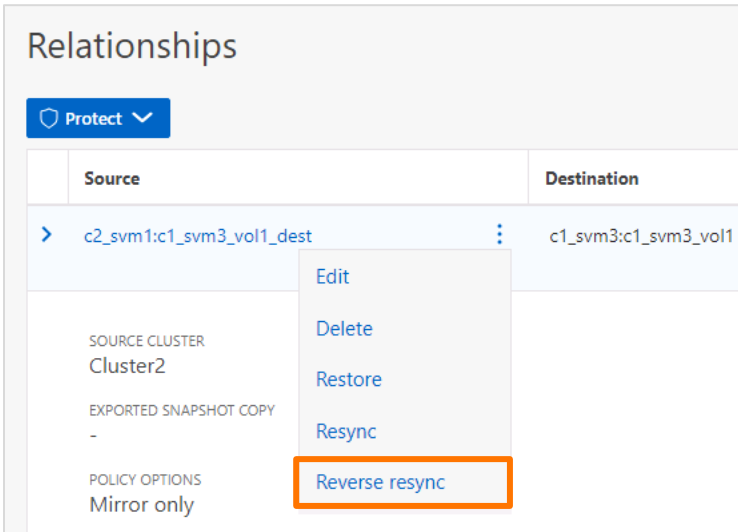
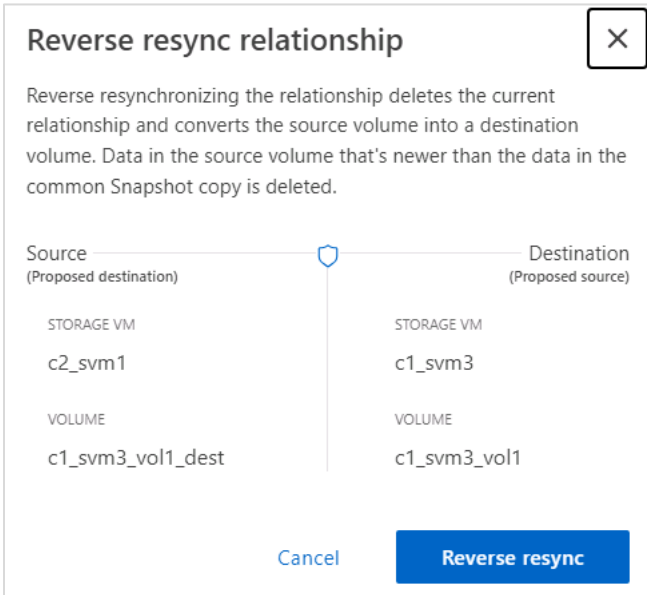
Step	Action				
6-1	 You start to restore the original relationship by updating the reversed SnapMirror relationship because clients might have written to the SnapMirror reversed secondary volume. After you update the reversed relationship, you break the relationship. You must verify that the transfer status is idle before you break the reversed mirroring relationship.				
6-2	In the System Manager navigation pane for Cluster1, select Protection > Relationships .				
6-3	In the Relationships pane, click the three vertical dots to the right of the source volume and then select Update. Relationships Protect <table><thead><tr><th>Source</th><th>Destination</th></tr></thead><tbody><tr><td>c2_svm1:c1_svm3_vol1_dest</td><td>c1_svm3:c1_svm3_vol1</td></tr></tbody></table> Edit Delete Update Pause Restore Test failover Break	Source	Destination	c2_svm1:c1_svm3_vol1_dest	c1_svm3:c1_svm3_vol1
Source	Destination				
c2_svm1:c1_svm3_vol1_dest	c1_svm3:c1_svm3_vol1				

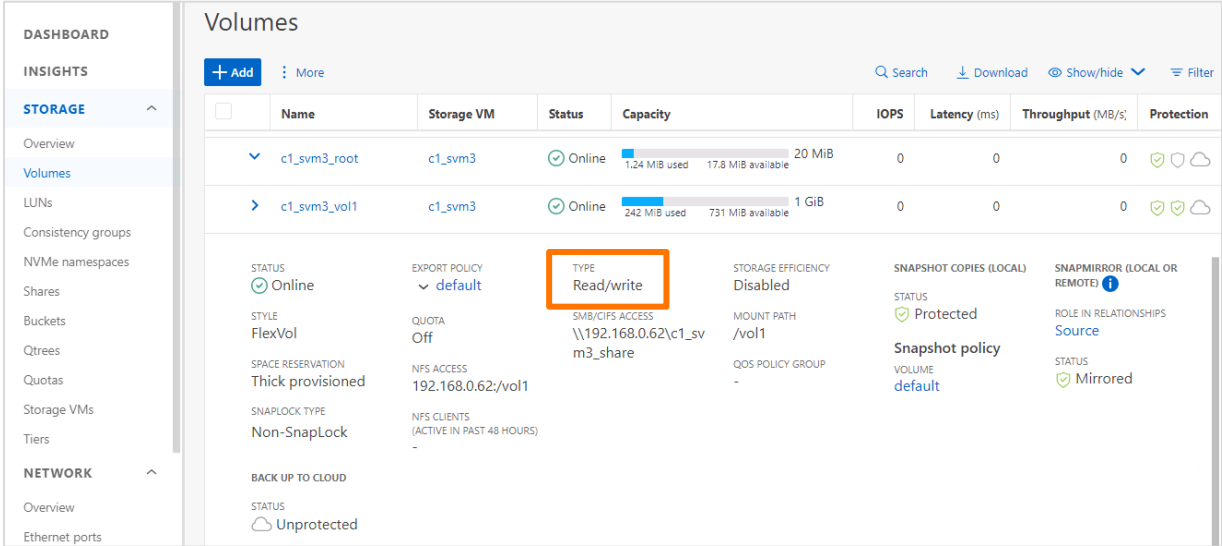
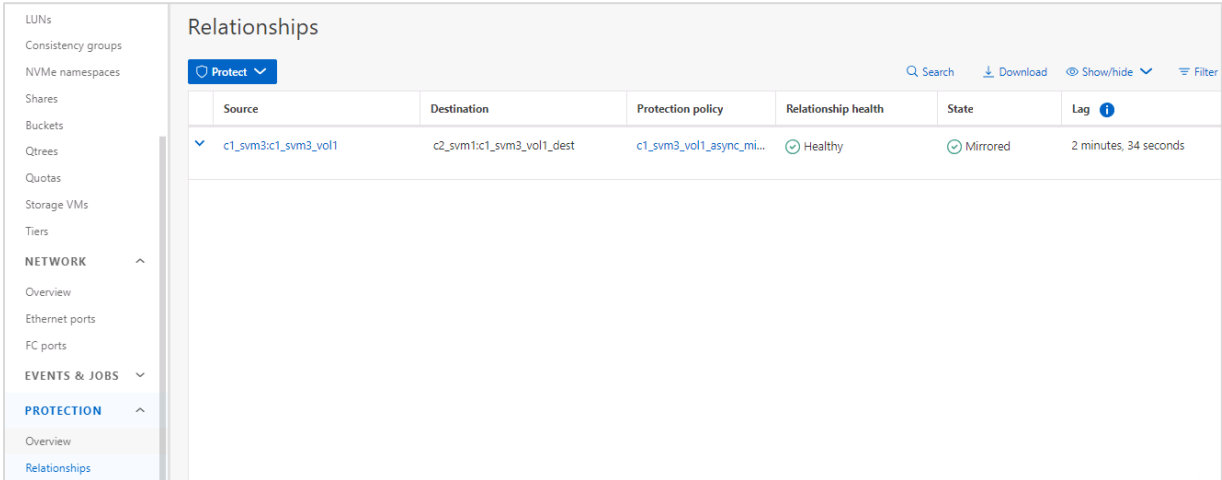
Step	Action
6-4	In the Update relationship dialog box, accept the default configuration setting and then click Update. 
6-5	 The update operation is necessary to write any new data that was written to the reversed secondary volume to the primary volume after the Reverse resync operation. The on-demand configuration setting transfers data according to the newest shared Snapshot copy on the source and destination volumes.
6-6	In File Explorer, access the mapped drive c1_svm3_share (\\192.168.0.62), and then verify that the new file that you created on the destination is now available on the source. 

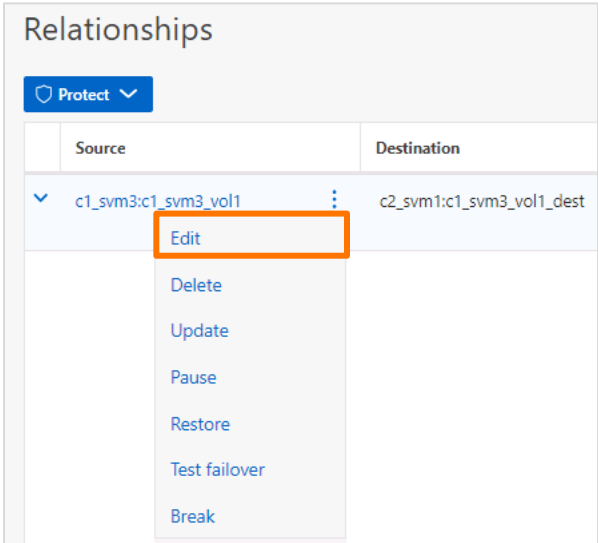
Step	Action										
6-7	 After you update the reversed relationship, you pause and break the relationship. You must verify that data is not transferring before you break the reversed mirroring relationship.										
6-8	Verify that the Transfer Status of the relationship is Success. Relationships Protect Search <table><thead><tr><th>Source</th><th>Destination</th><th>Protection policy</th><th>Relationship health</th><th>State</th></tr></thead><tbody><tr><td> > c2_svm1:c1_svm3_vol1_dest</td><td>c1_svm3:c1_svm3_vol1</td><td>c1_svm3_vol1_async_mirror</td><td> Healthy </td><td> Mirrored </td></tr></tbody></table> SOURCE CLUSTER Cluster2 EXPORTED SNAPSHOT COPY snapmirror.190ce37c-e5e9-11ee-834b-005056b0cce2_2153183180.20-24-03-24_120853 POLICY OPTIONS Mirror only DESTINATION CLUSTER Cluster1 POLICY TYPE Asynchronous TRANSFER SCHEDULE daily (Derived from policy) TRANSFER STATUS Success	Source	Destination	Protection policy	Relationship health	State	> c2_svm1:c1_svm3_vol1_dest	c1_svm3:c1_svm3_vol1	c1_svm3_vol1_async_mirror	Healthy	Mirrored
Source	Destination	Protection policy	Relationship health	State							
> c2_svm1:c1_svm3_vol1_dest	c1_svm3:c1_svm3_vol1	c1_svm3_vol1_async_mirror	Healthy	Mirrored							
6-9	If the Transfer Status is Transferring, wait until it becomes Success.										
6-10	In the Relationships pane, click the three vertical dots to the right of the source volume and then select Pause. Relationships Protect <table><thead><tr><th>Source</th><th>Destination</th></tr></thead><tbody><tr><td> > c2_svm1:c1_svm3_vol1_dest</td><td>c1_svm3:c1_svm3_vol1</td></tr></tbody></table> SOURCE CLUSTER Cluster2 EXPORTED SNAPSHOT COPY snapmirror.190ce37c-e5e9-11ee-834b-005056b0cce2_2153183180.20-24-03-24_120853 POLICY OPTIONS Mirror only Edit Delete Update Pause Restore Test failover Break	Source	Destination	> c2_svm1:c1_svm3_vol1_dest	c1_svm3:c1_svm3_vol1						
Source	Destination										
> c2_svm1:c1_svm3_vol1_dest	c1_svm3:c1_svm3_vol1										

Step	Action
6-11	In the Pause relationship dialog box, review the provided information and then click Pause. Pause relationship ✕ Disables all data transfers between the source volume and the destination volume. Source Destination STORAGE VM c2_svm1 VOLUME c1_svm3_vol1_dest STORAGE VM c1_svm3 VOLUME c1_svm3_vol1 Cancel Pause

Step	Action										
6-13	In the Relationships pane, click the three vertical dots to the right of the source volume and then select Break. Relationships Protect <table><thead><tr><th>Source</th><th>Destination</th></tr></thead><tbody><tr><td>c2_svm1:c1_svm3_vol1_dest</td><td>c1_svm3:c1_svm3_vol1</td></tr></tbody></table> Edit Delete Resume Restore Break SOURCE CLUSTER Cluster2 EXPORTED SNAPSHOT COPY snapmirror.190ce37c-e6b0cce2_2153183180.20 24-03-24_120853 POLICY OPTIONS Mirror only DESTINATION CLUSTER Cluster1 POLICY TYPE Asynchronous	Source	Destination	c2_svm1:c1_svm3_vol1_dest	c1_svm3:c1_svm3_vol1						
Source	Destination										
c2_svm1:c1_svm3_vol1_dest	c1_svm3:c1_svm3_vol1										
6-14	In the Break relationship dialog box, review the provided information and then click Break. Break relationship Breaks the relationship permanently and converts the destination volume type from DP to RW. To re-create the relationship, perform the resync operation. Source Destination STORAGE VM c2_svm1 VOLUME c1_svm3_vol1_dest STORAGE VM c1_svm3 VOLUME c1_svm3_vol1 Cancel Break										
6-15	Verify that the relationship state is Broken off. Relationships Protect Search Download <table><thead><tr><th>Source</th><th>Destination</th><th>Protection policy</th><th>Relationship health</th><th>State</th></tr></thead><tbody><tr><td>c2_svm1:c1_svm3_vol1_dest</td><td>c1_svm3:c1_svm3_vol1</td><td>c1_svm3_vol1_asyn...</td><td>Healthy</td><td>Broken off</td></tr></tbody></table>	Source	Destination	Protection policy	Relationship health	State	c2_svm1:c1_svm3_vol1_dest	c1_svm3:c1_svm3_vol1	c1_svm3_vol1_asyn...	Healthy	Broken off
Source	Destination	Protection policy	Relationship health	State							
c2_svm1:c1_svm3_vol1_dest	c1_svm3:c1_svm3_vol1	c1_svm3_vol1_asyn...	Healthy	Broken off							

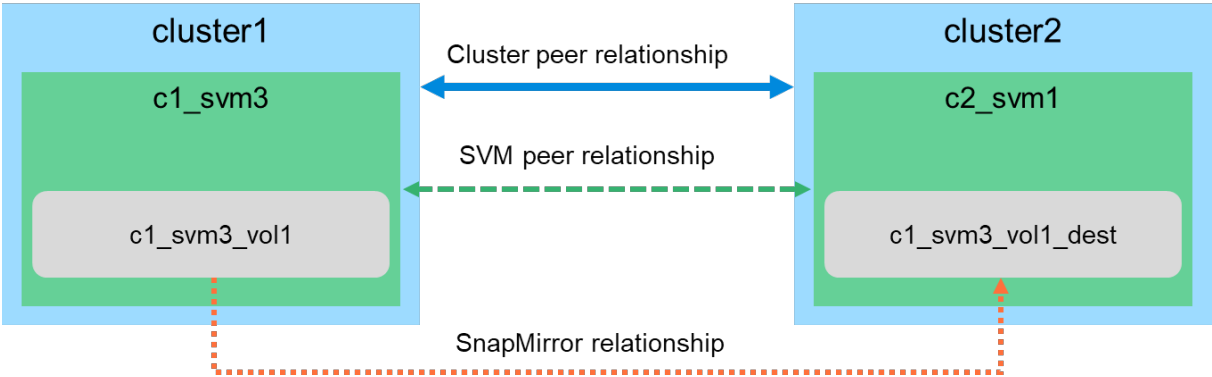
Step	Action
6-16	 In the next step, you re-establish the original SnapMirror relationship by using the Reverse resync operation.
6-17	In the Relationships pane, click the three vertical dots to the right of the source volume and then select Reverse resync. 
6-18	In the Reverse resync relationship dialog box, review the provided information and then click Reverse resync. 
6-19	 The SnapMirror relationship is removed from the relationships list.
6-20	In the System Manager navigation pane, select Storage > Volumes .

Step	Action
6-21	Expand c1_svm3_vol1 and verify that the value in the Type field is Read/write. 
6-22	 SnapMirror technology is always managed from the SnapMirror destination. Now that the SnapMirror relationship is reversed, you must log in to Cluster2 to verify and manage the SnapMirror relationship. You also verify that the SnapMirror policy and update schedule are retained.
6-23	In the System Manager navigation pane for Cluster2, select Protection > Relationships.
6-24	In the Relationships pane, verify that the original SnapMirror relationship is restored and healthy and that the relationship state is Mirrored. 

Step	Action
6-25	To verify that the original SnapMirror policy and schedule are retained, click the three vertical dots to the right of the source volume and then select Edit.  The screenshot shows a 'Relationships' window with a 'Protect' dropdown. Below it is a table with two columns: 'Source' and 'Destination'. The first row shows 'c1_svm3:c1_svm3_vol1' as the source and 'c2_svm1:c1_svm3_vol1_dest' as the destination. To the right of the source volume, there are three vertical dots. A context menu is open, showing options: 'Edit' (highlighted with an orange box), 'Delete', 'Update', 'Pause', 'Restore', 'Test failover', and 'Break'.
6-26	In the Edit relationship dialog box, verify that the original policy and schedule are retained, and then click Cancel.
6-27	 To manually update the SnapMirror relationship, you can click the three vertical dots to the right of the source volume, and then select Update.

Task 7: Review the Exercise Environment

In this task, you review the exercise environment that you created in Task 6.

Step	Action
7-1	Verify that you successfully returned the SnapMirror relationship to its original configuration. 

End of exercise

Exercise 2(a): Configuring SnapMirror for Unified Replication

In this exercise, you configure and initialize a vaulting relationship between two volumes. This relationship meets the following requirements:

- Daily scheduled backups with a retention policy of seven Snapshot copies
- Weekly scheduled backups with a retention policy of 52 Snapshot copies

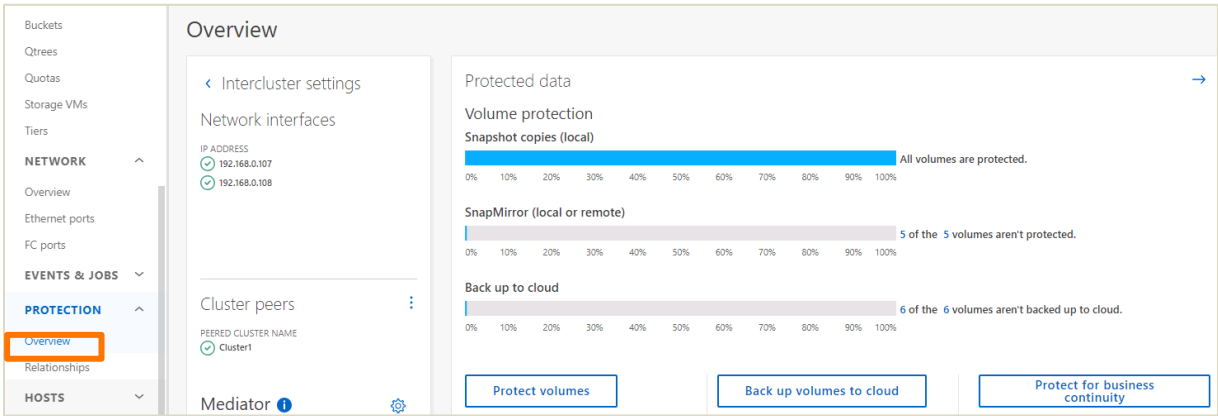
Objectives

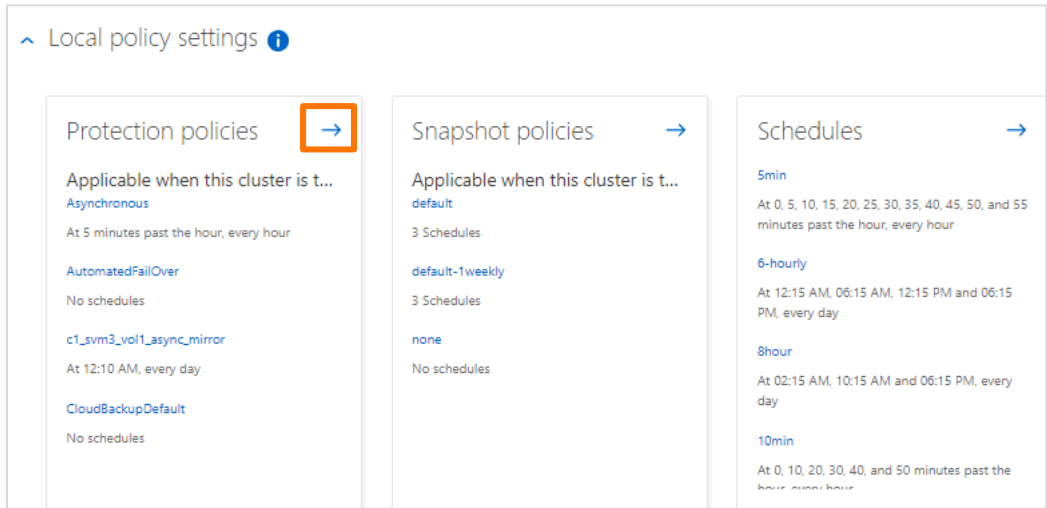
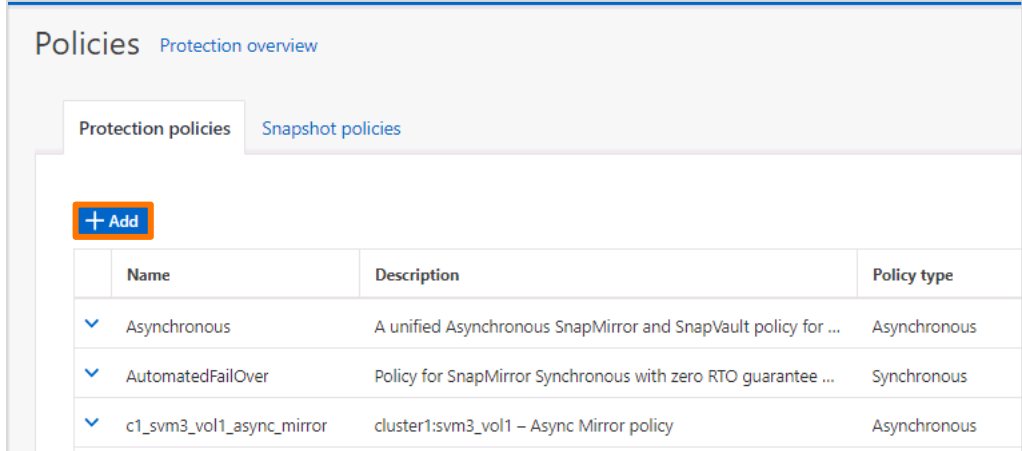
This exercise focuses on enabling you to do the following:

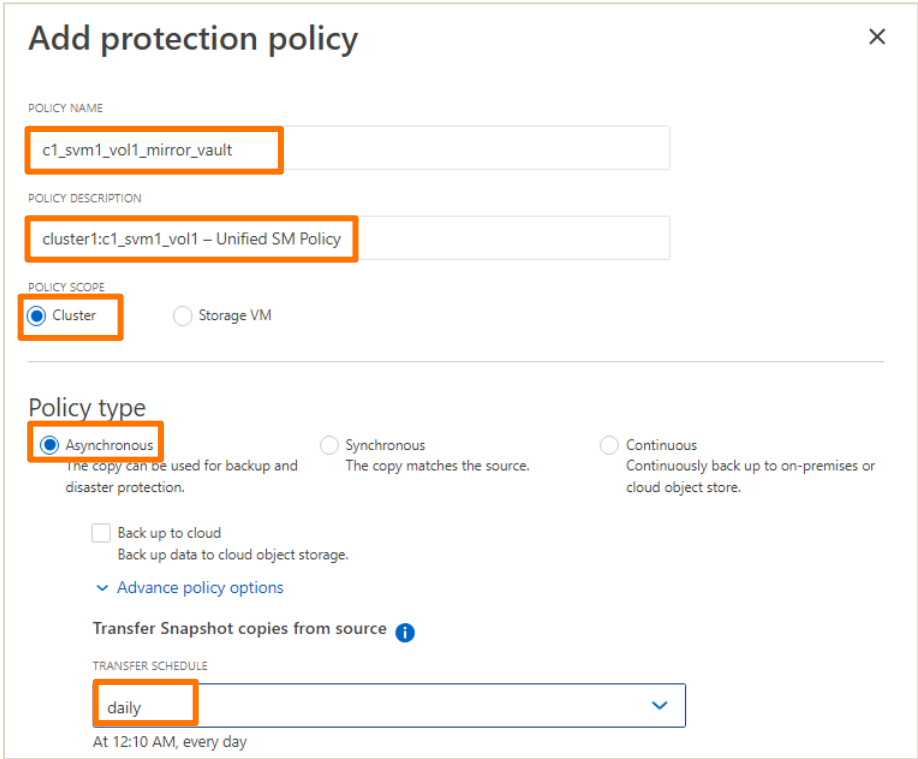
- Create a custom protection policy
- Create a SnapMirror unified replication relationship
- Perform a manual SnapMirror update
- Verify data transfer

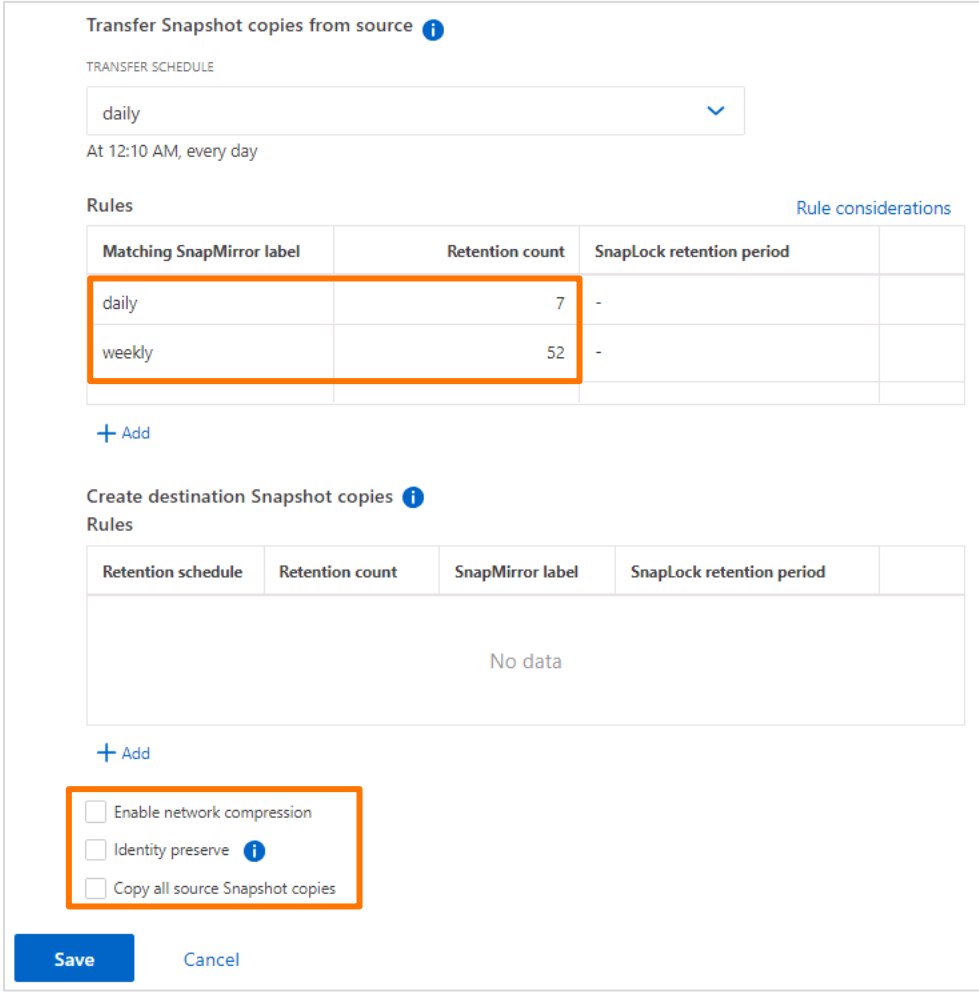
Task 1: Create a Custom Protection Policy and a SnapMirror Relationship

In this task, you create a custom unified replication policy and then create the SnapMirror relationship.

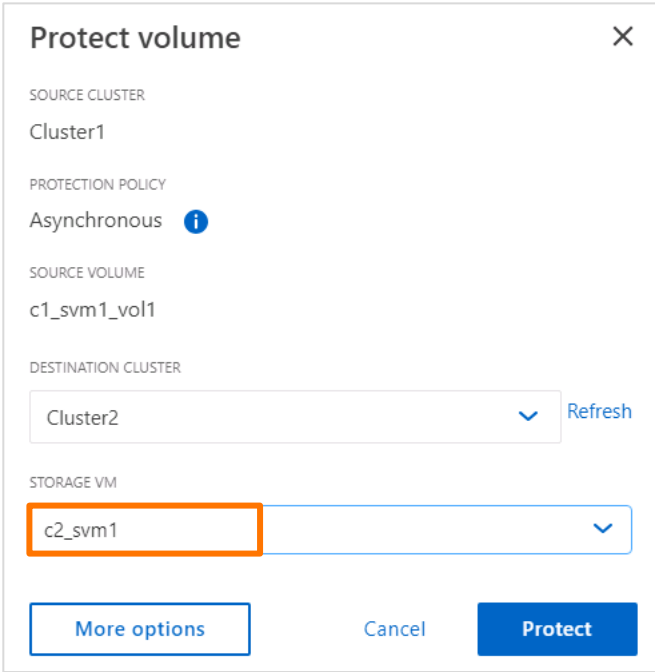
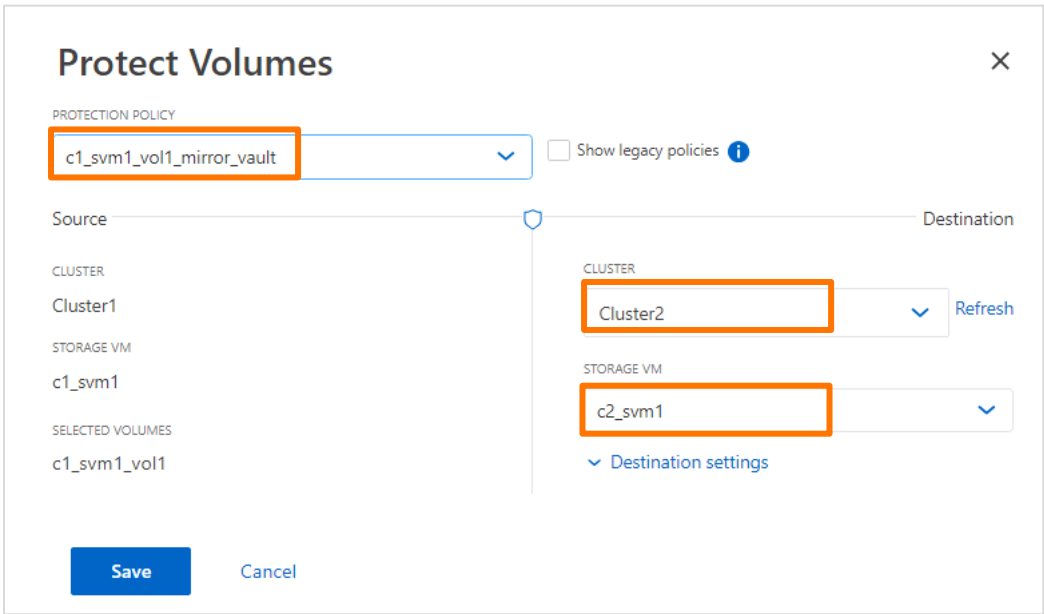
Step	Action
1-1	 Cluster2 is the destination cluster for the protection relationship. You create the custom policy on the destination cluster.
1-2	In the System Manager navigation pane for Cluster2, select Protection > Overview. 

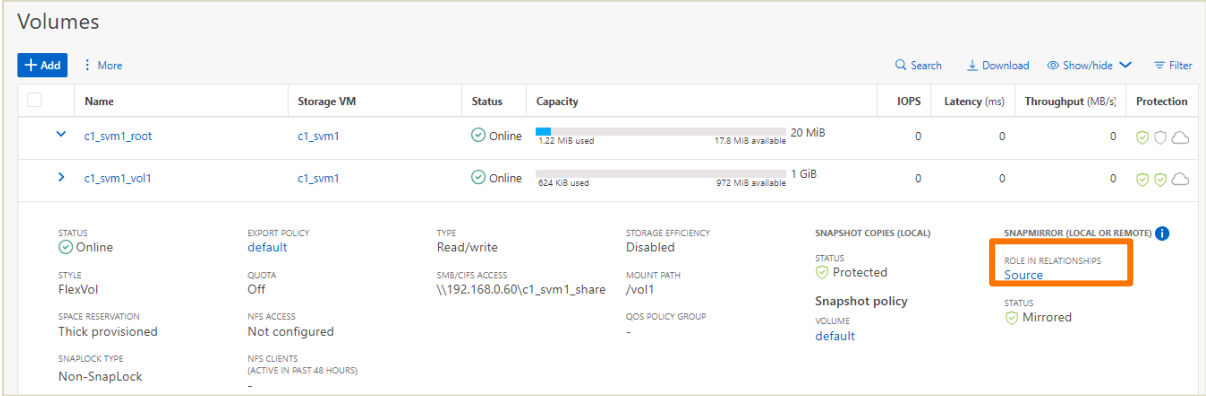
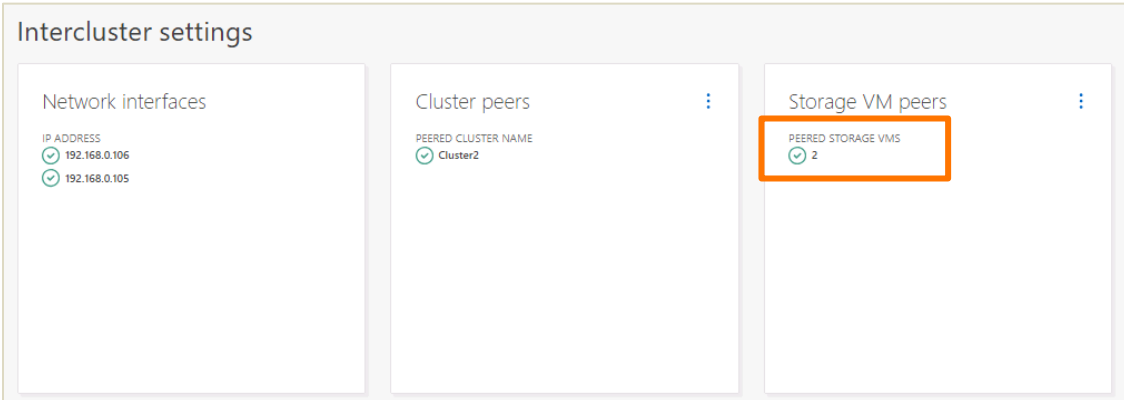
Step	Action																
1-3	In the Overview pane, scroll down and expand Local policy settings, and then click the arrow in the upper-right corner of the Protection policies pane.  The screenshot shows the 'Local policy settings' section with three sub-panels: 'Protection policies', 'Snapshot policies', and 'Schedules'. The 'Protection policies' panel is highlighted with an orange box and an arrow pointing to its right-pointing arrow icon in the top right corner.																
1-4	In the Protection policies pane, click Add to create a policy.  The screenshot shows the 'Policies' section with tabs for 'Protection policies' and 'Snapshot policies'. The 'Protection policies' tab is active, and the '+ Add' button is highlighted with an orange box. <table><tr><th></th><th>Name</th><th>Description</th><th>Policy type</th></tr><tr><td>▼</td><td>Asynchronous</td><td>A unified Asynchronous SnapMirror and SnapVault policy for ...</td><td>Asynchronous</td></tr><tr><td>▼</td><td>AutomatedFailOver</td><td>Policy for SnapMirror Synchronous with zero RTO guarantee ...</td><td>Synchronous</td></tr><tr><td>▼</td><td>c1_svm3_vol1_async_mirror</td><td>cluster1:svm3_vol1 – Async Mirror policy</td><td>Asynchronous</td></tr></table>		Name	Description	Policy type	▼	Asynchronous	A unified Asynchronous SnapMirror and SnapVault policy for ...	Asynchronous	▼	AutomatedFailOver	Policy for SnapMirror Synchronous with zero RTO guarantee ...	Synchronous	▼	c1_svm3_vol1_async_mirror	cluster1:svm3_vol1 – Async Mirror policy	Asynchronous
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▼	c1_svm3_vol1_async_mirror	cluster1:svm3_vol1 – Async Mirror policy	Asynchronous														

Step	Action
1-5	In the Add protection policy dialog box, specify the following values: - Policy Name: c1_svm1_vol1_mirror_vault - Policy Description: cluster1:c1_svm1_vol1 – Unified SM Policy - Policy Scope: Cluster - Policy type: Asynchronous - Transfer Schedule: daily  The screenshot shows the 'Add protection policy' dialog box with the following configuration: POLICY NAME: c1_svm1_vol1_mirror_vault POLICY DESCRIPTION: cluster1:c1_svm1_vol1 – Unified SM Policy POLICY SCOPE: Cluster (selected) Policy type: Asynchronous (selected) Transfer Snapshot copies from source: (checked) TRANSFER SCHEDULE: daily

Step	Action
1-6	In the Add protection policy dialog box, scroll down and specify the following values: - In the Transfer Snapshot copies from source section, in the Rules area, click Add and then specify the following values: - Matching SnapMirror label: daily - Retention count: 7 - Click Add again and specify the following values: - Matching SnapMirror label: weekly - Retention count: 52 - Enable network compression: <i>Clear checkbox</i> - Identity preserve: <i>Clear checkbox</i> - Copy all source Snapshot copies: <i>Clear checkbox</i>  The screenshot shows the 'Add protection policy' dialog box. The 'Transfer Snapshot copies from source' section has a 'TRANSFER SCHEDULE' dropdown set to 'daily' with a note 'At 12:10 AM, every day'. Below this is a 'Rules' table with two entries: 'daily' with retention count '7' and 'weekly' with retention count '52'. The 'Create destination Snapshot copies' section is empty. At the bottom, three checkboxes are shown: 'Enable network compression', 'Identity preserve', and 'Copy all source Snapshot copies', all of which are unchecked. A 'Save' button is highlighted in blue.
1-7	Click Save .

Step	Action																								
1-8	Verify that the policy was created successfully. Policies Protection overview Protection policies Snapshot policies + Add <table><thead><tr><th></th><th>Name</th><th>Description</th><th>Policy type</th></tr></thead><tbody><tr><td>▼</td><td>Asynchronous</td><td>A unified Asynchronous SnapMirror and SnapVault policy for mirroring the latest active file sys...</td><td>Asynchronous</td></tr><tr><td>▼</td><td>AutomatedFailOver</td><td>Policy for SnapMirror Synchronous with zero RTO guarantee where client I/O will not be disrup...</td><td>Synchronous</td></tr><tr><td>▼</td><td>c1_svm1_vol1_mirror_vault</td><td>cluster1:c1_svm1_vol1 – Unified SM Policy</td><td>Asynchronous</td></tr><tr><td>▼</td><td>c1_svm3_vol1_async_mirror</td><td>cluster1:svm3_vol1 – Async Mirror policy</td><td>Asynchronous</td></tr><tr><td>▼</td><td>CloudBackupDefault</td><td>Vault policy with daily rule.</td><td>Asynchronous</td></tr></tbody></table>		Name	Description	Policy type	▼	Asynchronous	A unified Asynchronous SnapMirror and SnapVault policy for mirroring the latest active file sys...	Asynchronous	▼	AutomatedFailOver	Policy for SnapMirror Synchronous with zero RTO guarantee where client I/O will not be disrup...	Synchronous	▼	c1_svm1_vol1_mirror_vault	cluster1:c1_svm1_vol1 – Unified SM Policy	Asynchronous	▼	c1_svm3_vol1_async_mirror	cluster1:svm3_vol1 – Async Mirror policy	Asynchronous	▼	CloudBackupDefault	Vault policy with daily rule.	Asynchronous
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1-9	 Cluster1 is the source cluster in the protection relationship. Next, you protect the c1_svm1_vol1 volume on the c1_svm1 SVM with the SnapMirror policy that you created.																								
1-10	In the System Manager navigation pane for Cluster1, select Storage > Volumes .																								
1-11	Click the three vertical dots to the right of c1_svm1_vol1 and then select Protect. DASHBOARD INSIGHTS STORAGE Overview Volumes LUNs Consistency groups NVMe namespaces Shares Buckets Qtrees Quotas Storage VMs Tiers NETWORK Overview Ethernet ports FC ports EVENTS & JOBS Volumes + Add : More <table><thead><tr><th>☐</th><th>Name</th></tr></thead><tbody><tr><td>▼</td><td>c1_svm1_root</td></tr><tr><td>></td><td>c1_svm1_vol1</td></tr><tr><td>></td><td>c1_svm2_root</td></tr></tbody></table> STATUS Online STYLE FlexVol SPACE RESERVATION Thick provisioned SNAPLOCK TYPE Non-SnapLock BACK UP TO CLOUD STATUS Unprotected Edit Resize Delete Clone Manage tags Take offline Enable anti-ransomware Enable quota Move Protect Enable activity tracking Enable analytics	☐	Name	▼	c1_svm1_root	>	c1_svm1_vol1	>	c1_svm2_root																
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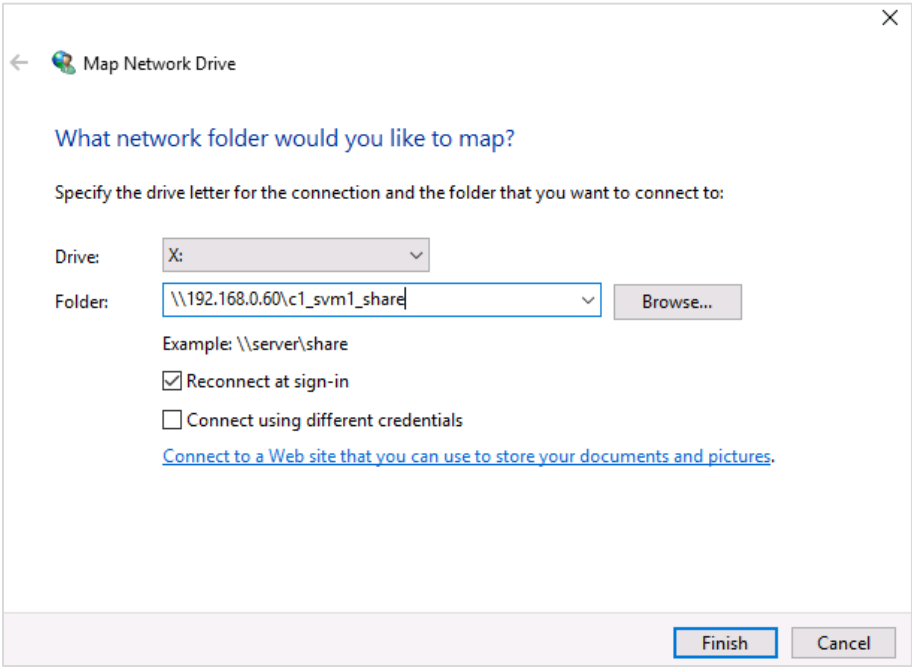
Step	Action
1-12	In the Protect volume dialog box, click the down arrow and select c2_svm1 as the destination storage VM. 
1-13	Click More options .
1-14	In the Protect Volumes pane, specify the following values: - Protection Policy: c1_svm1_vol1_mirror_vault - Destination > Cluster: Cluster2 - Destination > Storage VM: c2_svm1 
1-15	Click Save and wait for the relationship to be created.

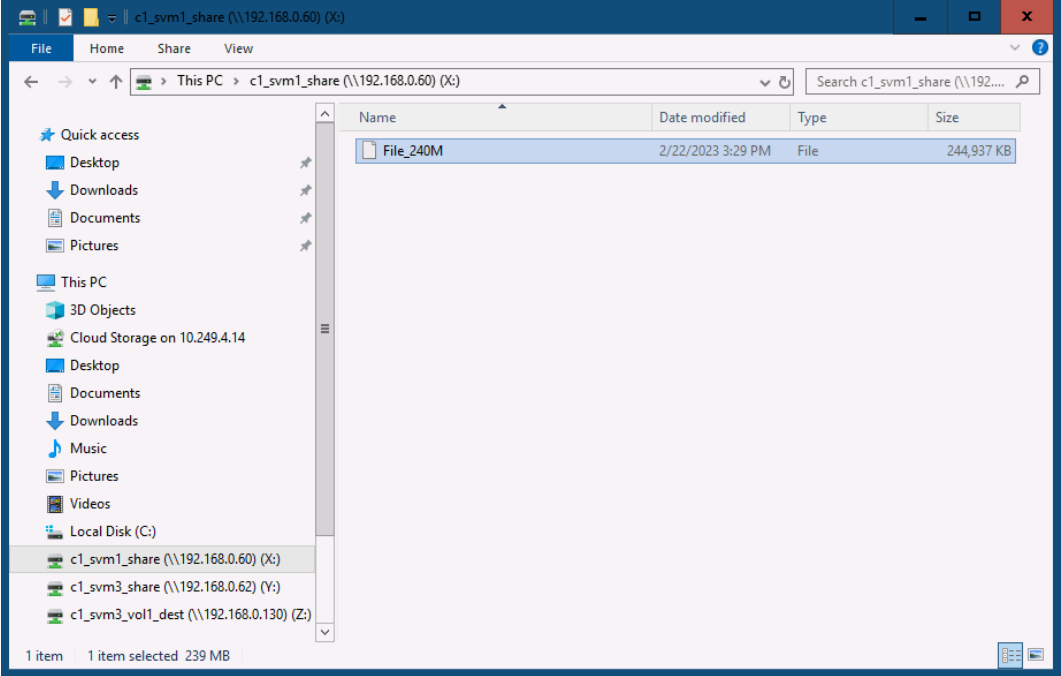
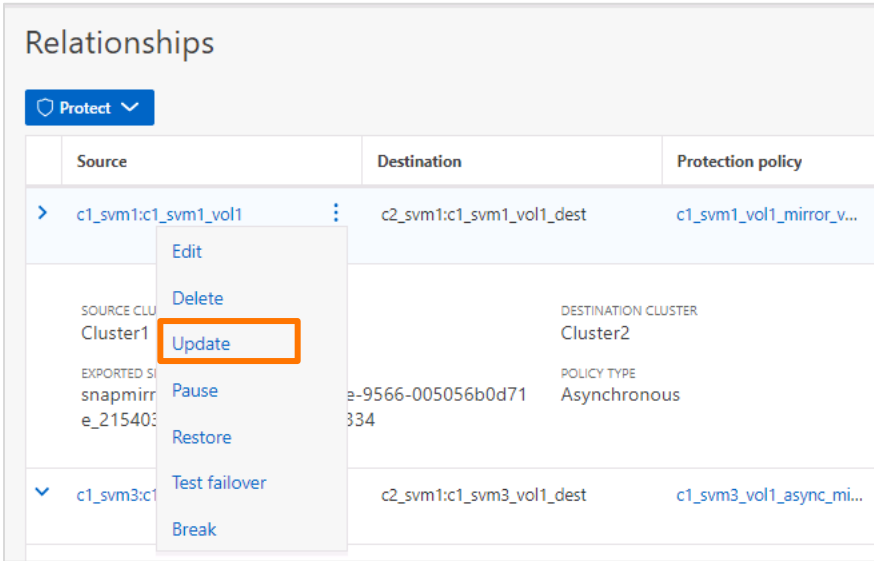
Step	Action
1-16	 Cluster1 and Cluster2 are currently in a peer relationship. On Cluster1, c1_svm1 is not in a peer relationship with c2_svm1 on Cluster2. However, after you click Save in the previous step, System Manager automatically peers the two SVMs and then creates the SnapMirror relationship.
1-17	In the Volumes pane, click the down arrow to the left of c1_svm1_vol1 to view details and verify that the volume is now protected on Cluster2. Click Source to see that the relationship is protected on Cluster2. 
1-18	In the System Manager navigation pane, select Cluster > Settings .
1-19	In the Settings pane, scroll down to the Intercluster settings section and, in the Storage VM peers pane, you see that the storage VMs are now peered. 

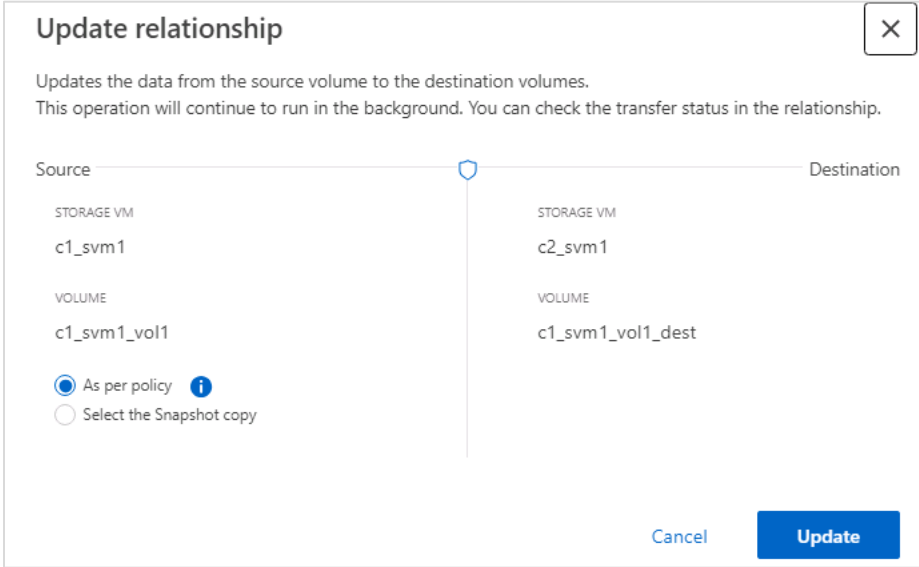
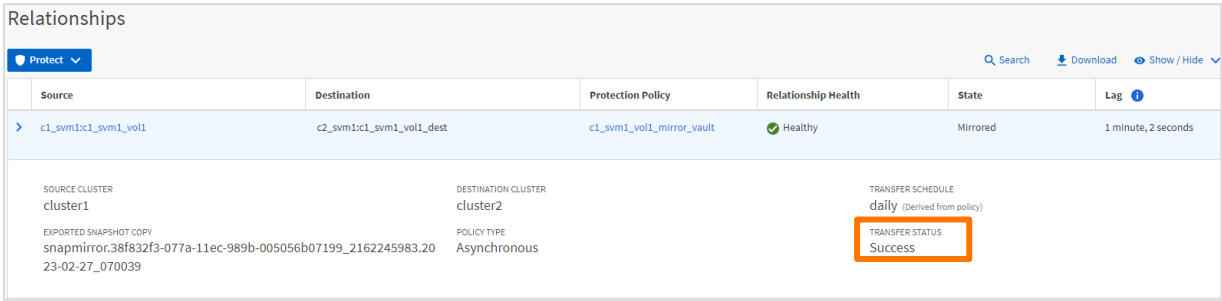
Step	Action																								
1-20	To view the details of the peered SVMs, in the Storage VM peers pane, click the three vertical dots in the upper-right corner and then select Manage storage VM peers. Intercluster settings Network interfaces IP ADDRESS ✓ 192.168.0.106 ✓ 192.168.0.105 Cluster peers PEERED CLUSTER NAME ✓ Cluster2 Storage VM peers PEERED STORAGE VMs ✓ 2 Peer storage VMs Manage storage VM peers Storage VM peers Cluster settings + Peer storage VMs <table><tr><th>Storage VM</th><th>Peered cluster</th><th>Peered storage VM</th><th>Status</th></tr><tr><td>c1_svm1</td><td>Cluster2</td><td>c2_svm1</td><td>Peered</td></tr><tr><td>c1_svm3</td><td>Cluster2</td><td>c2_svm1</td><td>Peered</td></tr></table>	Storage VM	Peered cluster	Peered storage VM	Status	c1_svm1	Cluster2	c2_svm1	Peered	c1_svm3	Cluster2	c2_svm1	Peered												
Storage VM	Peered cluster	Peered storage VM	Status																						
c1_svm1	Cluster2	c2_svm1	Peered																						
c1_svm3	Cluster2	c2_svm1	Peered																						
1-21	In the System Manager navigation pane for Cluster2, select Protection > Relationships .																								
1-22	DASHBOARD INSIGHTS STORAGE NETWORK EVENTS & JOBS PROTECTION Overview Relationships HOSTS CLUSTER Relationships Protect Search Download Show / Hide <table><tr><th>Source</th><th>Destination</th><th>Protection Policy</th><th>Relationship Health</th><th>State</th><th>Lag</th></tr><tr><td>> c1_svm1c1_svm1_voli</td><td>c2_svm1c1_svm1_voli_dest</td><td>c1_svm1_voli_mirror_vault</td><td>Healthy</td><td>Mirrored</td><td>9 minutes, 22 seconds</td></tr><tr><td colspan="6"> SOURCE CLUSTER cluster1 EXPORTED SNAPSHOT COPY snapmirror:38f832f3-077a-11ec-989b-005056b07199_2162245983.20-23-02-23_203738 DESTINATION CLUSTER cluster2 POLICY TYPE Asynchronous TRANSFER SCHEDULE daily (Derived from policy) TRANSFER STATUS Success </td></tr><tr><td>< c1_svm3c1_svm3_voli</td><td>c2_svm1c1_svm3_voli_dest</td><td>c1_svm3_voli_async_mirror</td><td>Healthy</td><td>Mirrored</td><td>33 minutes, 16 seconds</td></tr></table>	Source	Destination	Protection Policy	Relationship Health	State	Lag	> c1_svm1c1_svm1_voli	c2_svm1c1_svm1_voli_dest	c1_svm1_voli_mirror_vault	Healthy	Mirrored	9 minutes, 22 seconds	SOURCE CLUSTER cluster1 EXPORTED SNAPSHOT COPY snapmirror:38f832f3-077a-11ec-989b-005056b07199_2162245983.20-23-02-23_203738 DESTINATION CLUSTER cluster2 POLICY TYPE Asynchronous TRANSFER SCHEDULE daily (Derived from policy) TRANSFER STATUS Success						< c1_svm3c1_svm3_voli	c2_svm1c1_svm3_voli_dest	c1_svm3_voli_async_mirror	Healthy	Mirrored	33 minutes, 16 seconds
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1-23	To verify the SnapMirror status, expand the relationship that is listed.																								

Task 2: Verify Data Transfer

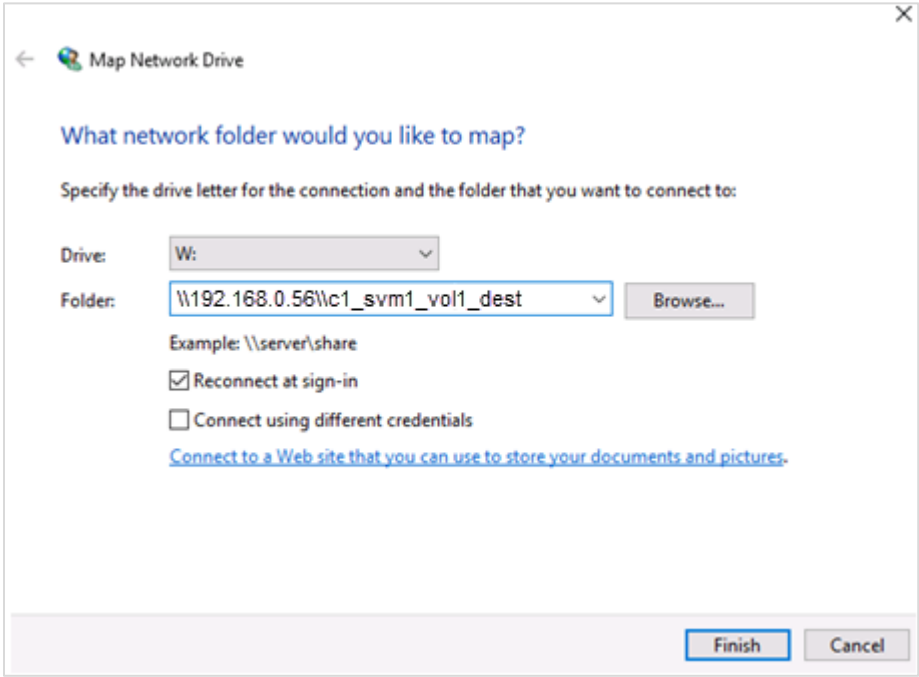
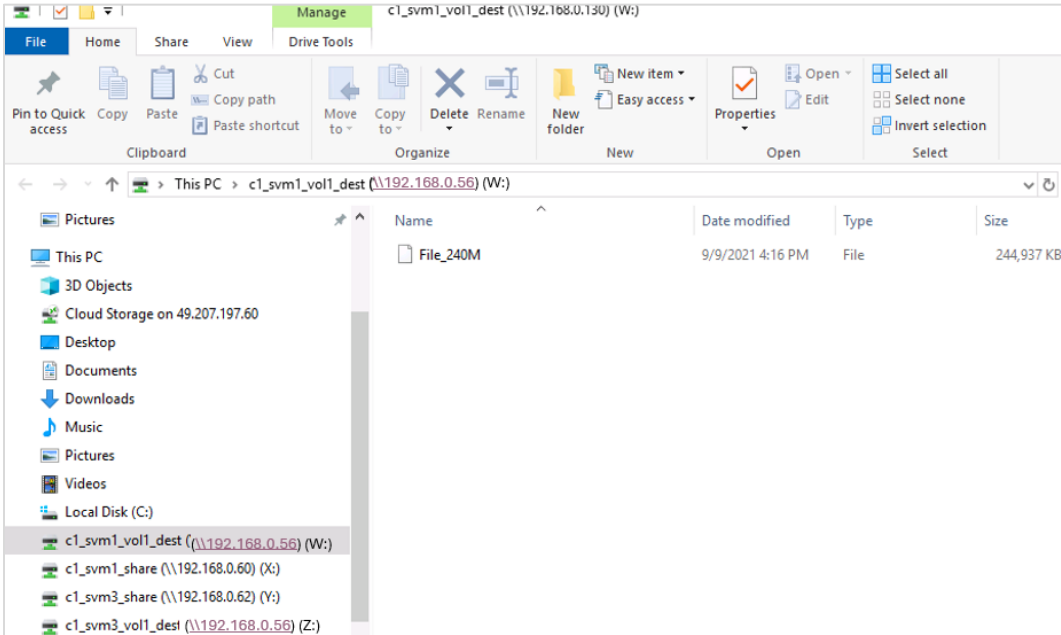
In this task, you verify that the SnapMirror relationship can successfully transfer data from the primary volume to the secondary volume. To test the replication, you open File Explorer and map a drive letter to the share on svm1.

Step	Action
2-1	In File Explorer, open This PC , click the Computer tab, and then select Map network drive .
2-2	 A share called c1_svm1_share is set up for the source volume c1_svm1_vol1.
2-3	On the jump host, use File Explorer to map a drive to the source volume on cluster1\svm1, and use the folder path \\192.168.0.60\c1_svm1_share. 
2-4	Verify that the mapped drive c1_svm1_share (\\192.168.0.60) is empty.

Step	Action
2-5	Copy File_240M from Desktop\CourseFiles to c1_svm1_share (\\192.168.0.60). 
2-6	Instead of waiting for the SnapMirror update schedule, initiate a manual update.
2-7	In the System Manager navigation pane for Cluster2, select Protection > Relationships .
2-8	In the Relationships pane, click the three vertical dots to the right of the source and then select Update. 

Step	Action
2-9	In the Update relationship dialog box, click Update. 
2-10	Wait for the Transfer Status to change from Transferring to Success. 
2-11	 Next, you mount the destination volume created on c2_svm1 in Cluster2.
2-12	In the System Manager navigation pane, select Storage > Volumes.

Step	Action																																																																
2-13	In the Volumes pane, click the three vertical dots to the right of c1_svm1_vol1_dest, and then select Edit > Mount path. Volumes + Add More <table><thead><tr><th></th><th>Name</th><th></th><th>Storage VM</th><th>Status</th><th>Capacity</th></tr></thead><tbody><tr><td>▼</td><td>c1_svm1_vol1_dest</td><td>⋮</td><td>c2_svm1</td><td>Online</td><td>364 KiB used164 MiB available173 MiB</td></tr><tr><td>▼</td><td>c1_s</td><td>⋮</td><td></td><td>Online</td><td>243 MiB used56.3 MiB available314 MiB</td></tr><tr><td>▼</td><td>c2sv</td><td>⋮</td><td></td><td>Online</td><td>1.14 MiB used17.9 MiB available20 MiB</td></tr><tr><td>▼</td><td>c2_s</td><td>⋮</td><td></td><td>Online</td><td>1.01 MiB used18 MiB available20 MiB</td></tr><tr><td>▼</td><td>c2_s</td><td>⋮</td><td>c2_svm4</td><td>Online</td><td>1.19 MiB used17.8 MiB available20 MiB</td></tr><tr><td>▼</td><td>Flex</td><td>⋮</td><td>c2_svm2</td><td>Online</td><td>526 MiB used8.99 GiB available10 GiB</td></tr><tr><td>▼</td><td>svm0_root</td><td>⋮</td><td>c2_svm1</td><td>Online</td><td>1.05 MiB used17.9 MiB available20 MiB</td></tr></tbody></table> <tr><td>2-14</td><td> In the Edit mount path pane, select the Mount checkbox, note the mount path, and then click Save. Edit mount path VOLUME c1_svm1_vol1_dest ☒ Mount PATH /c1_svm1_vol1_dest Browse Save Cancel <tr><td>2-15</td><td> In the System Manager navigation pane, select Storage > Shares. </td></tr><tr><td>2-16</td><td> In the Shares pane, click Add and then select Share. Shares + Add Share <table><thead><tr><th></th><th>Storage VM</th></tr></thead><tbody><tr><td>Home directory</td><td>c3_svm2</td></tr><tr><td>▼ ipc\$</td><td>c3_svm2</td></tr><tr><td>▼ c\$</td><td>c3_svm1</td></tr><tr><td>▼ ipc\$</td><td>c3_svm1</td></tr></tbody></table> </td></tr></td></tr>		Name		Storage VM	Status	Capacity	▼	c1_svm1_vol1_dest	⋮	c2_svm1	Online	364 KiB used164 MiB available173 MiB	▼	c1_s	⋮		Online	243 MiB used56.3 MiB available314 MiB	▼	c2sv	⋮		Online	1.14 MiB used17.9 MiB available20 MiB	▼	c2_s	⋮		Online	1.01 MiB used18 MiB available20 MiB	▼	c2_s	⋮	c2_svm4	Online	1.19 MiB used17.8 MiB available20 MiB	▼	Flex	⋮	c2_svm2	Online	526 MiB used8.99 GiB available10 GiB	▼	svm0_root	⋮	c2_svm1	Online	1.05 MiB used17.9 MiB available20 MiB	2-14	In the Edit mount path pane, select the Mount checkbox, note the mount path, and then click Save. Edit mount path VOLUME c1_svm1_vol1_dest ☒ Mount PATH /c1_svm1_vol1_dest Browse Save Cancel <tr><td>2-15</td><td> In the System Manager navigation pane, select Storage > Shares. </td></tr> <tr><td>2-16</td><td> In the Shares pane, click Add and then select Share. Shares + Add Share <table><thead><tr><th></th><th>Storage VM</th></tr></thead><tbody><tr><td>Home directory</td><td>c3_svm2</td></tr><tr><td>▼ ipc\$</td><td>c3_svm2</td></tr><tr><td>▼ c\$</td><td>c3_svm1</td></tr><tr><td>▼ ipc\$</td><td>c3_svm1</td></tr></tbody></table> </td></tr>	2-15	In the System Manager navigation pane, select Storage > Shares.	2-16	In the Shares pane, click Add and then select Share. Shares + Add Share <table><thead><tr><th></th><th>Storage VM</th></tr></thead><tbody><tr><td>Home directory</td><td>c3_svm2</td></tr><tr><td>▼ ipc\$</td><td>c3_svm2</td></tr><tr><td>▼ c\$</td><td>c3_svm1</td></tr><tr><td>▼ ipc\$</td><td>c3_svm1</td></tr></tbody></table>		Storage VM	Home directory	c3_svm2	▼ ipc\$	c3_svm2	▼ c\$	c3_svm1	▼ ipc\$	c3_svm1
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Step	Action
2-20	To map a drive to the destination volume, use the following path: \\192.168.0.56\c1_svm1_vol1_dest 
2-21	In File Explorer, open the mapped drive c1_svm1_vol1_dest (\\192.168.0.56) and verify that the SnapMirror relationship transferred the file from the source. 

End of exercise

Exercise 2(b): Restoring Data from a Snapshot Copy

In this exercise, you simulate data loss or corruption on the source volume. You then restore an entire volume to the most recent Snapshot copy.

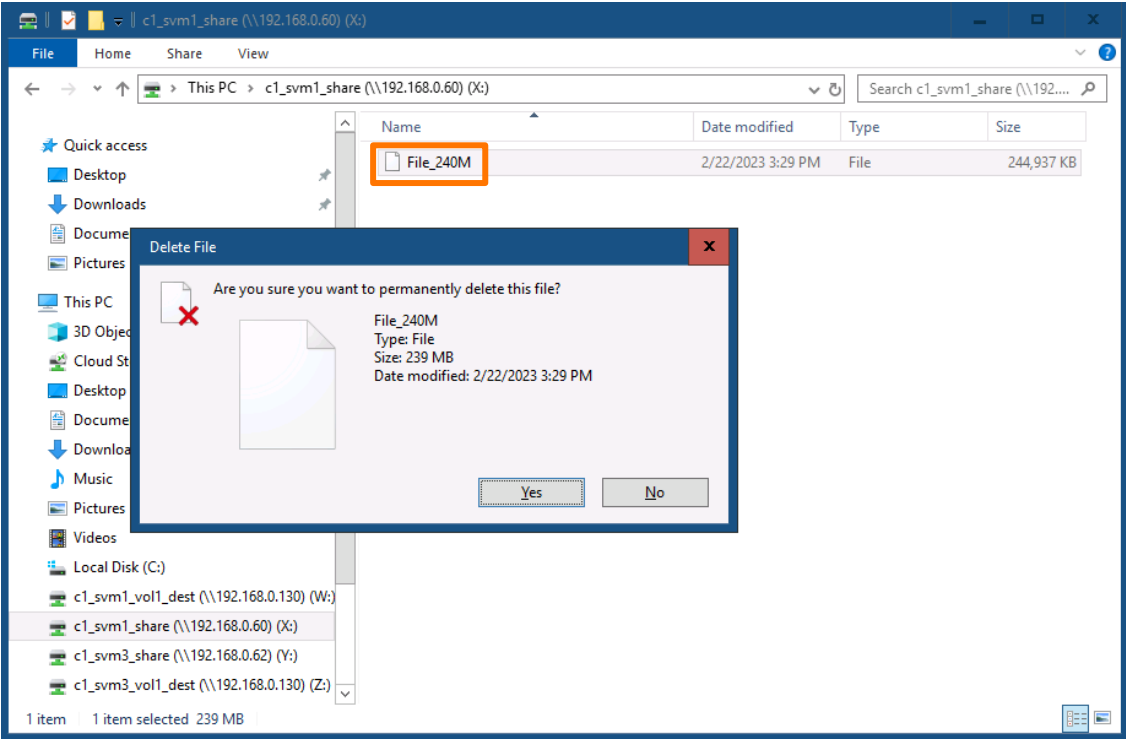
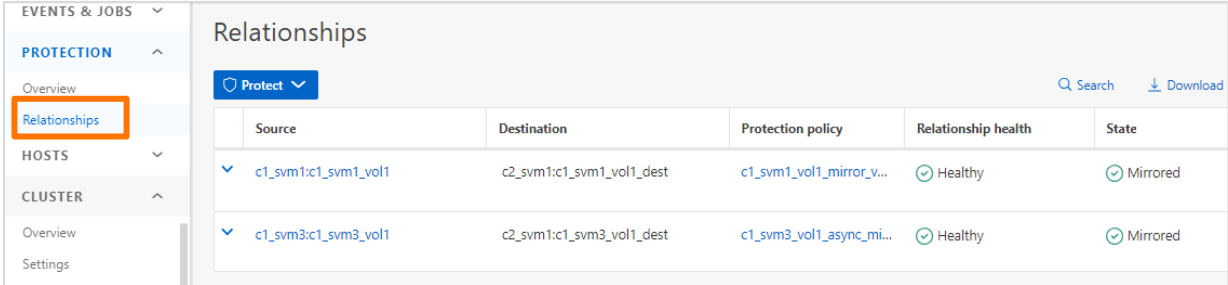
Objectives

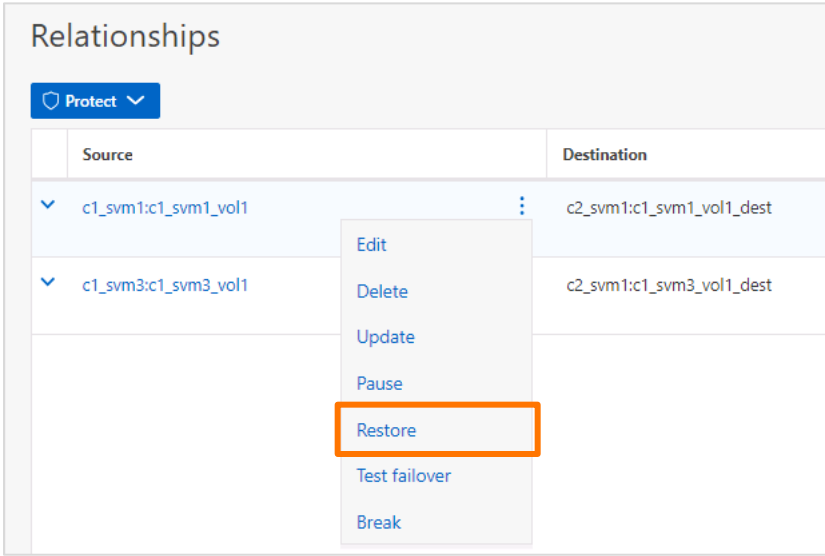
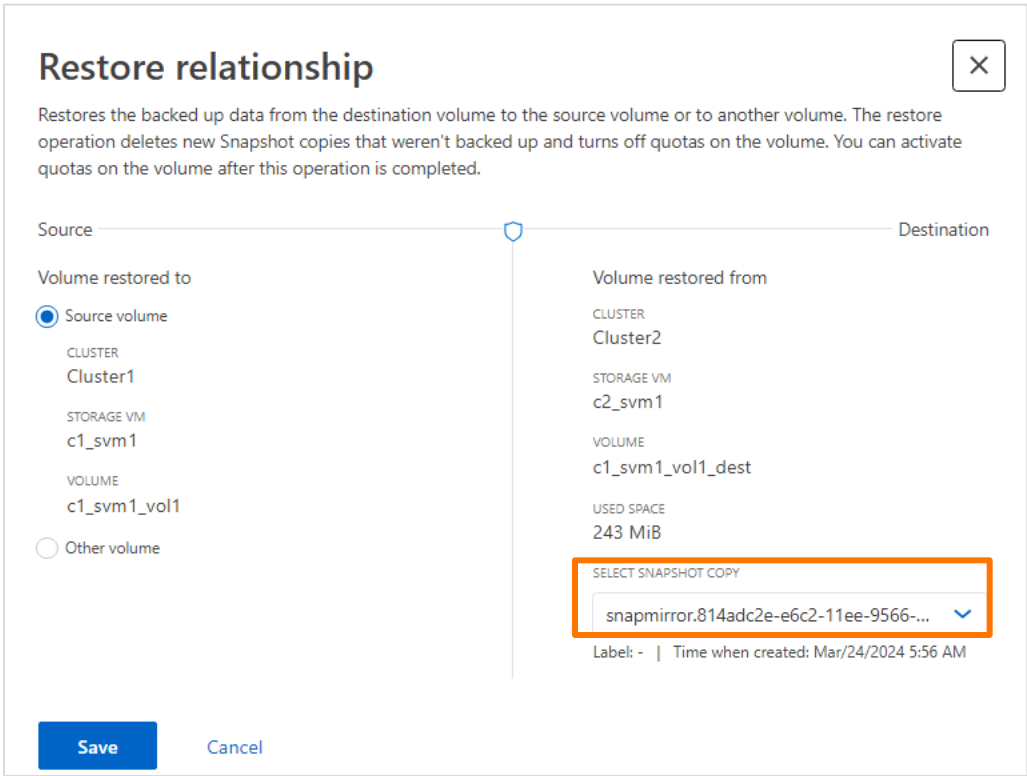
This exercise focuses on enabling you to do the following:

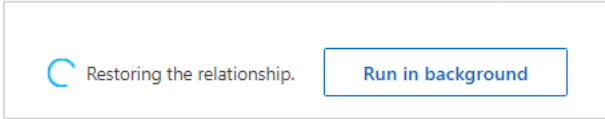
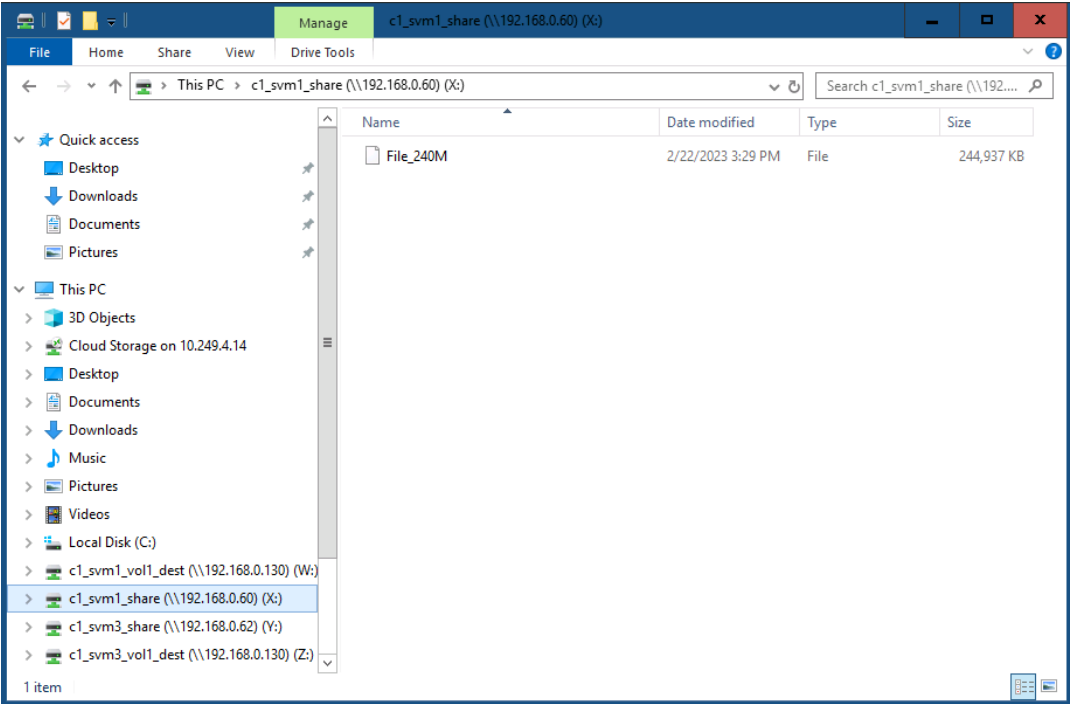
- Simulate a disaster
- Use SnapMirror to recover your data

Task 1: Simulate a Disaster and Recover Data

In this task, you delete a file on the source volume to simulate data loss or corruption. From the destination cluster, you then roll back your data by restoring from an earlier Snapshot copy.

Step	Action
1-1	In File Explorer, open the mapped drive c1_svm1_share (\\192.168.0.60) (X:), and simulate data loss by deleting File_240M. 
1-2	In the System Manager navigation pane for Cluster2, select Protection > Relationships. 

Step	Action
1-3	In the Relationships pane, in the mirror-vault relationship, click the three vertical dots to the right of the source, and then select Restore.  The screenshot shows the 'Relationships' pane with a 'Protect' dropdown. Below it is a table with 'Source' and 'Destination' columns. The first row shows 'c1_svm1:c1_svm1_vol1' as the source and 'c2_svm1:c1_svm1_vol1_dest' as the destination. A context menu is open for the first source entry, listing options: Edit, Delete, Update, Pause, Restore (highlighted with an orange box), Test failover, and Break.
1-4	In the Restore relationship pane, retain the default settings and verify that the latest Snapshot copy is selected.  The screenshot shows the 'Restore relationship' pane. It includes a description of the restore operation. Below, there are sections for 'Source' and 'Destination'. The 'Source' section shows 'Volume restored to' with radio buttons for 'Source volume' (selected) and 'Other volume'. The 'Destination' section shows 'Volume restored from' with details for Cluster2, Storage VM c2_svm1, and Volume c1_svm1_vol1_dest. A 'SELECT SNAPSHOT COPY' dropdown is highlighted with an orange box, showing 'snapmirror.814adc2e-e6c2-11ee-9566-...' as the selected option. At the bottom are 'Save' and 'Cancel' buttons.
1-5	 The policy that is assigned to the SnapMirror relationship has a job schedule that is configured for daily and weekly updates, so you need to restore from the most recent Snapshot copy.

Step	Action
1-6	Click Save and wait for the restore operation to finish. 
1-7	In File Explorer, return to c1_svm1_share (\\192.168.0.60) and verify that file File_240M is restored. 
1-8	If necessary, refresh your screen until you see that the file is restored.

End of exercise

Exercise 3(a): Configuring Cluster and Storage VM Peering for a Cascade Relationship (Optional)

In Exercise 1 of Module 1, you created a cluster peering between Cluster1 and Cluster 2 and a storage VM peering between two storage VMs on the two clusters.

In this exercise, you configure cluster peering between Cluster 2 and Cluster 3 to establish a cascading relationship. You also configure a storage VM peering between the storage VMs in the two clusters.

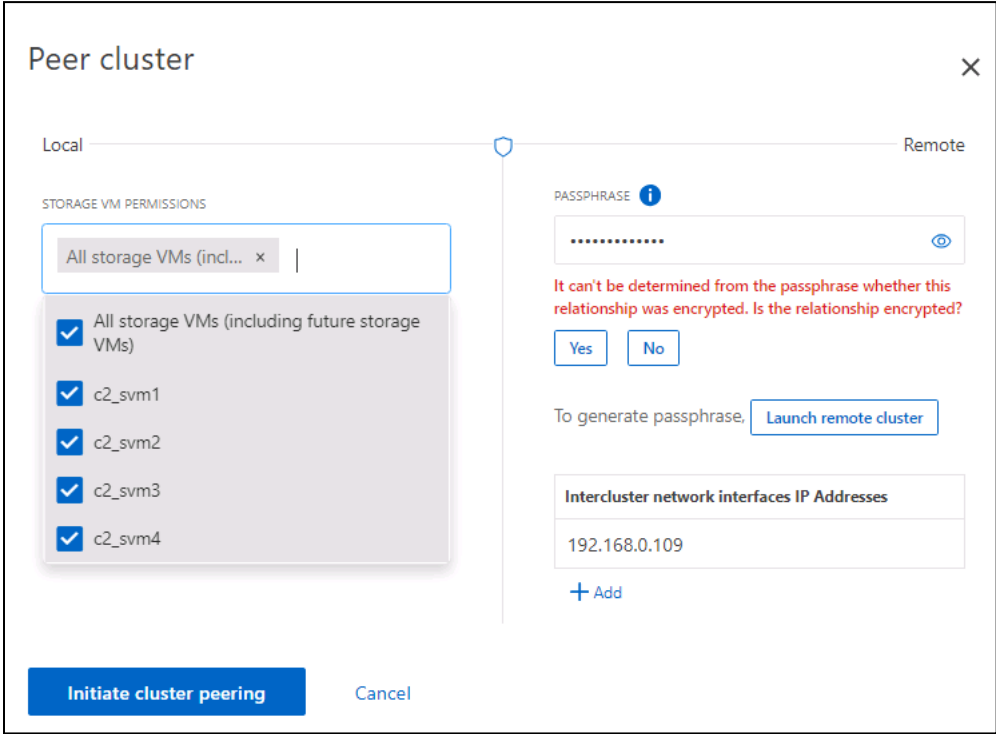
Objectives

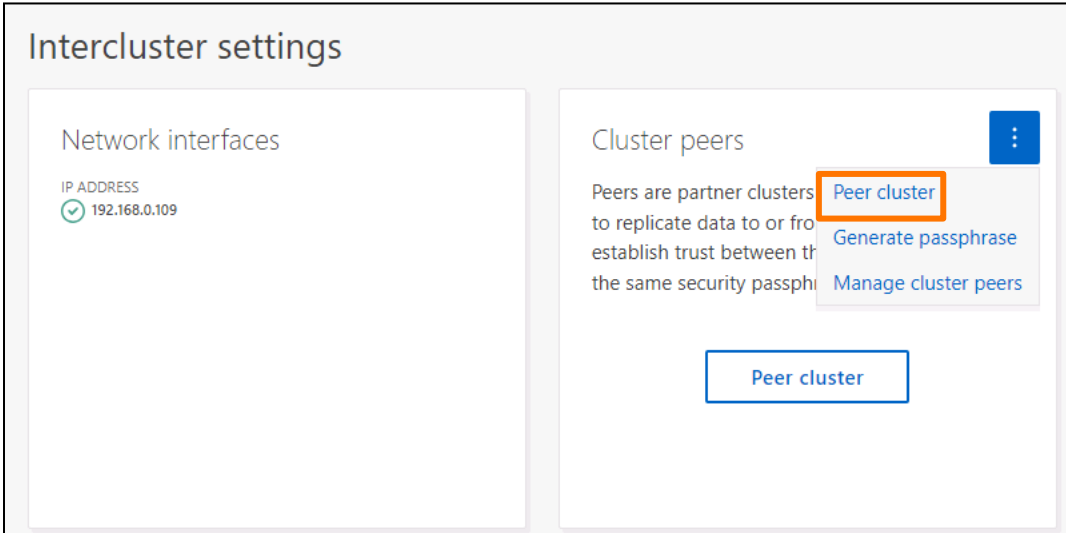
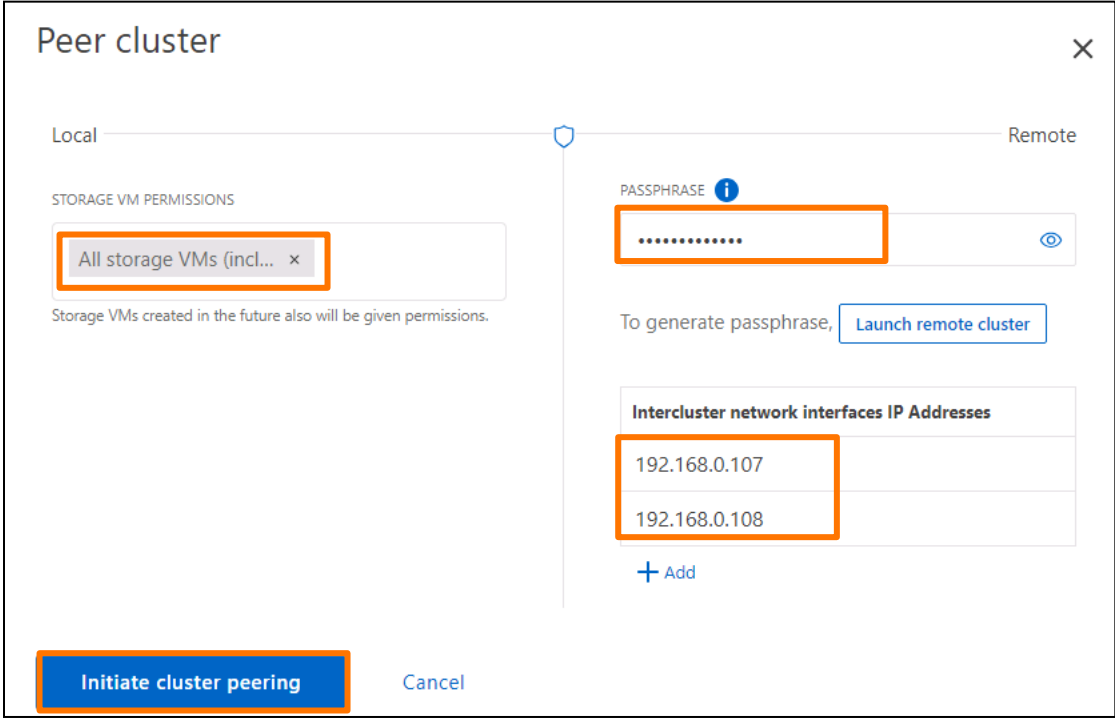
This exercise focuses on enabling you to do the following:

- Configure cluster peering
- Configure storage VM peering

Task 1: Configure Cluster Peering

Step	Action
1-1	Create an intercluster LIF in Cluster3 with the following details: ▪ Name: Intercluster_LIF1 ▪ Home Node: Cluster3-01 ▪ IP Address: 192.168.0.109
1-2	 Review the tasks of Exercise 1 of Module 1 for the detailed steps to configure intercluster peering between Cluster2 and Cluster3.
1-3	In the System Manager navigation pane for Cluster2, navigate to Cluster > Settings > Intercluster settings > Cluster peers, click the three vertical dots, and then select Peer cluster. Intercluster settings Network interfaces IP ADDRESS ✓ 192.168.0.108 ✓ 192.168.0.107 Cluster peers PEERED CLUSTER NAME ✓ Cluster1 Peer cluster Generate passphrase Manage cluster peers Storage VM peers PEERED STORAGE VMs ✓ 2

Step	Action
1-4	In the Peer cluster dialog box, perform the following steps: - For Passphrase, enter dpadmincourse. - In response to the question “Is the relationship encrypted?”, click Yes. - Click Add. - In the Intercluster Network Interfaces IP Addresses field, enter 192.168.0.109. - Click outside the text box. - In the Storage VM Permissions field, select the All storage VMs (including future storage VMs) checkbox. 
1-5	Click Initiate cluster peering .

Step	Action
1-6	In the System Manager navigation pane for Cluster3, navigate to Cluster > Settings > Intercluster settings > Cluster peers, click the three vertical dots, and then select Peer cluster. 
1-7	In the Peer cluster dialog box, perform the following tasks: - For Passphrase, enter dpadmincourse. - For relationship encryption, click Yes. - Add two Intercluster network interfaces IP Addresses: 192.168.0.107 and 192.168.0.108. - In the Storage VM Permissions field, select the All storage VMs (including future storage VMs) checkbox. 
1-8	Click Initiate cluster peering .

Step	Action
1-9	In the Intercluster settings > Cluster peers pane, wait a few seconds for the peering operation to finish and verify that Cluster2 appears, indicating that the cluster peering is successful on Cluster3. Intercluster settings Network interfaces IP ADDRESS  192.168.0.109 Cluster peers PEERED CLUSTER NAME  Cluster2 Storage VM peers PEERED STORAGE VMS 0
1-10	In the Intercluster settings > Cluster peers pane for Cluster2, wait for the status icon to turn green and verify that cluster peering is successful on Cluster2. Intercluster settings Network interfaces IP ADDRESS  192.168.0.108  192.168.0.107 Cluster peers PEERED CLUSTER NAME  Cluster1  Cluster3 Storage VM peers PEERED STORAGE VMS  2
1-11	 Next, you create a peer relationship between c2_svm1 on Cluster2 and c3_svm1 on Cluster3.

Task 2: Configure Storage VM Peering

Step	Action																
2-1	Create a storage VM relationship between c2_svm1 and c3_svm1 in Cluster2 and Cluster3 respectively. - Cluster2 Storage VM: c2_svm1 - Cluster: Cluster3 - Cluster3 Storage VM: c3_svm1 Peer storage VMs Local CLUSTER Cluster2 STORAGE VM c2_svm1 Remote CLUSTER Cluster3 STORAGE VM c3_svm1 Peer storage VMs Cancel																
2-2	i Review the tasks of the previous exercise for the detailed steps to configure storage VM peering between Cluster2 and Cluster3.																
2-3	On the Storage VM peers page, verify that the storage VM peer relationship is listed and that the status is Peered. Storage VM peers Cluster settings + Peer storage VMs <table><tr><th>Storage VM</th><th>Peered cluster</th><th>Peered storage VM</th><th>Status</th></tr><tr><td>c2_svm1</td><td>Cluster3</td><td>c3_svm1</td><td>Peered</td></tr><tr><td>c2_svm1</td><td>Cluster1</td><td>c1_svm3</td><td>Peered</td></tr><tr><td>c2_svm1</td><td>Cluster1</td><td>c1_svm1</td><td>Peered</td></tr></table>	Storage VM	Peered cluster	Peered storage VM	Status	c2_svm1	Cluster3	c3_svm1	Peered	c2_svm1	Cluster1	c1_svm3	Peered	c2_svm1	Cluster1	c1_svm1	Peered
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Storage VM	Peered cluster	Peered storage VM	Status														
c3_svm1	Cluster2	c2_svm1	Peered														

End of exercise

Exercise 3(b): Configuring a Cascade Relationship (Optional)

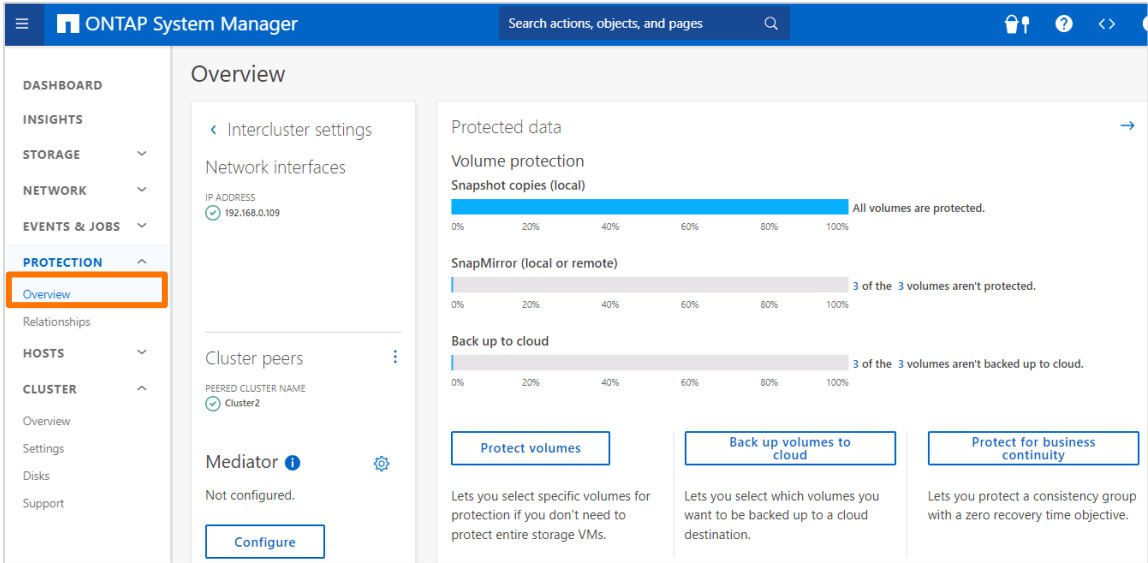
In Exercise 1 of Module 2, you created a SnapMirror relationship between a source volume on Cluster1 and a destination volume on Cluster2. In this optional exercise, you configure a cascade relationship by creating a SnapMirror relationship between the secondary volume on Cluster2 and a tertiary volume on Cluster3.

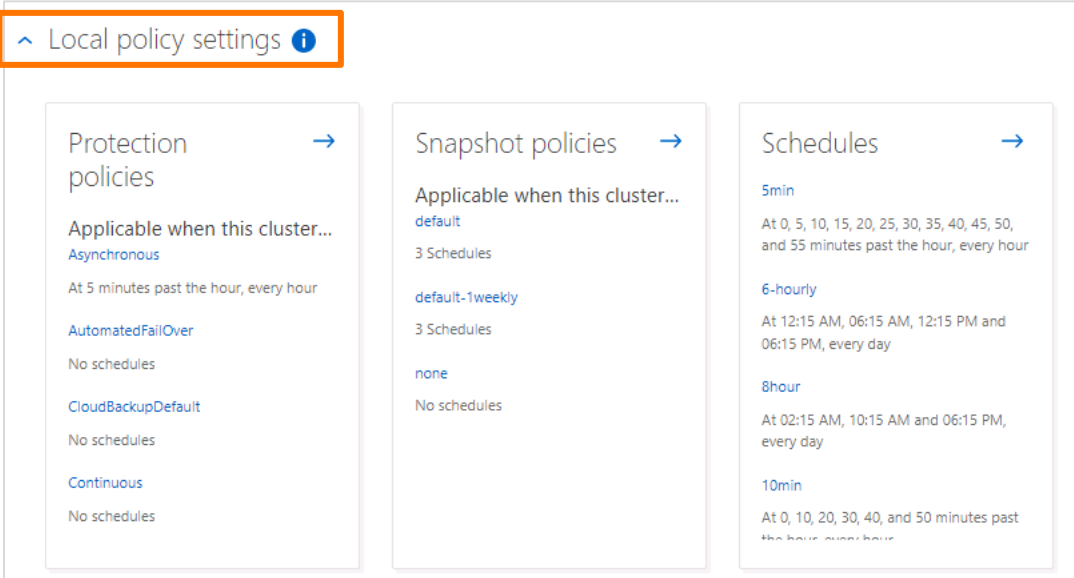
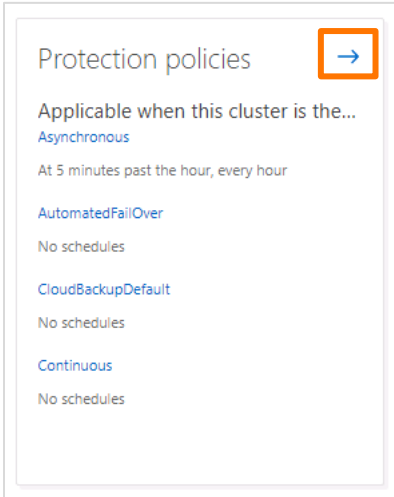
Objectives

This exercise focuses on enabling you to do the following:

- Create a SnapMirror policy
- Create a SnapMirror cascade relationship
- Verify data transfer

Task 1: Configure a SnapMirror Policy

Step	Action
1-1	 Cluster3 is the destination cluster for the protection relationship. You create the custom SnapMirror policy on the destination cluster.
1-2	In the System Manager navigation pane for Cluster3, select Protection > Overview. 

Step	Action
1-3	Scroll down to the Local policy settings section and expand Local policy settings. 
1-4	In the Local policy settings > Protection policies pane, click the arrow in the upper-right corner. 

Step

Action

1-5

In the Policies > Protection policies pane, click **Add** to create a policy.

Policies

Protection overview

Protection policies

Snapshot policies

+ Add

	Name	Description
✓	Asynchronous	A unified Asynchronous SnapMirror and SnapVault policy for m...
✓	AutomatedFailOver	Policy for SnapMirror Synchronous with zero RTO guarantee wh...
✓	CloudBackupDefault	Vault policy with daily rule.

Step	Action
1-6	In the Add protection policy dialog box, specify the following values: - Policy Name: c1_svm3_vol1_mirror_vault_tertiary - Policy Description: cluster1:c1_svm3_vol1 – Unified SM Policy for Cascade - Policy Scope: Cluster - Policy type: Asynchronous - Transfer Schedule: daily

Add protection policy ×

POLICY NAME

POLICY DESCRIPTION

POLICY SCOPE

☒ Cluster
 ☐ Storage VM

Policy type

☒ **Asynchronous**
 The copy can be used for backup and disaster protection.

☐ Synchronous
 The copy matches the source.

☐ Continuous
 Continuously back up to on-premises or cloud object store.

☐ Back up to cloud
 Back up data to cloud object storage.

[Advance policy options](#)

Transfer Snapshot copies from source ⓘ

TRANSFER SCHEDULE

▼

At 12:10 AM, every day

1-7

Scroll down and specify the following values:

In the Transfer Snapshot copies from source section, in the Rules area, click **Add**, and then specify the following values:

Matching SnapMirror label: **weekly**

Retention count: **250**

Enable network compression: *Clear checkbox*

Identity preserve: *Clear checkbox*

Copy all source Snapshot copies: *Clear checkbox*

Rules

Rule considerations

Matching SnapMirror label	Retention count	SnapLock retention period	
weekly	250	-	

+ Add

Create destination Snapshot copies ⓘ

Rules

Retention schedule	Retention count	SnapMirror label	SnapLock retention period	
No data				

+ Add

☐ Enable network compression

☐ Identity preserve ⓘ

☐ Copy all source Snapshot copies

Save

Cancel

1-8

Click **Save**.

Step

Action

1-9

Verify that the policy was created successfully.

Policies

Protection overview

Protection policies

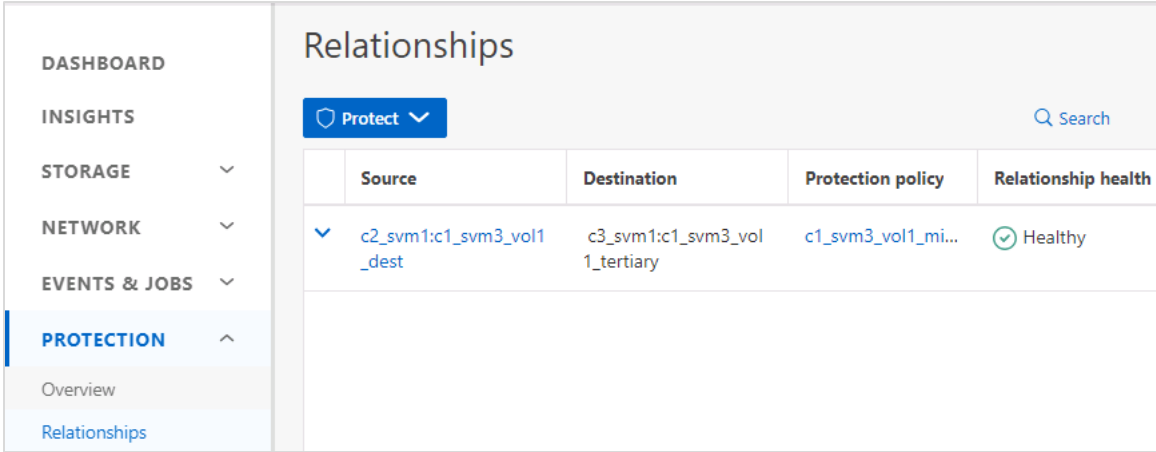
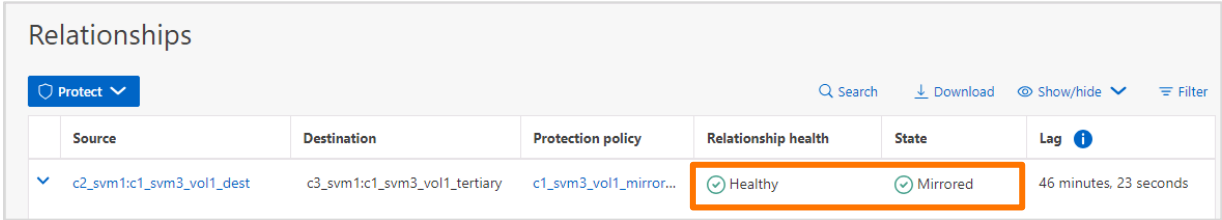
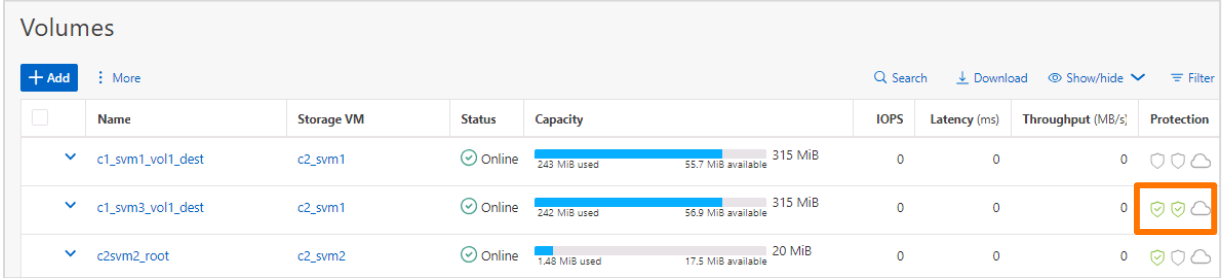
Snapshot policies

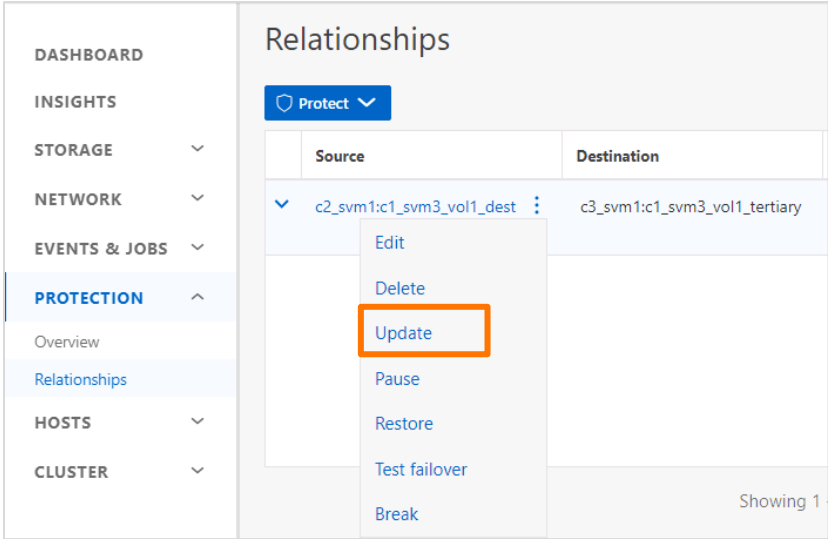
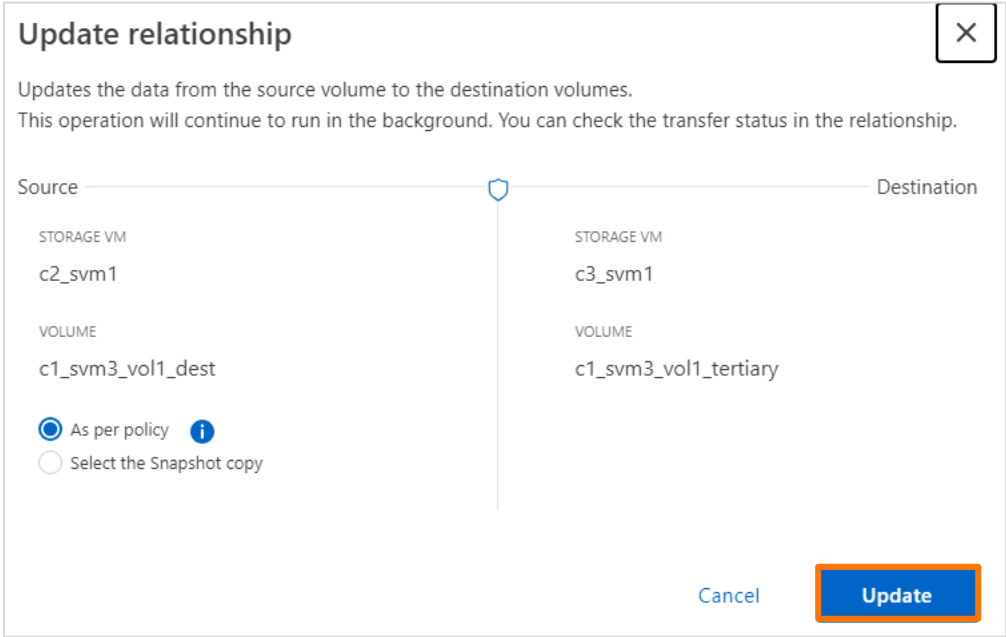
+ Add

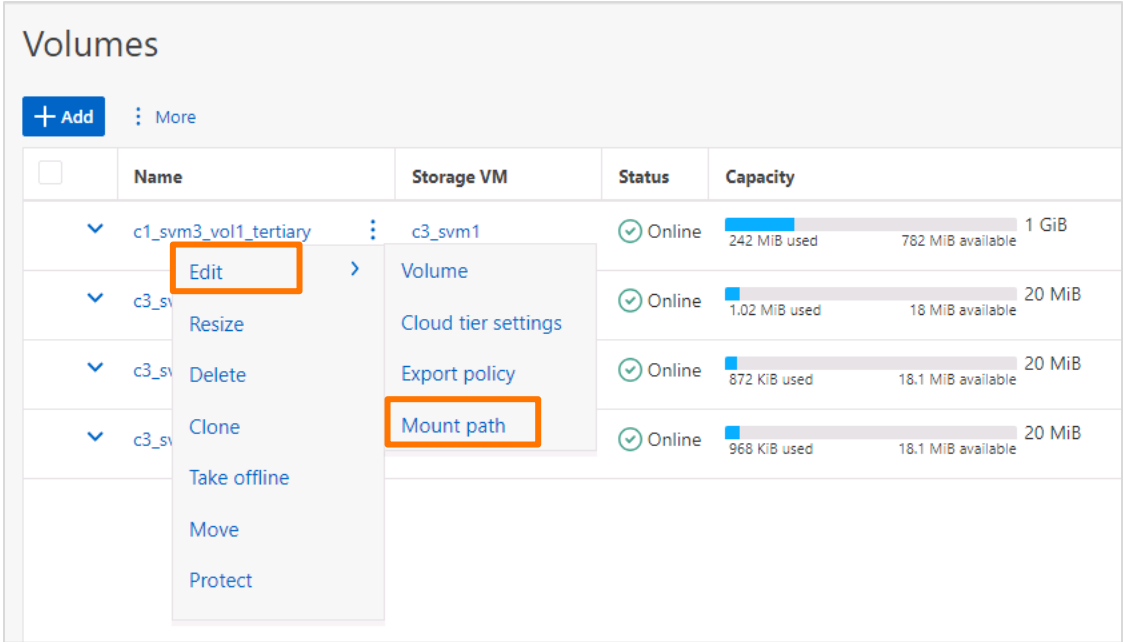
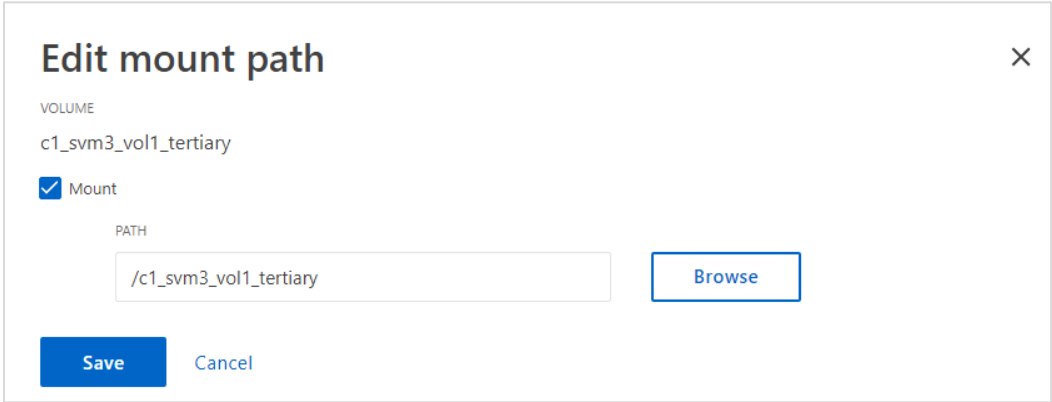
	Name	Description	Policy type
✓	Asynchronous	A unified Asynchronous SnapMirror and SnapVault policy for mirroring the latest ...	Asynchronous
✓	AutomatedFailOver	Policy for SnapMirror Synchronous with zero RTO guarantee where client I/O will ...	Synchronous
✓	c1_svm3_vol1_mirror_vault_tertiary	cluster1:c1_svm3_vol1 – Unified SM Policy for Cascade	Asynchronous
✓	CloudBackupDefault	Vault policy with daily rule.	Asynchronous

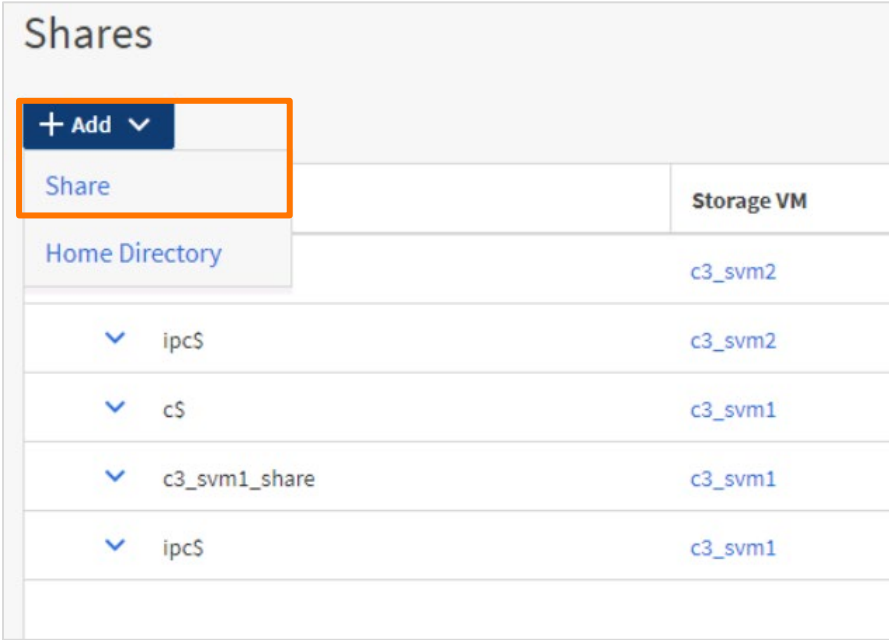
Task 2: Configure a SnapMirror Cascade Relationship

Step	Action
2-1	 Cluster2 is the secondary cluster and is the source in this protection relationship. Next, you protect c1_svm3_vol1_dest on c2_svm1 with the SnapMirror policy that you created. You perform the tasks in the CLI. The System Manager UI supports creating cascade relationships only for SnapMirror Cloud and Cloud Backup service.
2-2	Log in to the PuTTY session for Cluster3 with these credentials: - Login: admin - Password: Netapp1!
2-3	Use the <code>volume create</code> command to create a destination volume that is the same or greater in size than the source volume. <pre>volume create -vserver c3_svm1 -volume c1_svm3_vol1_tertiary -aggregate Cluster3_01_FC_1 -type DP -size 1GB</pre> Sample output: <pre>volume create -vserver c3_svm1 -volume c1_svm3_vol1_tertiary -aggregate Cluster3_01_FC_1 -type DP -size 1GB [Job 60] Job succeeded: Successful</pre>
2-4	Use the <code>snapmirror create</code> command to create the SnapMirror relationship between the source volume c1_svm3_vol1_dest on Cluster2 and the destination volume c1_svm3_vol1_tertiary on Cluster3. <pre>snapmirror create -source-path c2_svm1:c1_svm3_vol1_dest -destination- path c3_svm1:c1_svm3_vol1_tertiary -type XDP -policy c1_svm3_vol1_mirror_vault_tertiary</pre> Sample output: <pre>snapmirror create -source-path c2_svm1:c1_svm3_vol1_dest -destination-path c3_svm1:c1_svm3_vol1_tertiary -type XDP -policy c1_svm3_vol1_mirror_vault_tertiary Operation succeeded: snapmirror create for the relationship with destination "c3_svm1:c1_svm3_vol1_tertiary".</pre>
2-5	Use the <code>snapmirror initialize</code> command to initialize the relationship with the secondary cluster. <pre>snapmirror initialize -source-path c2_svm1:c1_svm3_vol1_dest -destination-path c3_svm1:c1_svm3_vol1_tertiary</pre> Sample output: <pre>snapmirror initialize -source-path c2_svm1:c1_svm3_vol1_dest -destination-path c3_svm1:c1_svm3_vol1_tertiary Operation is queued: snapmirror initialize of destination "c3_svm1:c1_svm3_vol1_tertiary".</pre>
2-6	Use the <code>snapmirror show</code> command to view the SnapMirror relationship.

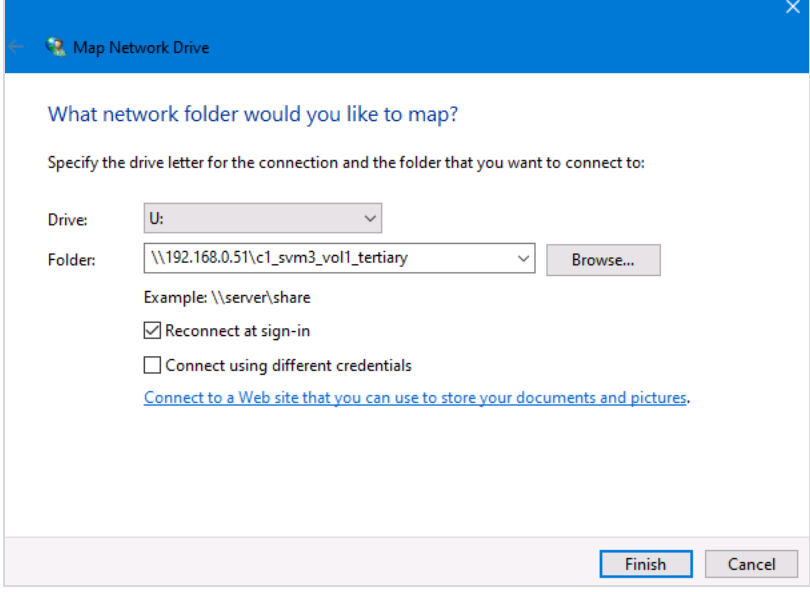
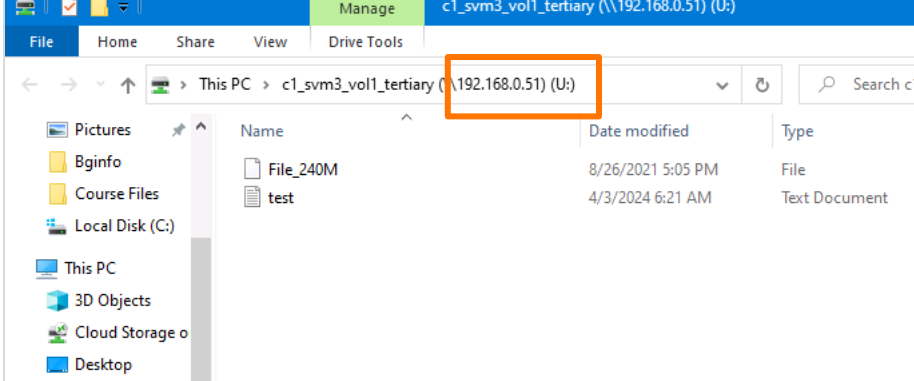
Step	Action
2-7	Switch to System Manager for Cluster3 and, in the navigation pane, select Protection > Relationships. 
2-8	On the Relationships page, verify that the SnapMirror relationship is listed and that it is healthy and that the state is Mirrored. 
2-9	In the System Manager navigation pane for Cluster2, navigate to Storage > Volumes and review the protection status of c1_svm3_vol1_dest. 

Step	Action
2-10	In the Protection > Relationships pane for Cluster3, click the three vertical dots to the right of the source volume, and then select Update to perform a manual update.  The screenshot shows the 'Relationships' pane in the 'Protection' section. A table lists a relationship between source volume 'c2_svm1:c1_svm3_vol1_dest' and destination volume 'c3_svm1:c1_svm3_vol1_tertiary'. A context menu is open for the source volume, with the 'Update' option highlighted by an orange rectangle. Other options include Edit, Delete, Pause, Restore, Test failover, and Break. The left sidebar shows navigation options like Dashboard, Insights, Storage, Network, Events & Jobs, Protection (selected), Hosts, and Cluster.
2-11	In the Update relationship dialog box, retain the default values and click Update.  The screenshot shows the 'Update relationship' dialog box. It contains a description: 'Updates the data from the source volume to the destination volumes. This operation will continue to run in the background. You can check the transfer status in the relationship.' Below this, it shows the source and destination volumes: 'c2_svm1' and 'c1_svm3_vol1_dest' for the source, and 'c3_svm1' and 'c1_svm3_vol1_tertiary' for the destination. There are two radio buttons: 'As per policy' (selected) and 'Select the Snapshot copy'. At the bottom right, the 'Update' button is highlighted with an orange box, next to a 'Cancel' button. A close button (X) is in the top right corner.
2-12	Navigate to Storage > Volumes.

Step	Action
2-13	Click the three vertical dots to the right of c1_svm3_vol1_tertiary and then select Edit > Mount path. 
2-14	In the Edit mount path pane, select the Mount checkbox, review the mount path, and then click Save. 
2-15	In the System Manager navigation pane, select Storage > Shares.

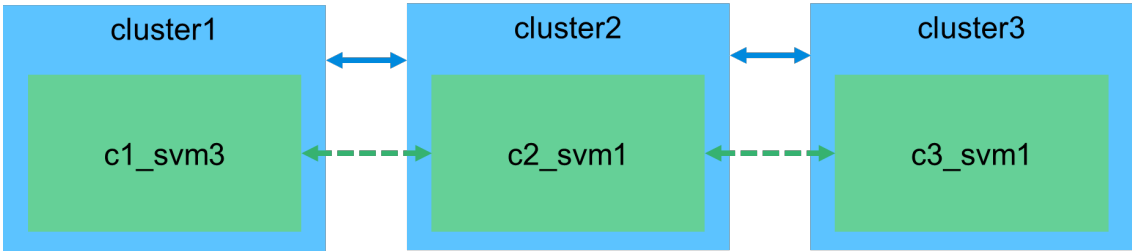
Step	Action														
2-16	In the Shares pane, click Add and then select Share. <table><thead><tr><th colspan="2">Shares</th></tr><tr><th></th><th>Storage VM</th></tr></thead><tbody><tr><td>Home Directory</td><td>c3_svm2</td></tr><tr><td>▼ ipc\$</td><td>c3_svm2</td></tr><tr><td>▼ c\$</td><td>c3_svm1</td></tr><tr><td>▼ c3_svm1_share</td><td>c3_svm1</td></tr><tr><td>▼ ipc\$</td><td>c3_svm1</td></tr></tbody></table>	Shares			Storage VM	Home Directory	c3_svm2	▼ ipc\$	c3_svm2	▼ c\$	c3_svm1	▼ c3_svm1_share	c3_svm1	▼ ipc\$	c3_svm1
Shares															
	Storage VM														
Home Directory	c3_svm2														
▼ ipc\$	c3_svm2														
▼ c\$	c3_svm1														
▼ c3_svm1_share	c3_svm1														
▼ ipc\$	c3_svm1														

Step	Action								
2-17	In the Add share pane, specify the following values: - Share Name: c1_svm3_vol1_tertiary - Storage VM: c3_svm1 - Folder Name: /c1_svm3_vol1_tertiary (or click Browse and select the folder) - Description: <i>Clear field</i> Add share × SHARE NAME c1_svm3_vol1_tertiary STORAGE VM c3_svm1 ▼ FOLDER NAME /c1_svm3_vol1_tertiary Browse DESCRIPTION ACCESS PERMISSION <table><tr><th>User/group</th><th>User type</th><th>Access permission</th><th></th></tr><tr><td colspan="4">No data</td></tr></table> ONTAP adds a default "Full-control" permission for "Everyone" when you either don't specify any permissions or specify for other than "Everyone" user or group. + Add ☐ Enable continuous availability Enable this function to have uninterrupted access to shares that contain Hyper-V and SQL Server over SMB. ☐ Encrypt data while accessing this share Encrypts data using SMB 3.0 to prevent unauthorized file access on this share. Save Cancel	User/group	User type	Access permission		No data			
User/group	User type	Access permission							
No data									
2-18	Click Save and then verify that the new share is listed in the Shares pane.								
2-19	i Next, on the jump host, you map a drive to the destination volume.								

Step	Action
2-20	To map a drive to the tertiary volume on cluster3\c3_svm1, use the folder path \\192.168.0.51\c1_svm3_vol1_tertiary.
	
2-21	Open the mapped drive c1_svm3_vol1_tertiary (\\192.168.0.51) and verify that the SnapMirror relationship transferred the file to the tertiary volume.
	
2-22	 Optionally, you can simulate a disaster on Cluster1 and Cluster2 and attempt to recover data from Cluster3. For example, you can delete a file from the primary volume on Cluster1, take the secondary volume offline, and then attempt to restore the deleted file from the tertiary volume.

Task 3: Review the Exercise Environment

In this task, you review the exercise environment that you created in the previous tasks.

Step	Action
3-1	Verify that the following actions occurred: - You configured intercluster LIFs on each node of the two ONTAP clusters. - You established peer relationships between the Cluster2 and Cluster3 ONTAP clusters. - You created a peer relationship between the storage VMs c2_svm1 and c3_svm1. - You configured a cascading relationship from Cluster1 to Cluster3. 

End of exercise